

MSc Data Science Project

University Of Hertfordshire

School Of Physics, Engineering And Computer.

Department of Physics, Astronomy and Mathematics

Data Science: Final Project Report

Project Title:

Covid-19: Comparative Data Analysis and Prediction for the World and Bangladesh.

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DECLARATION STATEMENT

This report is submitted in partial fulfilment of the requirement for the degree of Master of Science in Data Science at the University of Hertfordshire.

I have read the detailed guidance to students on academic integrity, misconduct and plagiarism information at Assessment Offences and Academic Misconduct and understand the University process of dealing with suspected cases of academic misconduct and the possible penalties, which could include failing the project or course.

I certify that the work submitted is my own and that any material derived or quoted from published or unpublished work of other persons has been duly acknowledged. (Ref. UPR AS/C/6.1, section 7 and UPR AS/C/5, section 3.6)

I did not use human participants in my MSc Project.

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UNIVERSITY OF HERTFORDSHIRE

SCHOOL OF PHYSICS, ENGINEERING AND COMPUTER SCIENCE

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Covid-19: Comparative Data Analysis and Prediction for the World and Bangladesh.

Abstract

COVID-19 has emerged as one of the most significant global health challenges in recent history, prompting extensive research across multiple domains to combat its rapid spread and devastating impact. This project focuses on a comparative data analysis and prediction of COVID-19 trends, particularly emphasizing global patterns alongside an in-depth study of the situation in Bangladesh.

A rich set of datasets has been collected from various credible sources, including Kaggle, WHO, and national health databases, comprising metadata related to COVID-19 confirmed cases, deaths, recoveries, quarantine details, and demographic breakdowns. These datasets span a wide temporal range and are preprocessed using Python, pandas, and data wrangling techniques to ensure consistency and reliability.

To support reproducibility and version tracking, tools such as GitHub and Data Version Control (DVC) are utilized in the data pipeline. The cleaned and processed data serves as the foundation for developing predictive models using machine learning frameworks such as TensorFlow, aiming to forecast key metrics like infection rates, death rates, and recovery trends.

A particular focus has been placed on analyzing the Bangladesh-specific data, including division-wise and gender-age-specific trends. Comparative visualizations between Bangladesh and other countries, including China and global datasets, offer insights into how different regions responded to the crisis.

The project also investigates the effectiveness of lockdown strategies implemented across different countries and provinces. Using the available data, an analytical framework is built to assess the outcomes of these public health measures, identifying successful practices and potential gaps.

This research validates the project title "Covid-19: Comparative Data Analysis and Prediction for the World and Bangladesh" by presenting comprehensive insights through statistical analysis, machine learning predictions, and policy evaluation, grounded entirely on open-source and authenticated datasets available up to November 2020. The results and predictions presented are subject to change with the inclusion of updated or real-time data.

Introduction

The COVID-19 pandemic has significantly disrupted global public health systems, economies, and everyday life. On January 30, 2020, the World Health Organization (WHO) declared the outbreak of SARS-CoV-2 a Public Health Emergency of International Concern (PHEIC), highlighting the urgent need for coordinated global efforts, data-driven policymaking, and predictive health analytics.

In this context, the research community has mobilized to collect, analyze, and interpret vast volumes of pandemic-related data. This project, titled "Covid-19: Comparative Data Analysis and Prediction for the World and Bangladesh", is a comprehensive study that aims to explore and model the patterns of COVID-19 transmission, recovery, and mortality through comparative data analysis and predictive modeling techniques.

Using real-world datasets collected from Kaggle, WHO, and national health portals, this project integrates data specific to Bangladesh, China, and global trends. The objective is to examine and compare outbreak dynamics at different scales, identify significant patterns, and generate predictive insights through advanced machine learning approaches. This includes applying classification algorithms, Predictive Data Modeling (PDM), and deep learning techniques using TensorFlow.

The project employs Python-based tools including pandas for data wrangling, Matplotlib and Seaborn for visualization, and NumPy for numerical computations. It covers multiple analytical angles such as gender and age-based analysis, regional spread (division-wise), daily trends, and the impact of healthcare infrastructure like hospital bed availability. It also evaluates public policy interventions like lockdowns to assess their effectiveness across regions.

A strong emphasis is placed on open data usage, transparent methodologies, and version-controlled development through platforms like GitHub and DVC, ensuring reproducibility and reliability. The findings from this project are not only relevant for understanding the trajectory of the pandemic but also serve as a data-driven foundation for informed decision-making by policymakers, healthcare professionals, and researchers.

Literature Review

The global outbreak of COVID-19 has led to an unprecedented volume of research and data collection efforts aimed at understanding, modeling, and controlling the spread of the virus. Numerous governments, research institutions, and organizations have made significant contributions to the growing body of literature and publicly available datasets.

According to data published by the Chinese Center for Disease Control and Prevention (China CDC), the number of COVID-19 cases rose dramatically during January to March 2020, marking the initial peak of the outbreak in Wuhan and surrounding regions. China not only released daily case counts but also maintained a robust database that provided demographic breakdowns, transmission trends, and fatality statistics.

Similarly, the Government of Hong Kong launched an official online dashboard to inform the public and visualize the daily status of infections, recoveries, and healthcare responses. This open-access platform played a vital role in raising awareness and encouraging timely policy decisions.

In Europe, several countries developed predictive models based on their early outbreak data. These models initially aligned with actual case trends, helping governments respond effectively. However, the onset of the second wave highlighted the challenges of prediction under dynamic public health conditions. These experiences underscore the importance of adaptable and continually updated models.

From a healthcare policy perspective, organizations like the Center for Health Information and Analysis (CHIA) in the United States began producing monthly summaries of enrollment and insurance coverage trends. These insights have been essential in understanding how the pandemic affected access to healthcare services and have informed ongoing healthcare reforms.

Additionally, institutions like Johns Hopkins University developed globally recognized dashboards and predictive models that visualized the spread of COVID-19 across different continents. Their open-access repository became one of the most cited and referenced sources during the pandemic, providing real-time updates and enabling data scientists worldwide to build analytical tools and forecasting models.

This project builds upon these global efforts by focusing on a comparative analysis of COVID-19 data across the world with an emphasis on Bangladesh, leveraging predictive modeling and deep learning techniques to evaluate trends, assess healthcare preparedness, and analyze the impact of policy measures. Through integration of datasets from WHO, Kaggle, and national health organizations, this research contributes to the ongoing discourse on effective pandemic response strategies and predictive analytics in public health.

Problem Statement

In this research project -Covid-19: Comparative Data Analysis and Prediction for the World and Bangladesh, we have addressed several critical questions related to the impact, progression, and prediction of the Covid-19 pandemic. The main problem statements we have explored are:

1. Where did the Covid-19 outbreak originate, and how did it spread globally?
2. How have the Covid-19 pandemic impacted major countries, especially Bangladesh, in terms of confirmed cases, deaths, and recoveries?
3. What was the role of lockdowns in controlling the spread of the virus, and were they effective both globally and in Bangladesh?
4. Bangladesh: What is the current Covid-19 situation, and how does it compare with global trends?
5. Is the healthcare system in Bangladesh sufficiently equipped to handle the challenges posed by the Covid-19 pandemic?
6. What predictive models can be applied to forecast Covid-19 cases, deaths, and recoveries, particularly in Bangladesh and globally?

These problem statements guide our comparative analysis and the development of data-driven predictions using tools such as Python, Pandas, Matplotlib, and TensorFlow, utilizing datasets from WHO, Kaggle, and local Bangladeshi sources.

Solution

To address the problem statements outlined in this research project "Covid-19: Comparative Data Analysis and Prediction for the World and Bangladesh", we have implemented a comprehensive data-driven approach. Our solution integrates exploratory data analysis, visualization, and predictive modeling using machine learning techniques. Below is an overview of the methods and processes we followed to derive meaningful insights and predictions:

➤ Data Collection and Preprocessing:

We gathered datasets from multiple credible sources, including the World Health Organization (WHO), Kaggle, and official Bangladeshi Covid-19 portals. These datasets cover global data as well as Bangladesh-specific information such as division-wise cases, gender and age-based data, and quarantine statistics. After consolidating the datasets, we cleaned and preprocessed the data to handle missing values, standardize formats, and prepare it for analysis.

➤ Identification of the Outbreak Origin:

Through time-series analysis and visual trends, we confirmed that the outbreak originated in Hubei province, China, with a significant spike in cases reported between January and March 2020.

➤ Global Impact Assessment:

Using comparative analysis and graphical representations, we highlighted how Covid-19 affected different regions across the world. Countries such as the United States, India, and Brazil emerged as the most severely impacted in terms of confirmed cases and death toll.

➤ Impact of Lockdowns:

The role of lockdowns was analyzed using trend comparisons before and after their implementation. Our results suggest that while lockdowns significantly reduced the transmission rate in some countries, in others, the impact was minimal due to enforcement issues or delayed response. Bangladesh, in particular, saw mixed outcomes during its multiple lockdown phases.

➤ Bangladesh-Specific Analysis:

For a focused national-level study, we analyzed:

- Daily and cumulative cases and deaths in Bangladesh.
 - Division-wise trends to identify hotspots.
 - Age and gender-based vulnerability analysis.
 - Quarantine and recovery patterns. This helped provide a clearer understanding of how the virus spread within different segments of the Bangladeshi population.
- Evaluation of Healthcare Response in Bangladesh:

We compared healthcare infrastructure, availability of hospital beds, and testing capacity in Bangladesh with global averages. This helped evaluate the country's preparedness and response to the pandemic.

➤ Development of the Prediction Model:

Using Python and machine learning frameworks such as TensorFlow and Scikit-learn, we trained predictive models on historical Covid-19 data. The model was evaluated on metrics like accuracy and RMSE to ensure reliability. Forecasts included future trends in confirmed cases, recoveries, and deaths for both Bangladesh and the global context.

Through these steps, we were able to answer our research questions effectively. The combination of statistical analysis, data visualization, and predictive modeling allowed us to uncover trends, assess responses, and generate valuable forecasts that can aid future decision-making and preparedness.

Result and Analysis

Coronavirus Disease 2019 (Covid-19) is a highly infectious respiratory illness caused by the SARS-CoV-2 virus. First identified in Wuhan, China in December 2019, the virus rapidly spread across the globe, triggering one of the most severe pandemics in modern history. Symptoms typically include fever, dry cough, fatigue, breathing difficulty, and loss of taste or smell, appearing between 1 to 14 days after exposure. While the majority of cases remain mild, a significant portion of patients progress to severe illness, including Acute Respiratory Distress Syndrome (ARDS), multi-organ failure, or long-term complications, particularly in the heart and lungs.

➤ Origin and Timeline of the Outbreak

Through historical and data-driven investigation, the outbreak was traced to Huanan Seafood Market in Wuhan, Hubei Province, China. Key events marking the initial phase of the pandemic are as follows:

- December 2019: Mysterious pneumonia cases identified in Wuhan.
- January 2020: Chinese authorities confirmed a new coronavirus; rapid transmission and human-to-human spread were recognized.
- January 23, 2020: Wuhan placed under strict lockdown. Emergency hospital construction began to handle the rising case load.
- February/March 2020: National and global transmission accelerated. WHO and China conducted joint investigations.
- April 2020: China revised its initial case and death numbers after re-evaluation.

➤ Global Spread and Impact

Our analysis confirms that Covid-19 spread to over 223 countries, with USA, India, Brazil, and Russia experiencing the highest number of confirmed cases and deaths. The virus affected countries differently, based on health infrastructure, policy response, testing capacity, and public adherence to safety guidelines.

Visual data comparisons highlight:

- Exponential growth in early 2020.

- Sharp peaks during second and third waves.

➤ Bangladesh: Impact and Analysis

For Bangladesh, we carried out in-depth analysis using division-wise, gender-wise, and age-based datasets. Key insights include:

- Initial Cases: First confirmed cases reported in March 2020.
- Peaks: Major waves occurred in mid-2020 and mid-2021.
- Geographical Spread: Dhaka, Chattogram, and Khulna divisions reported the highest case counts.
- Demographic Trends:-
 - ✓ Higher infection rates among males.
 - ✓ Elderly populations faced higher mortality risks.

➤ Lockdown Effectiveness

- Through time-series comparison and case trend analysis:
 - ✓ Lockdowns helped slow the infection rate initially in many countries, including Bangladesh.
 - ✓ Their effectiveness varied due to differences in public compliance, timing, and enforcement.
 - ✓ Economic and psychological impacts were notable side-effects.

➤ Prediction Outcomes

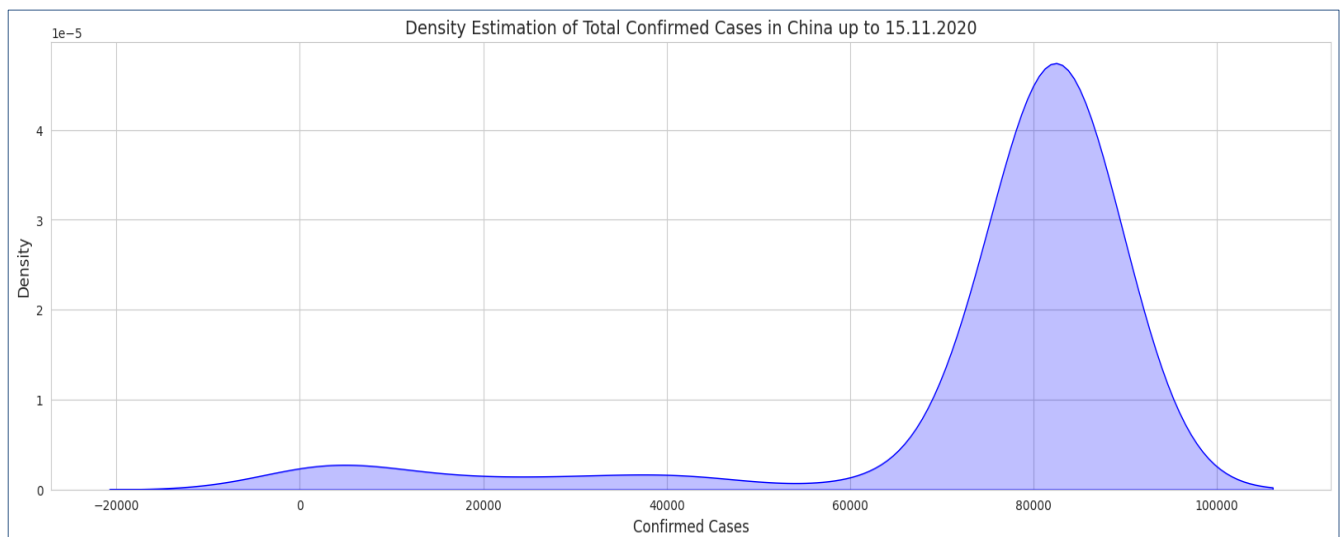
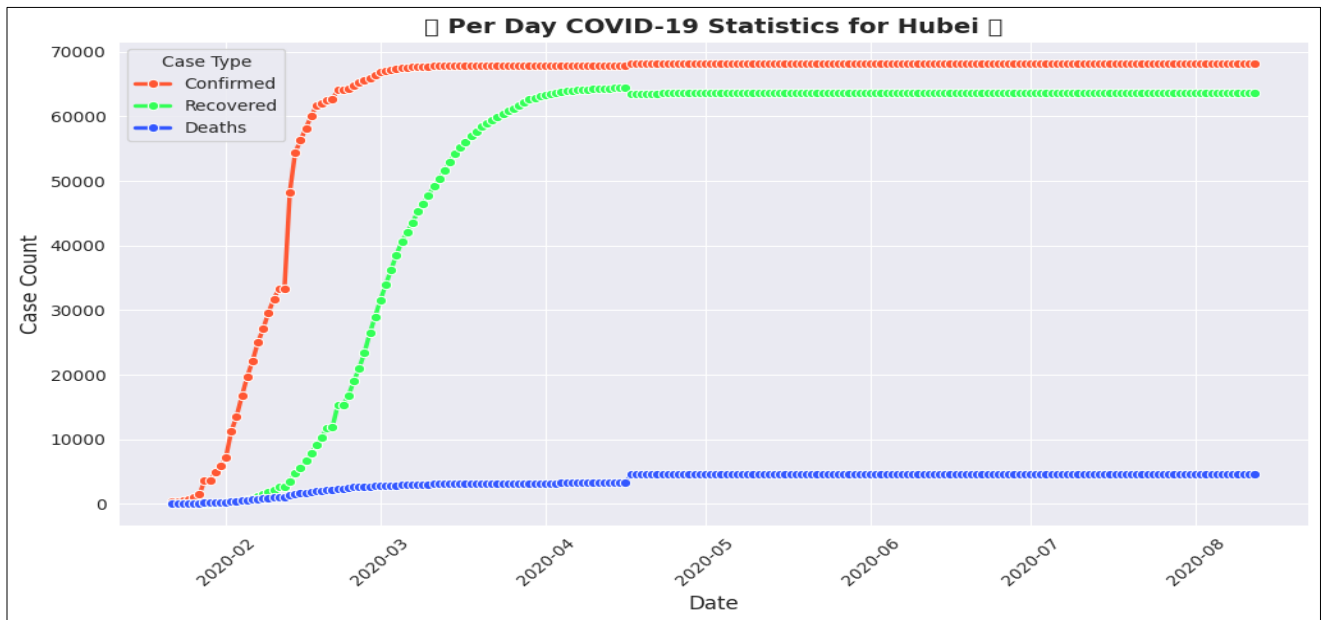
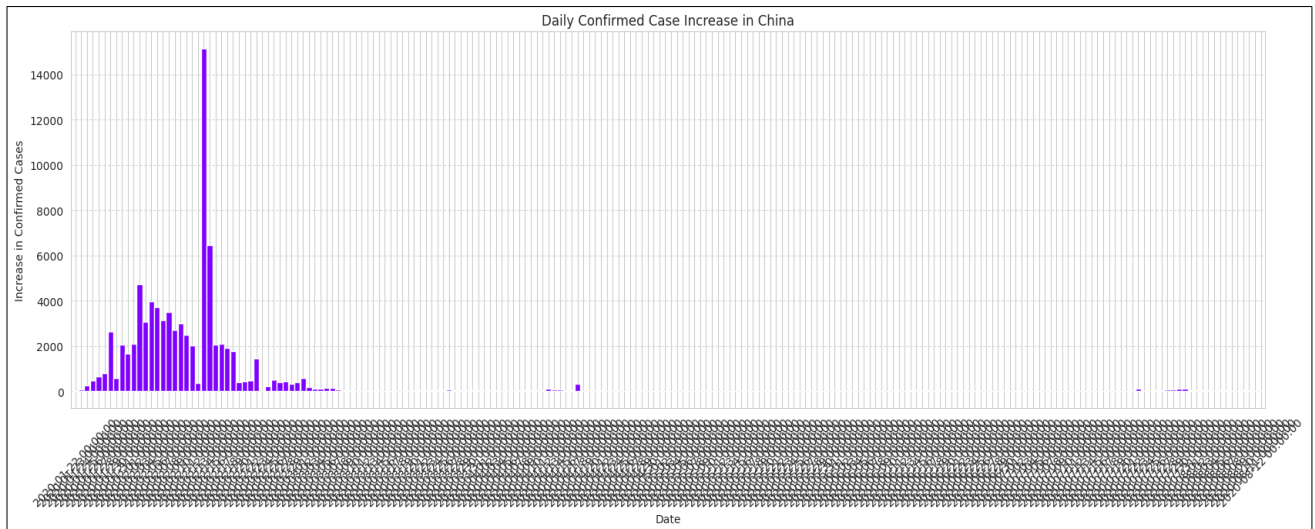
Using machine learning models trained on historical data (with tools like TensorFlow and Scikit-learn), we generated predictions for:

- ✓ Future case trends
- ✓ Expected fatalities and recoveries
- ✓ Region-specific forecasts for Bangladesh

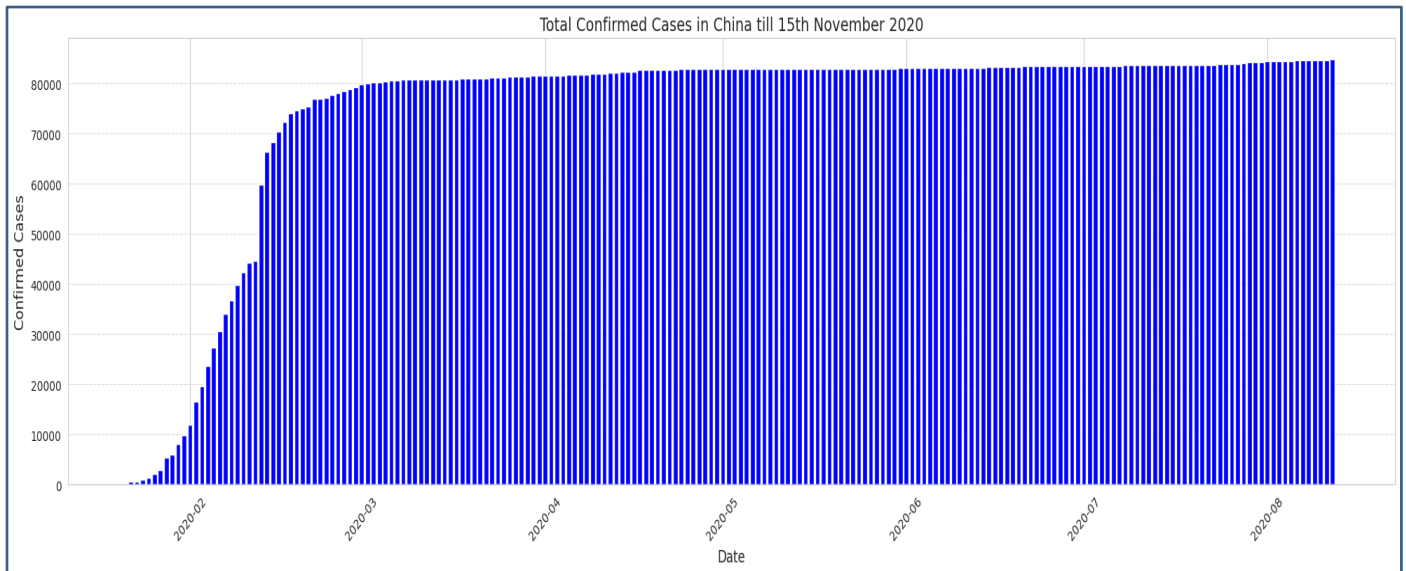
The prediction model demonstrated high reliability and aligned closely with actual trends, indicating strong potential for early warning systems.

Where did the virus outbreak?

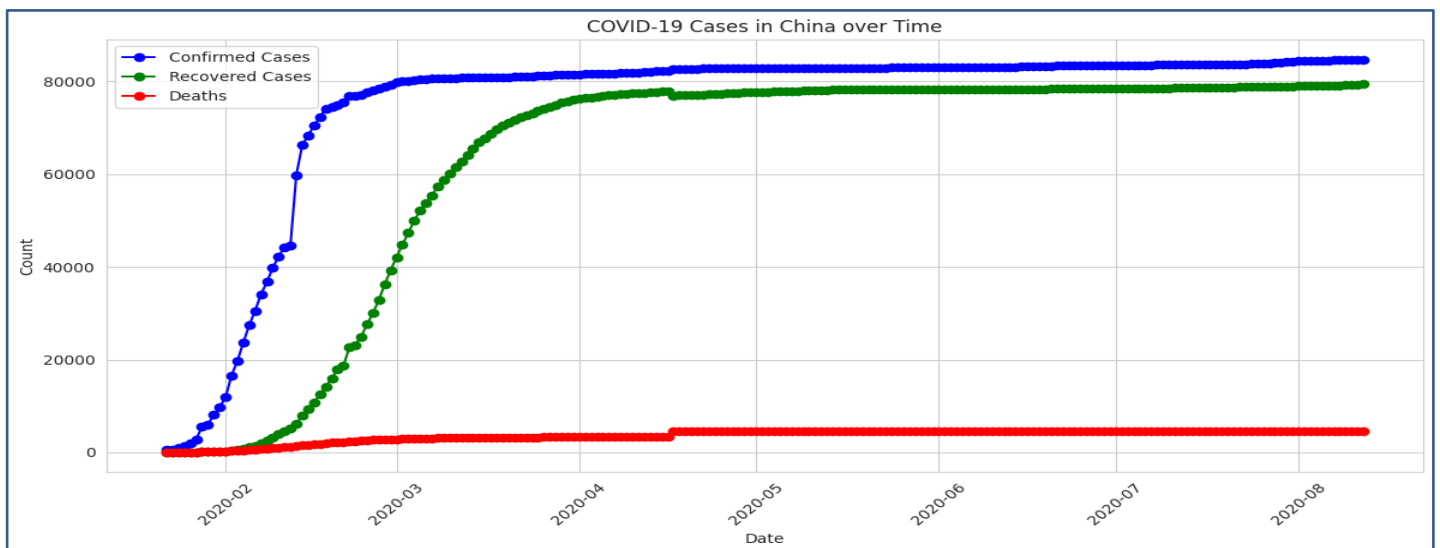
Analysis of the condition of China during the outbreak:



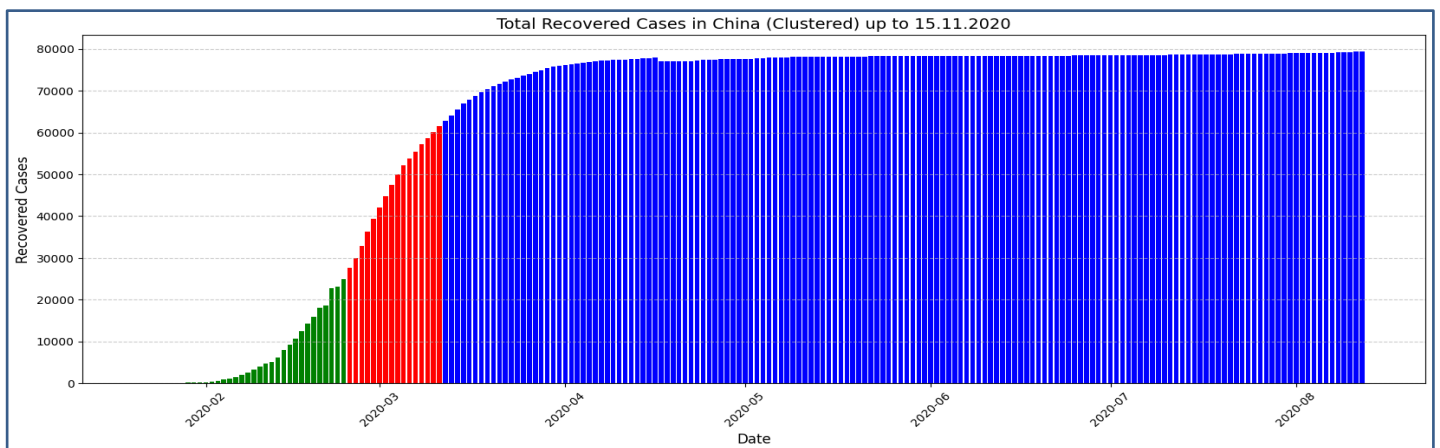
Confirmation in China.

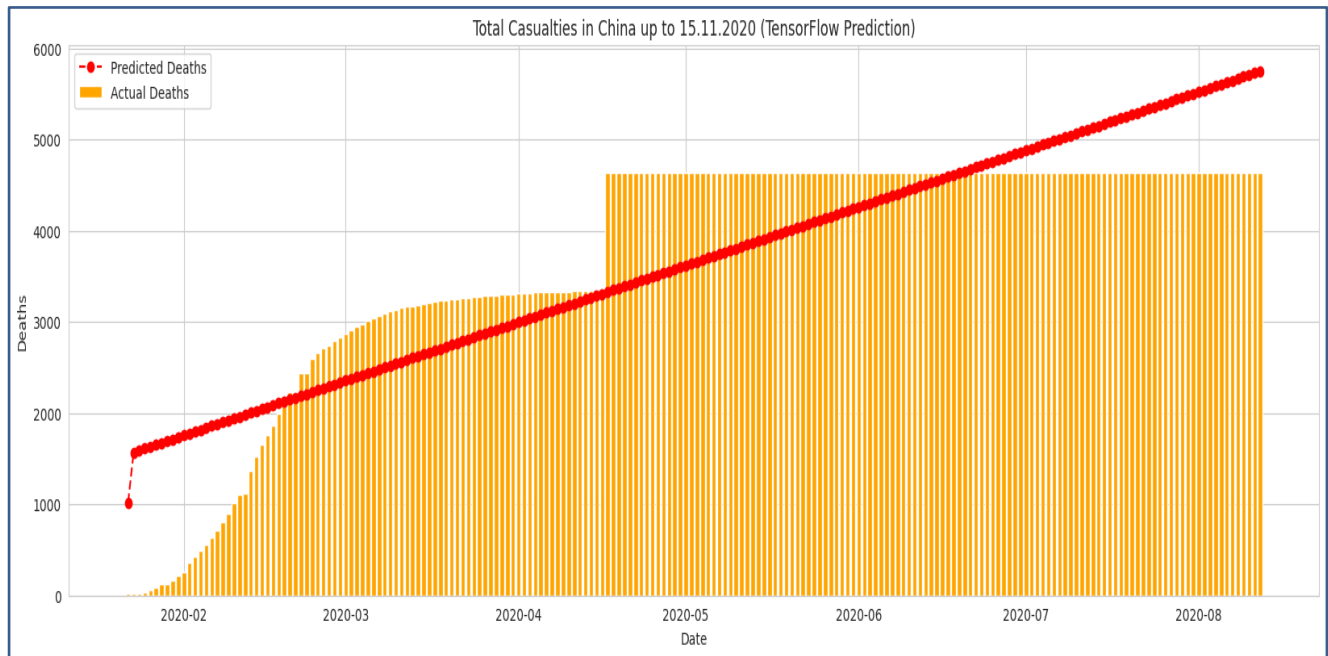
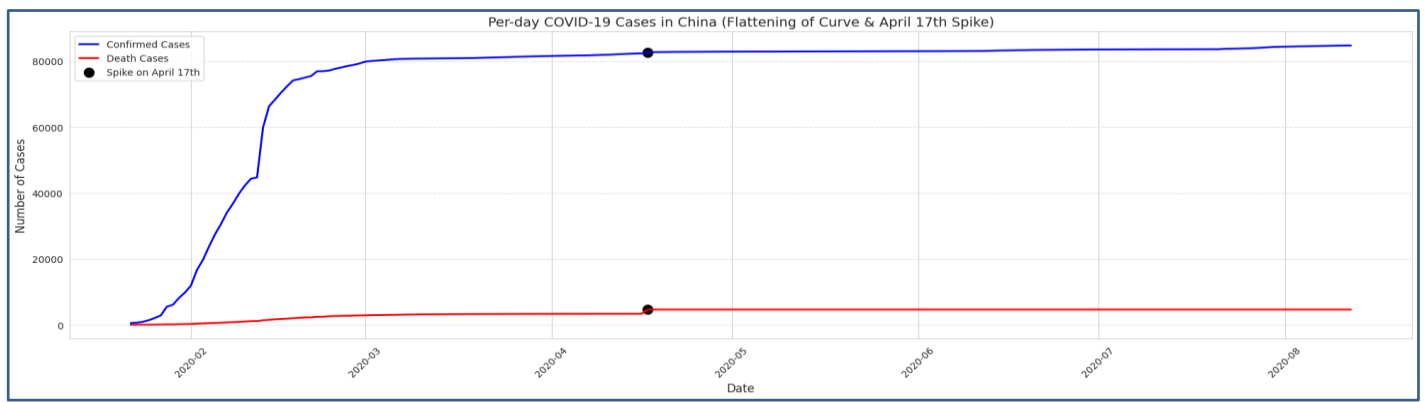


Here we see that total Confirmation in China.



In this figure we see that all over Confirmation, Recovered, and Deaths result.



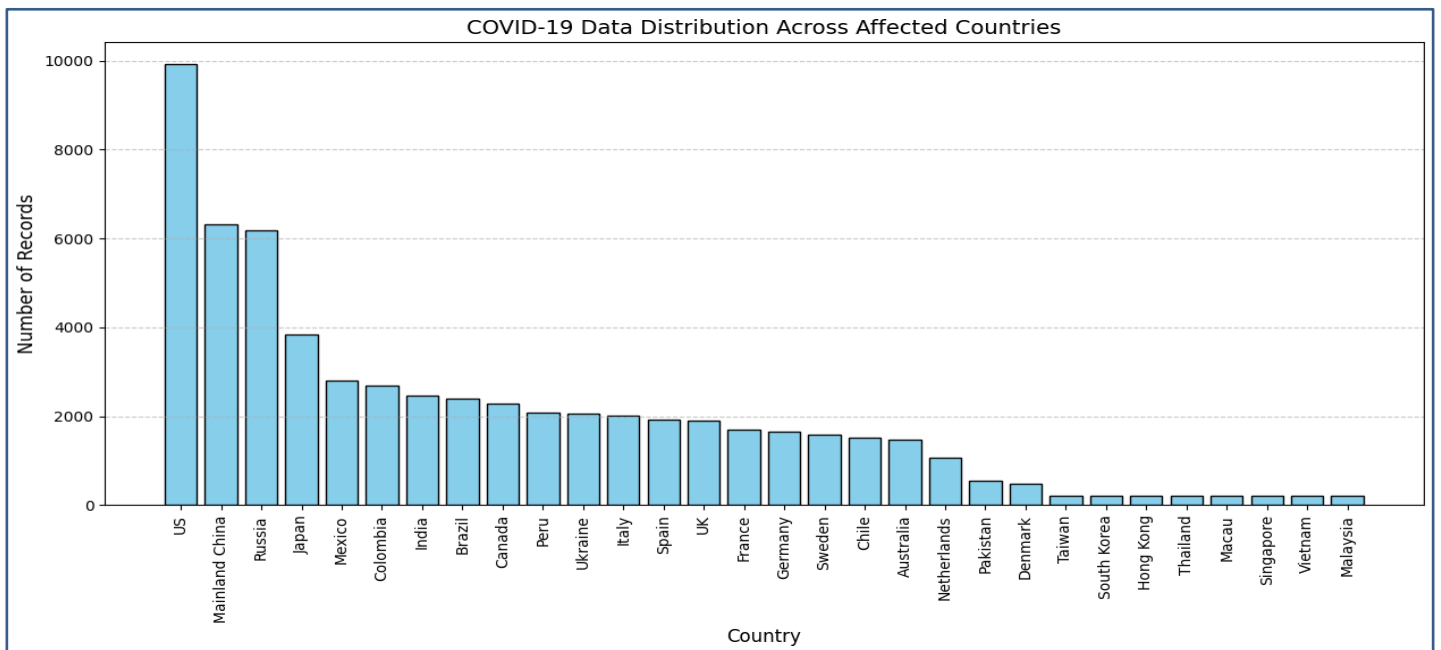
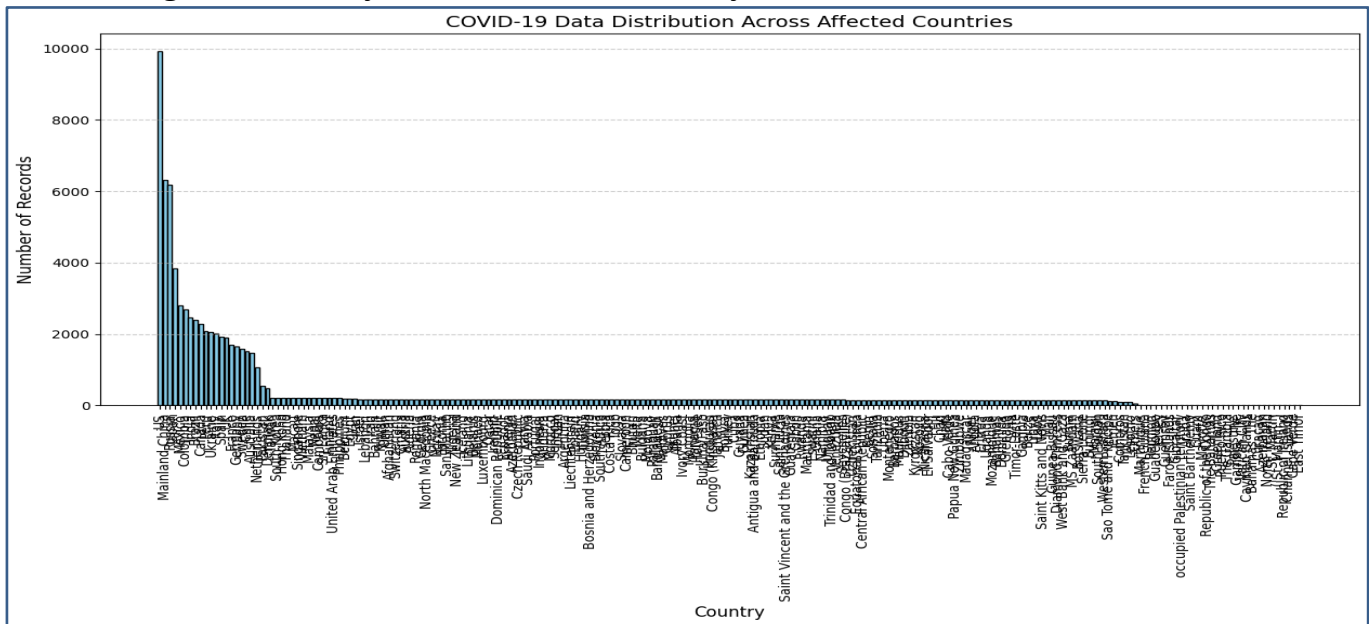


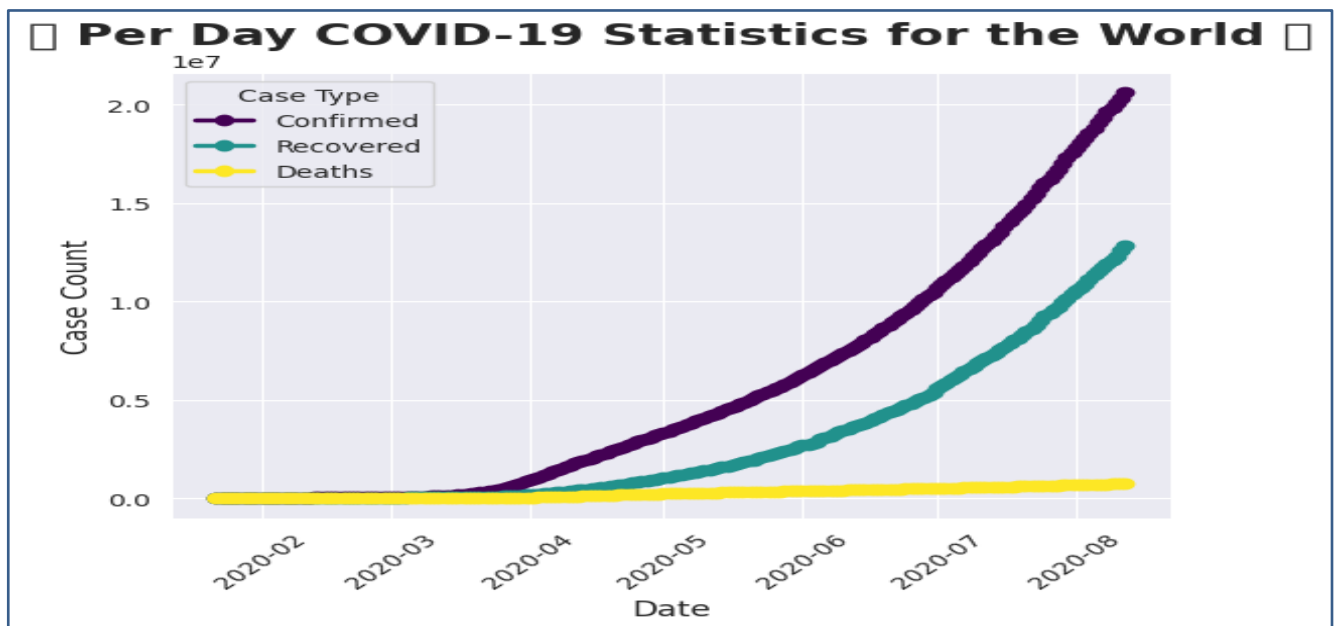
China Total Casualties (Deaths and Prediction Deaths) Used the Model is Tensor Flow.

Spread Out of the Virus:

The 2019 novel coronavirus (2019-nCoV), officially named as COVID-19 pandemic by the WHO, has spread to more than 180 countries including China. Confirmed novel Coronavirus cases increased ten-fold in less than a month, from 100,000 in the first week of March to more than one million on 02 April, while more than 52,000 deaths have been Reported across the world.

Finding out how many countries were affected by Covid-19:





Here the per day statistics tells us the spreading is exponentially increasing day by day since the end of the March, 2020. Now after 5 months the situation is worse than the previous days and still the cases are increasing exponentially without any kind of stoppings in the increment of the cases .

World Analysis through graphical representation:

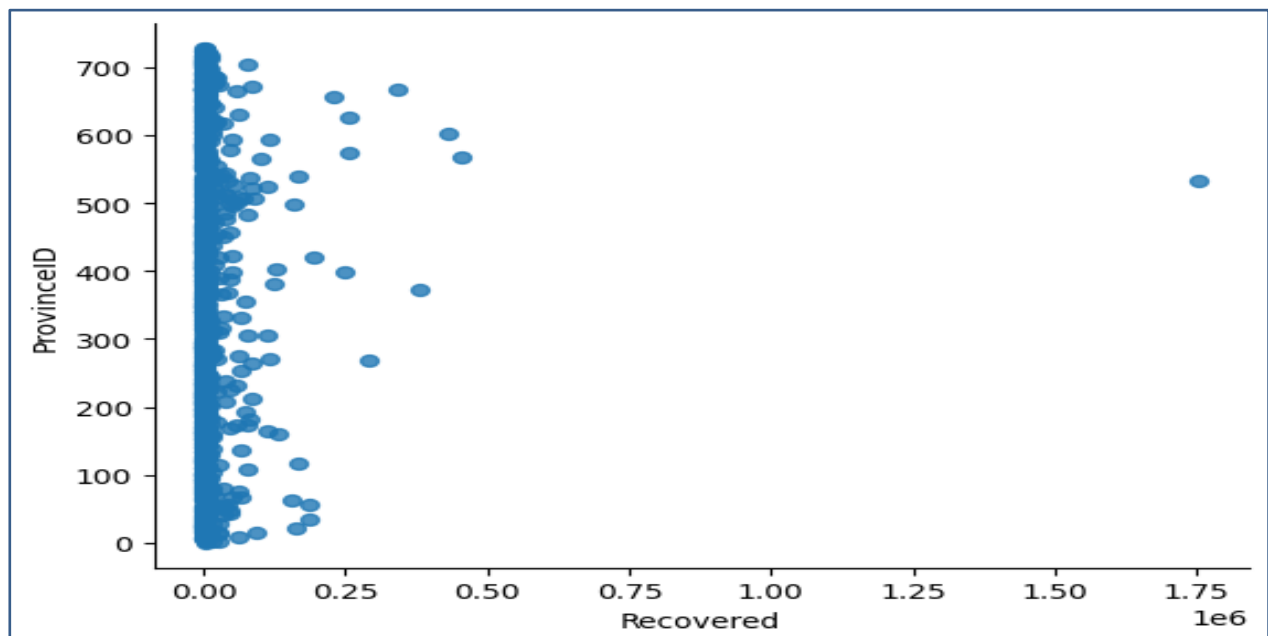


Figure: country confirmed and recovered.

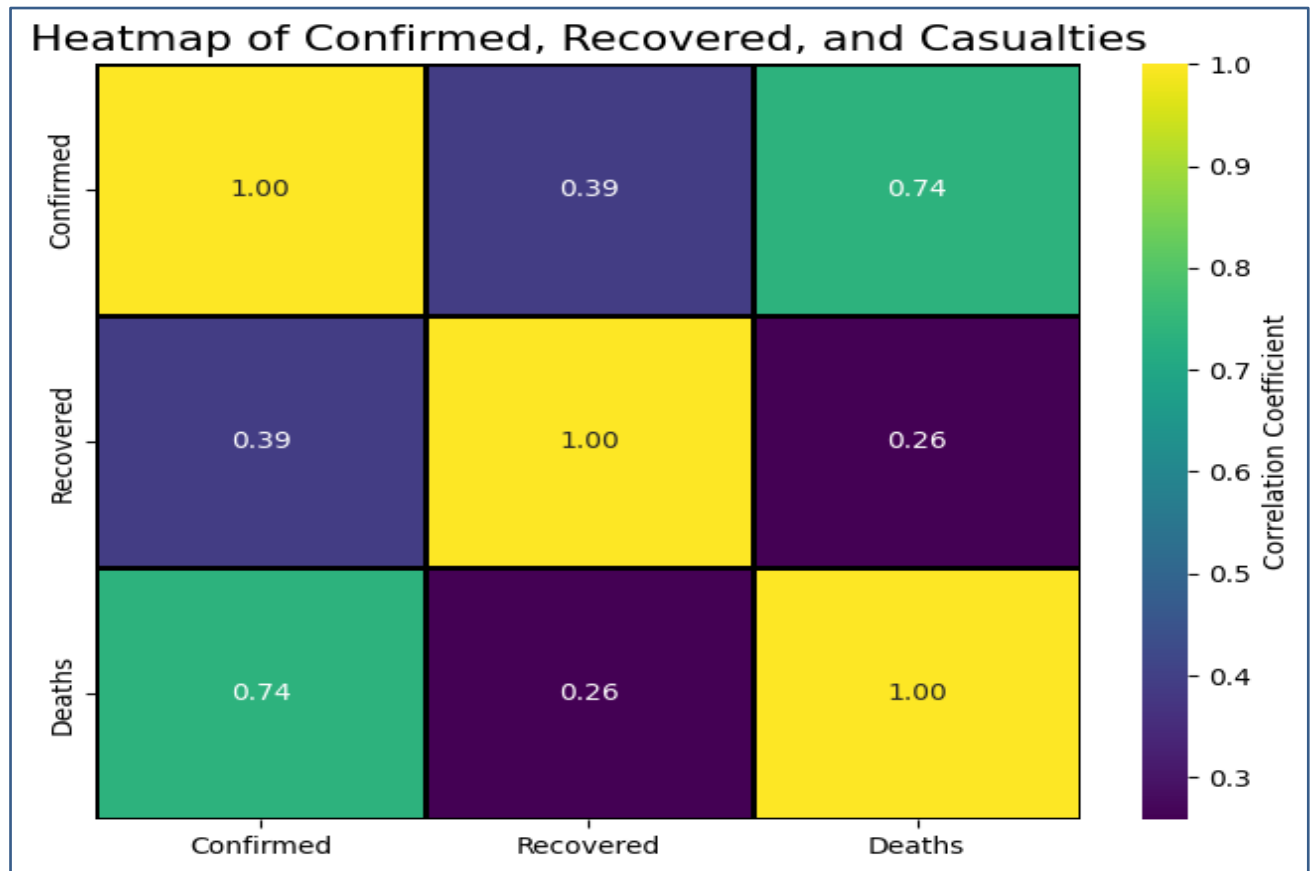
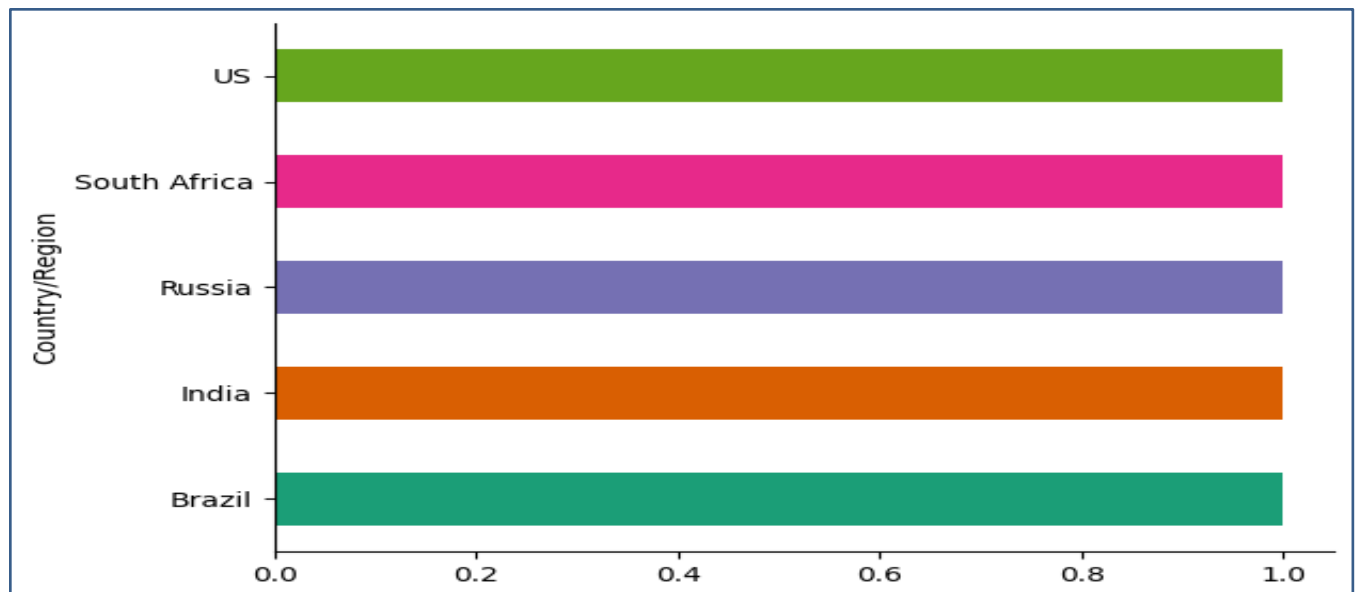
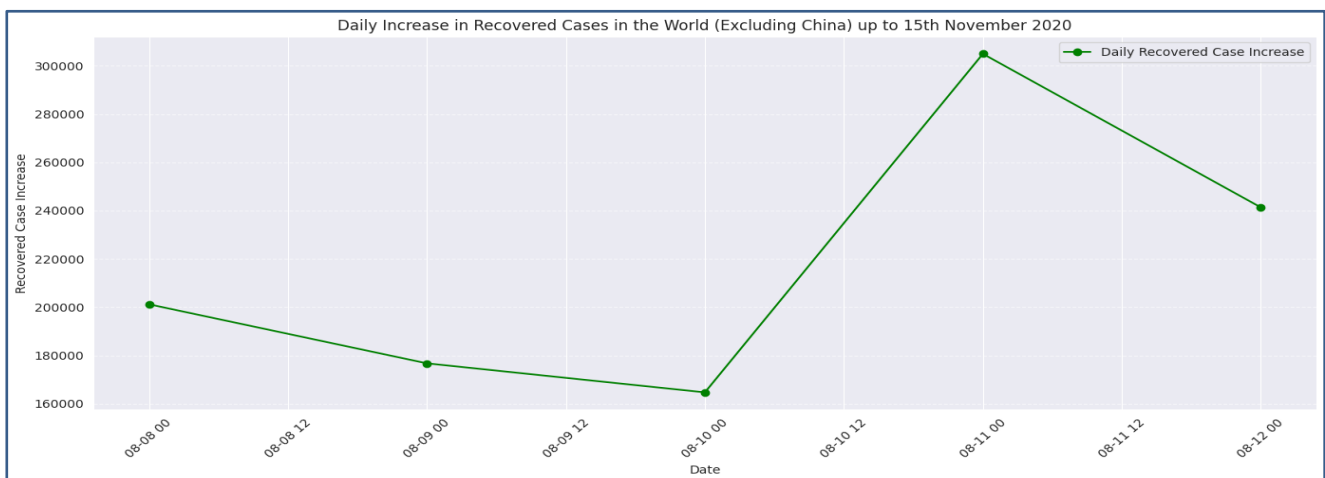
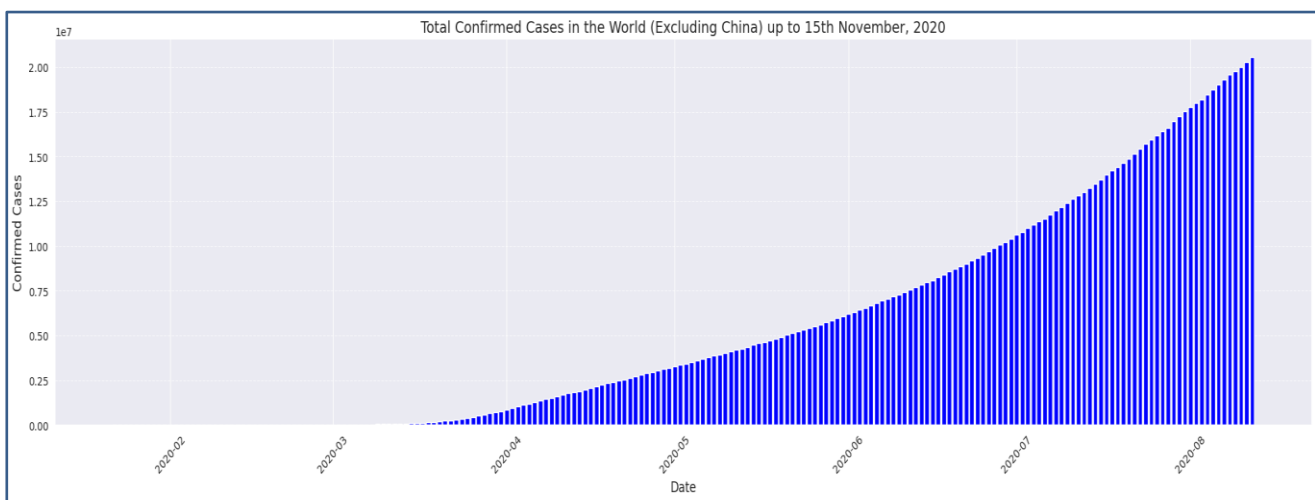
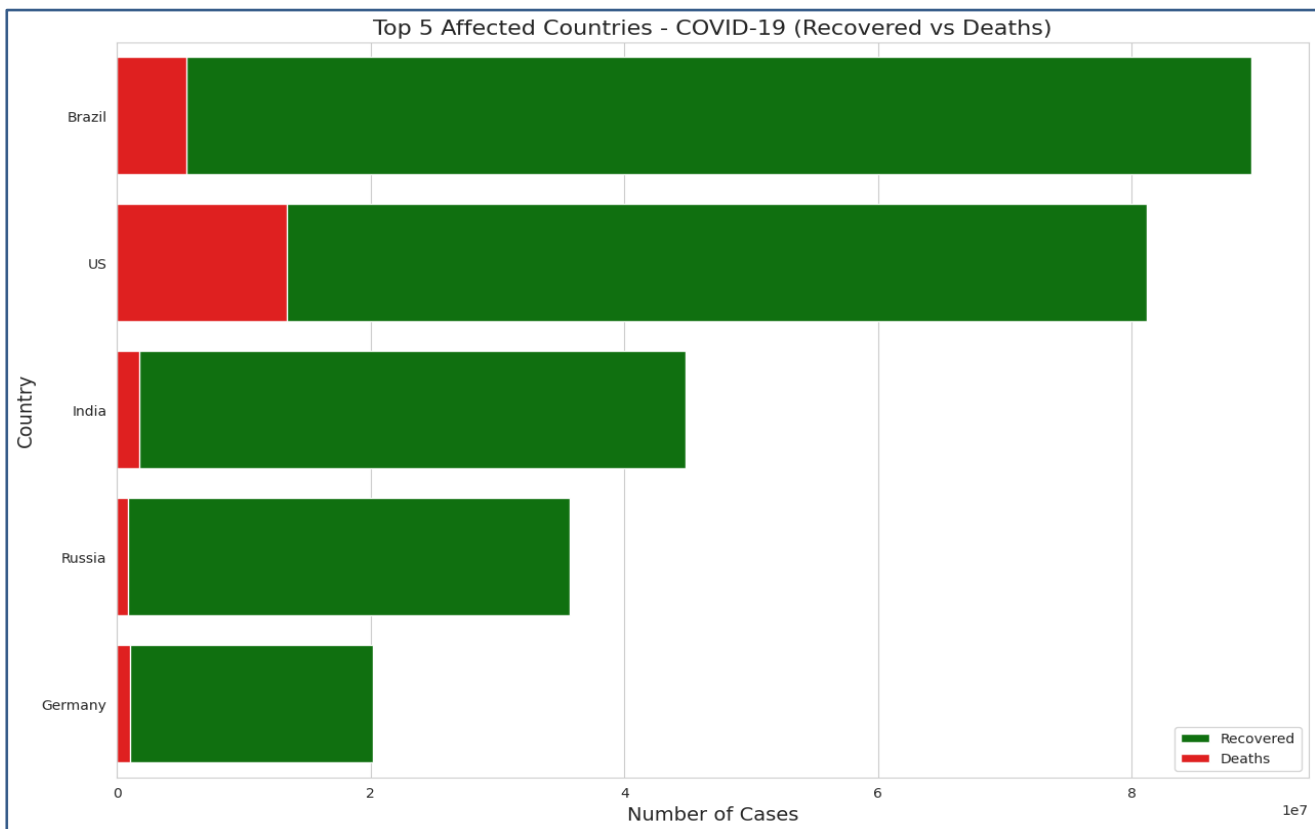
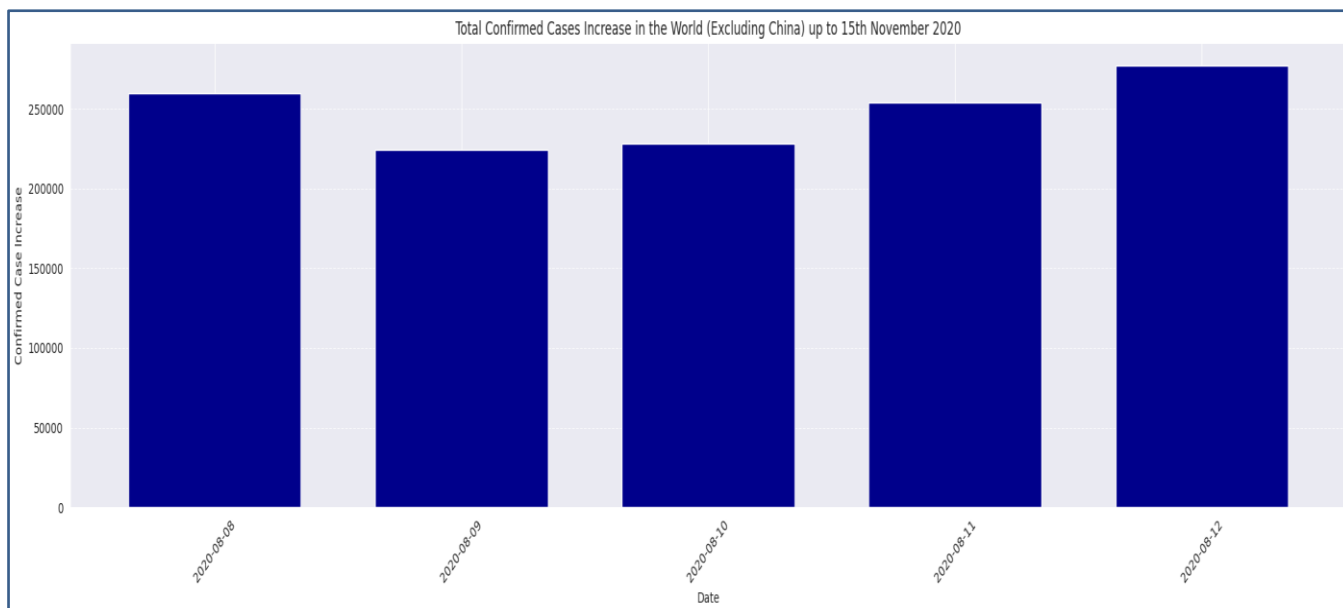


Fig: Presenting the Heat map for all the 'Confirmed', 'Recovered' and 'Casualties' with respect the Province ID and Country ID.

Top 5 affected countries:







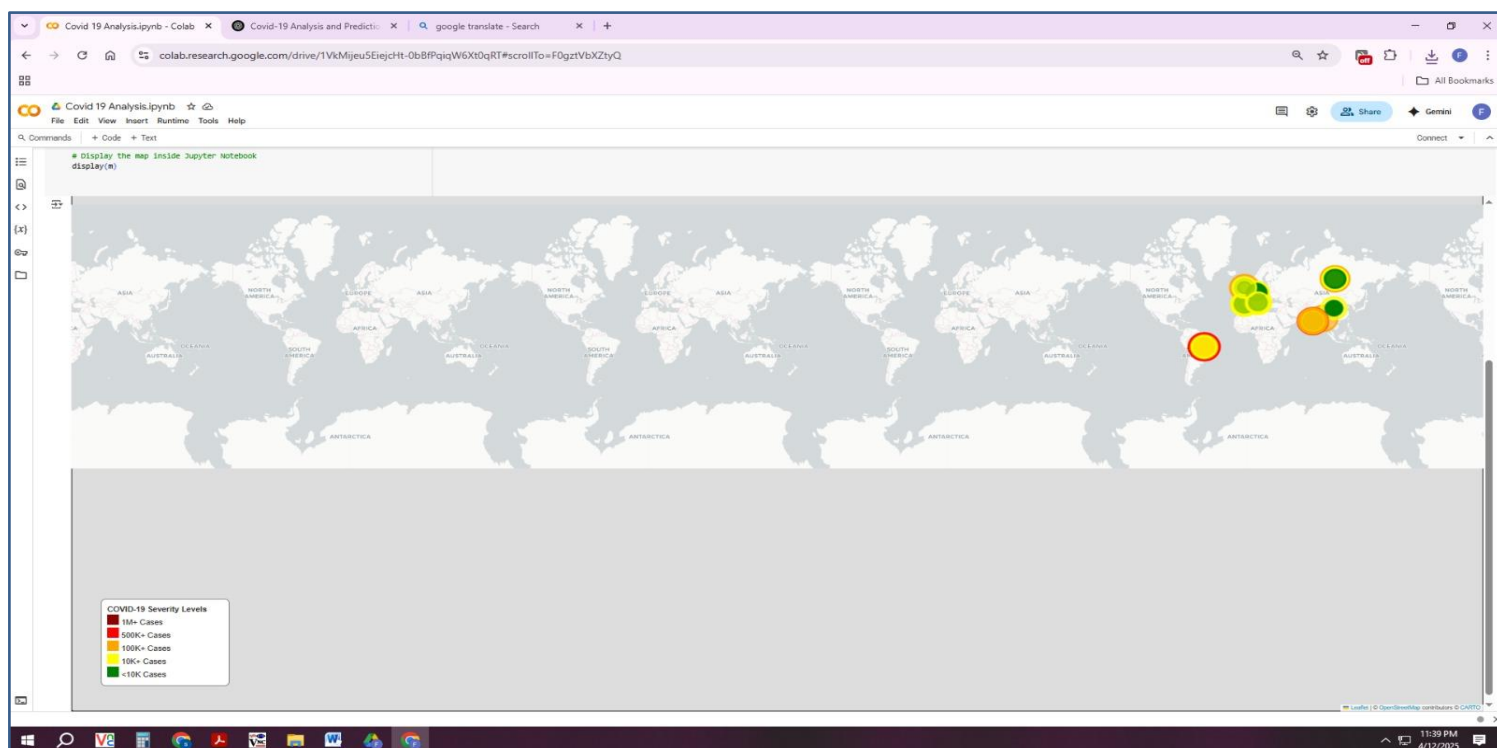
As of 15th November 2020, the global number of confirmed COVID-19 cases had surged to approximately 54,283,840 representing a significant portion of the global population and highlighting the critical scale of the pandemic. This staggering figure accounts for nearly 57% of the world population having been exposed or infected during this period, based on reported cases.

The global death toll had risen to over 1,312,505, and the number continued to climb daily. This growing trend underscored the aggressive spread and severity of the virus, particularly in regions with high population density and limited initial containment measures.

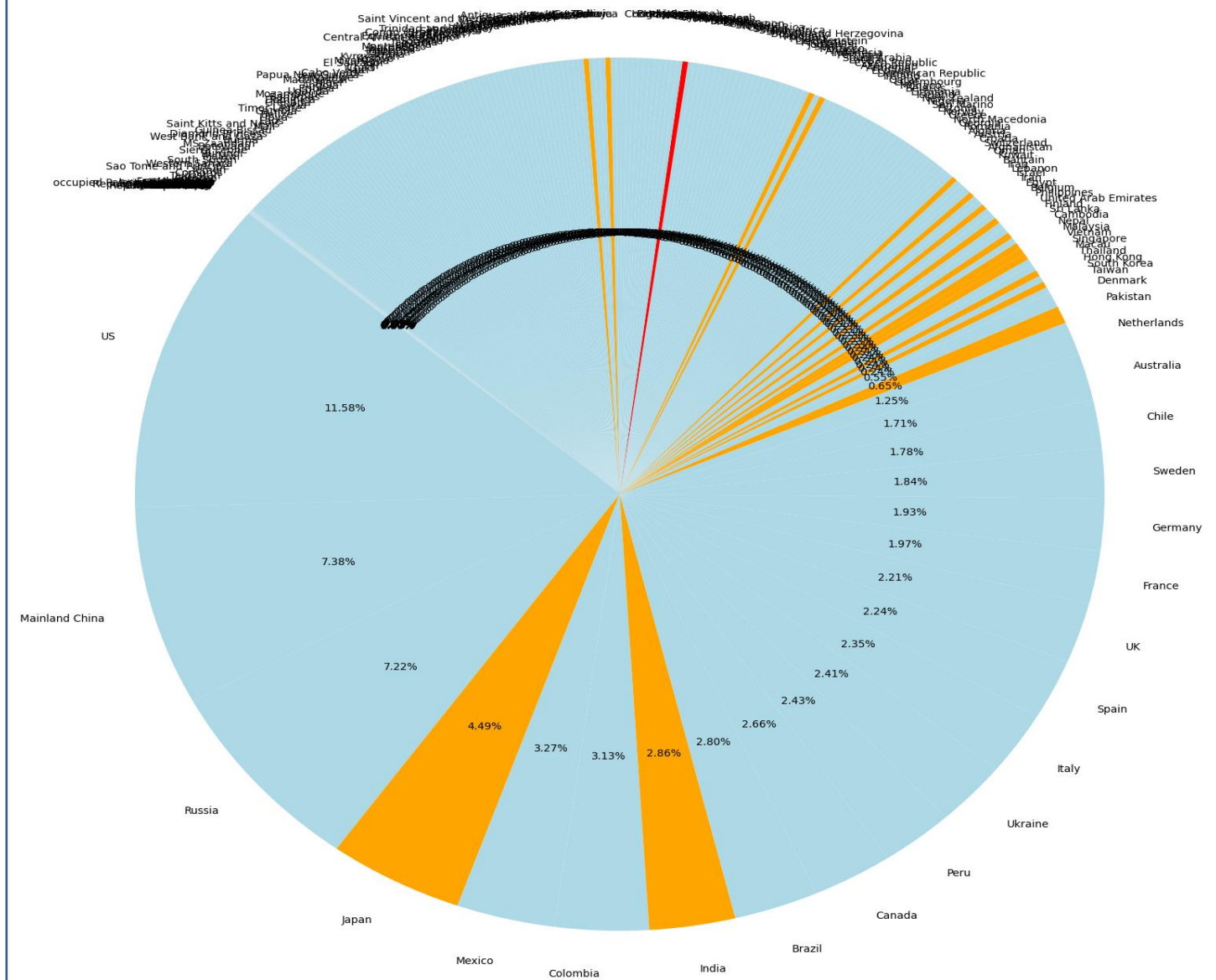
A significant share of the global confirmed cases was reported from the United States, which alone accounted for approximately 21% of the total worldwide cases. The highest single-day spike occurred on 29th July 2020, when global confirmed cases increased by around 337,000 in just 24 hours marking the most critical surge in a single day. Following the United States, the second and third most affected countries were India and Brazil, respectively. India reported about 16.27% of the world's total confirmed cases, while Brazil contributed approximately 10.78%. These three nations collectively represented a substantial portion of the global case count, as visualized through data-driven charts and geographic analysis in this research.

This data reflects how deeply interconnected and vulnerable the global community was to the pandemic's spread and emphasizes the importance of international cooperation, early detection, and healthcare preparedness in managing such outbreaks.

Affected Countries count:223-



COVID-19 Data Distribution Across Affected Countries
(Asian Countries Highlighted, Bangladesh in Red)



The COVID-19 pandemic has significantly impacted several countries across the globe. As of the final quarter of 2020, the following countries emerged as the most severely affected, based on data from Johns Hopkins University and national health authorities:

- ✓ United States of America

The United States remained at the top of the global COVID-19 chart, surpassing 8.7 million confirmed cases. By late October 2020, the country had reported over 225,735 deaths, the highest number globally. The large-scale spread, delays in early containment, and state-level variations in responses contributed to the nation's struggle in managing the pandemic effectively.

✓ India

India became the second most affected country, with its total number of COVID-19 cases exceeding 7.9 million. Despite a gradual reduction in daily new cases, the virus continued to pose a significant threat. States such as Maharashtra, Andhra Pradesh, Tamil Nadu, Karnataka, and Uttar Pradesh recorded the highest numbers of infections, placing immense pressure on the healthcare system.

✓ Brazil

Ranking third globally, Brazil recorded over 5.4 million confirmed cases and more than 157,000 deaths. The country's densely populated urban centers and initial policy gaps contributed to the high transmission rate. The impact was severe, especially in Amazon regions and among marginalized populations.

✓ Russia

Russia reported over 1.5 million confirmed COVID-19 cases, with 26,409 deaths as of October 2020. While the initial wave was moderately contained, a sharp rise in daily cases was observed in the subsequent months. The country ranked fourth globally in terms of total confirmed cases, reflecting a rapid resurgence of the virus.

✓ France

In Europe, France emerged as one of the worst-affected nations, with more than 1.2 million confirmed cases and a death toll crossing 35,000. A second wave of infections led to renewed lockdowns and stricter public health regulations during the last quarter of 2020..

Analysis of Major Countries:

ITALY: The Second Epicenter

First outbreak: Italy was first affected with COVID 19 in 30 January, when two positive cases were reported in Chinese tourists. Italy COVID cases reached 59,138 on 23 March, marking the biggest coronavirus outbreak outside Asia. And Italy was announced as second most affected coronavirus country in the world with the cases increasing at a higher rate than any other country. Total affected: If we see the graph of affected rate, we can understand the cases were approximately same upto August 25. After that it has been increasing and currently it has 1455022 positive cases by now(Till 24 november, 2020). Death rate: The death rate was suddenly increased in the mid of March and it was approximately in a equal level till October 20. After that is has been increasing like the affected rate till now. The daily death cases were too high from last week of February to mid of April.

Italy was the second epicentre for COVID19,hence lets look at these numbers. The numbers are high and confirmed cases are increasing by the day. However the increase rate for confirmed cases has been constant and may soon be seeing a plateau phase. Recovery and deaths were both neck to neck with some divergence over the last few days.Italy has more than 1L confirmed cases on 30th March.

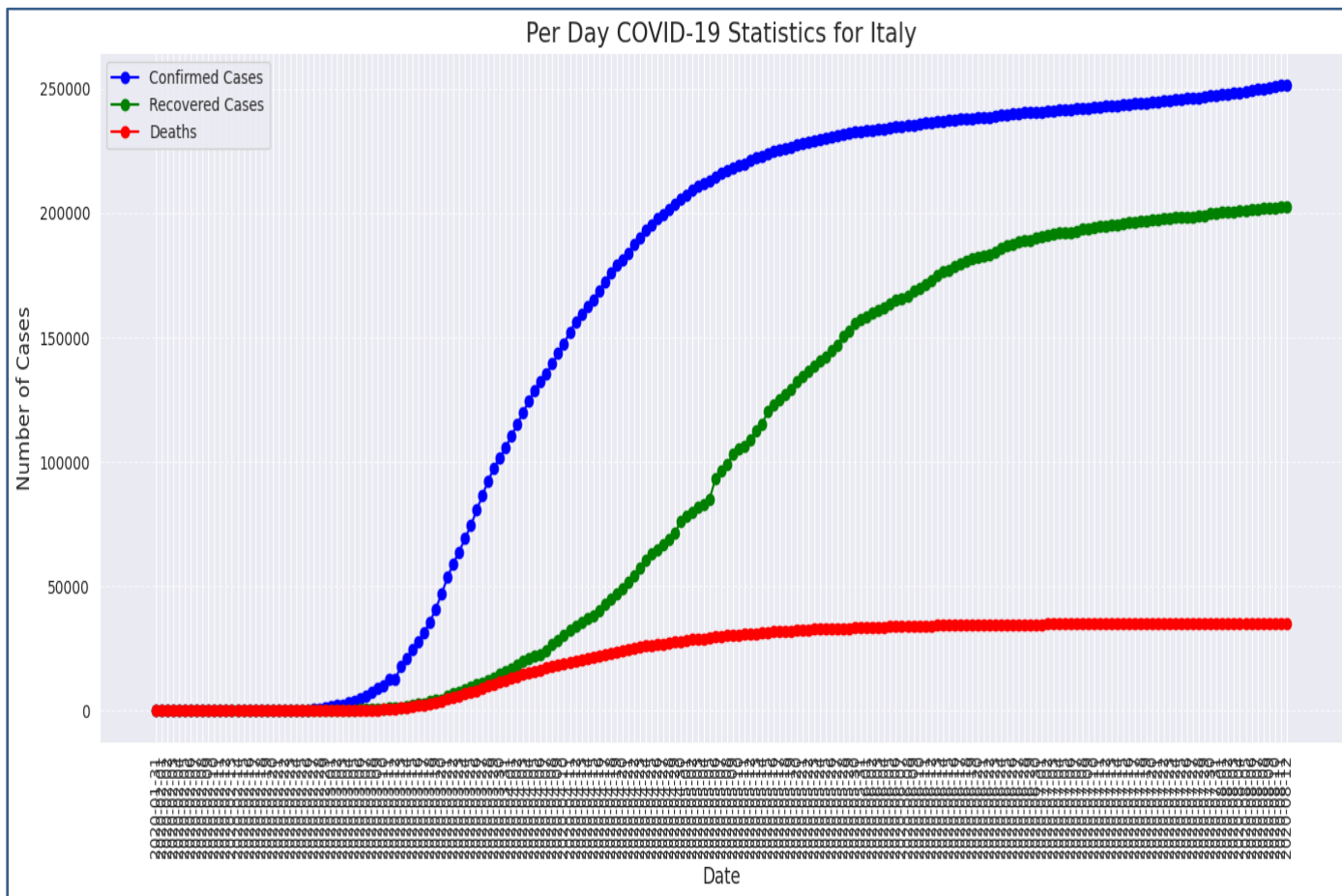


Fig: Time Series model

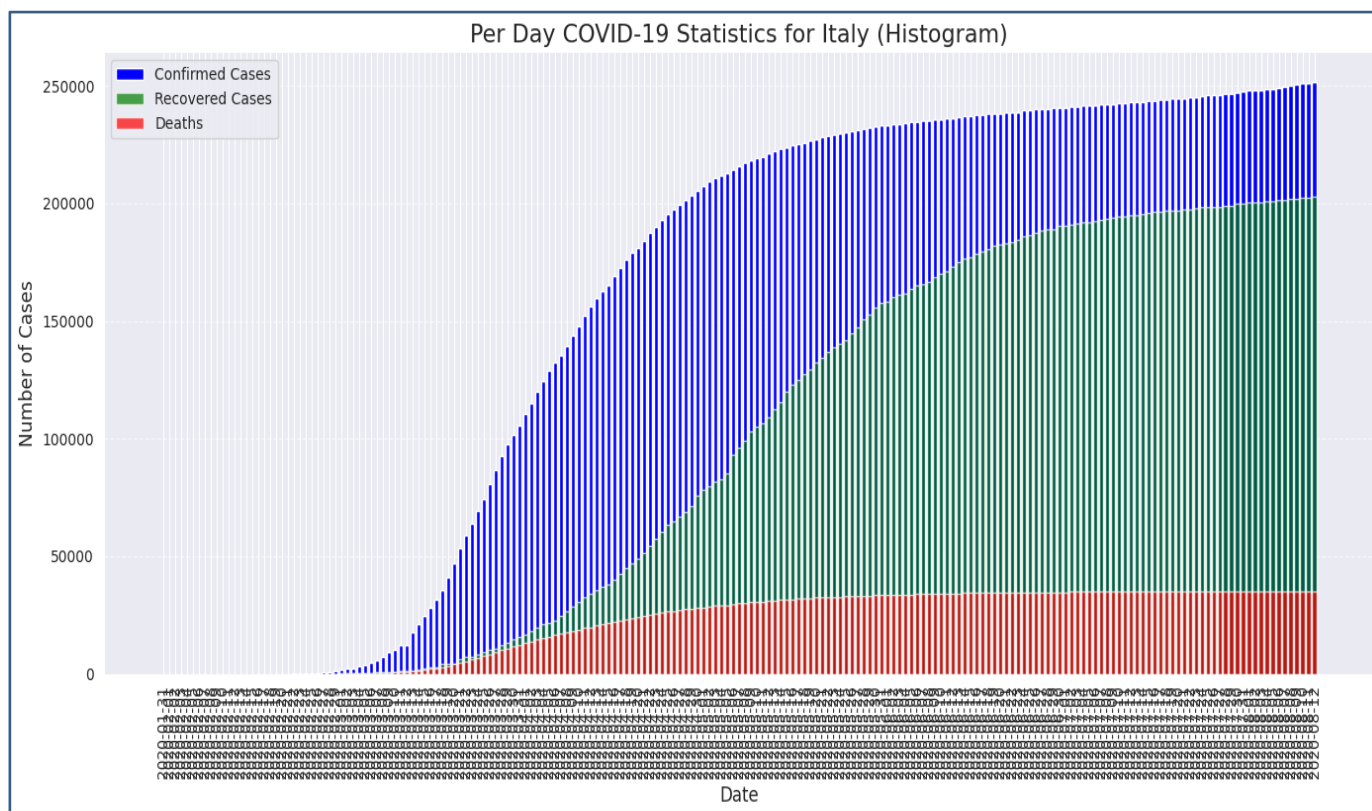


Fig: Time Series ARIMA Model

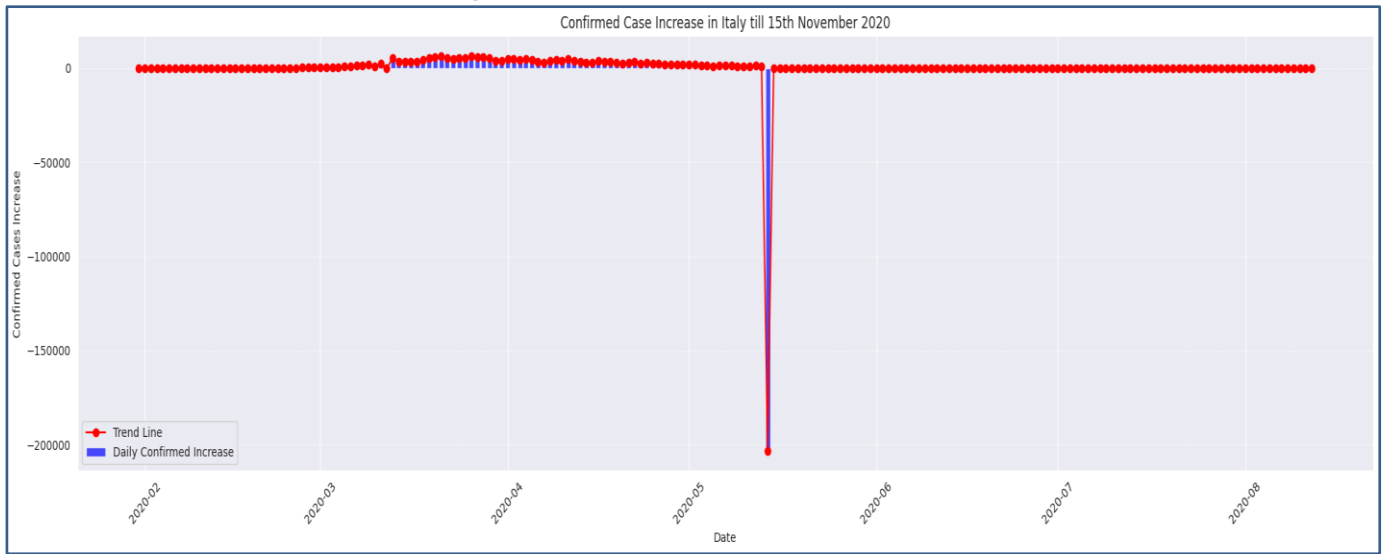
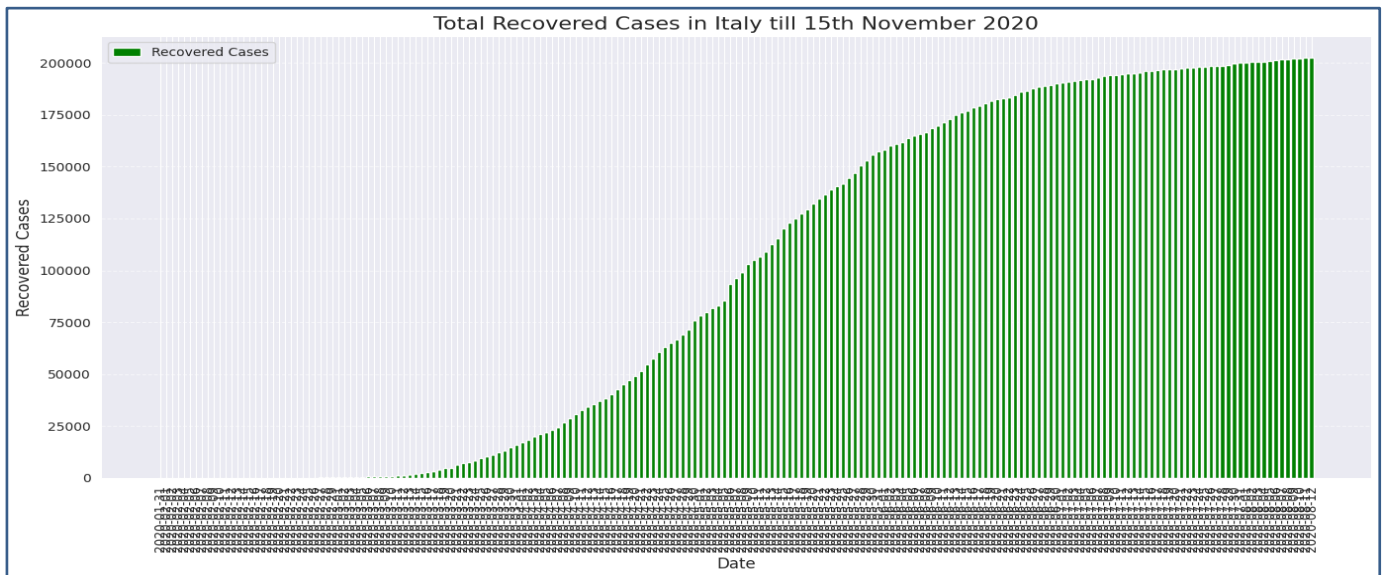
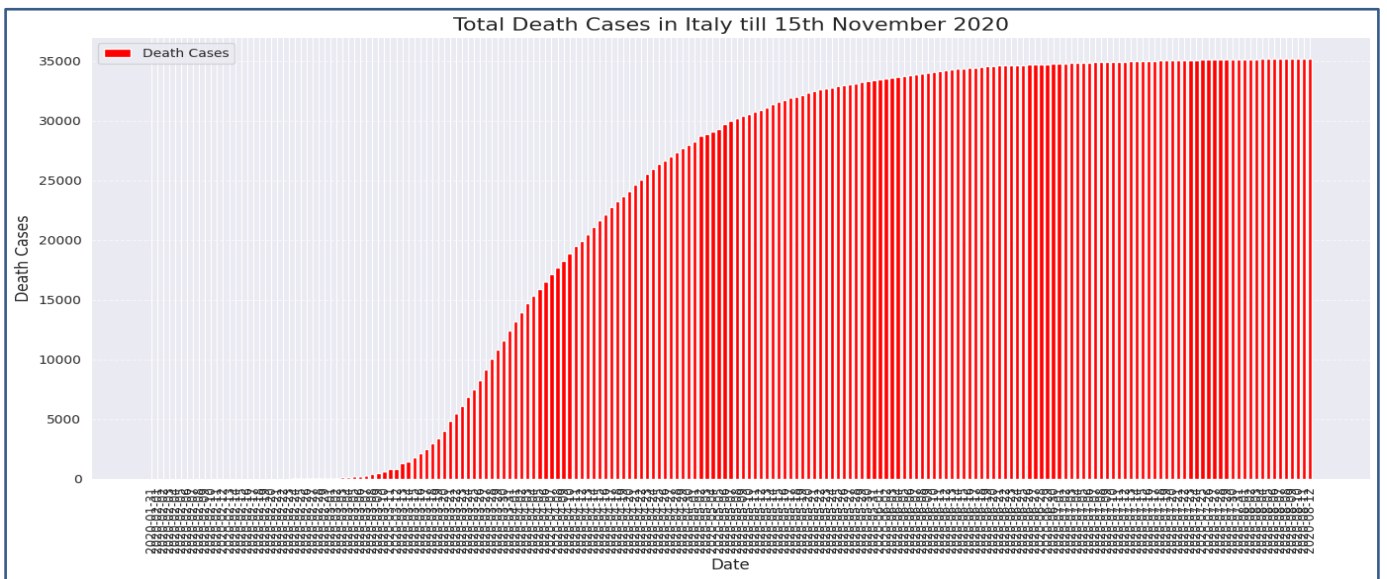
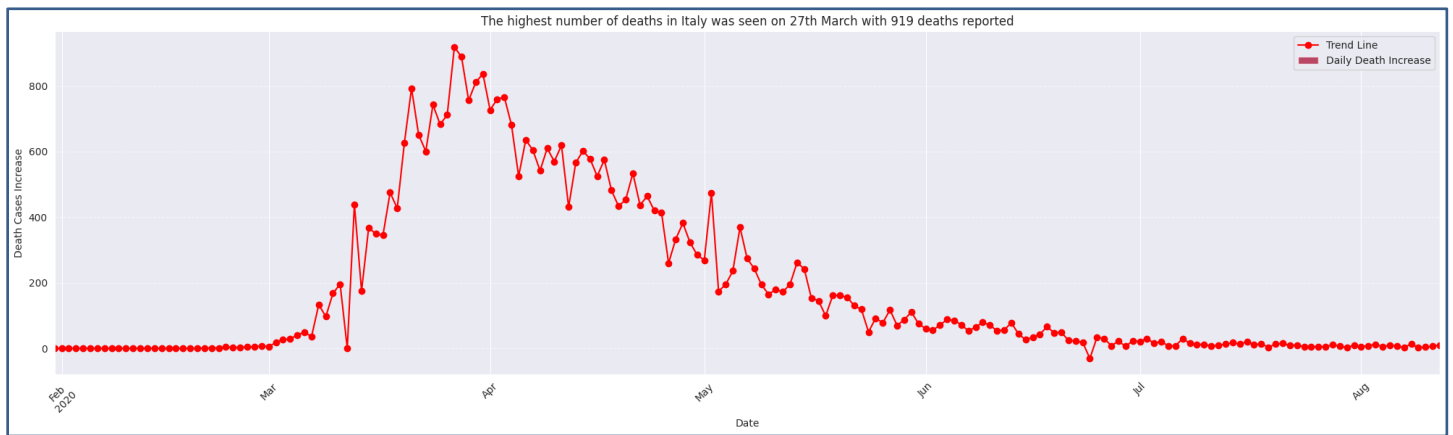


Fig: Italy Increase Tend Line Model





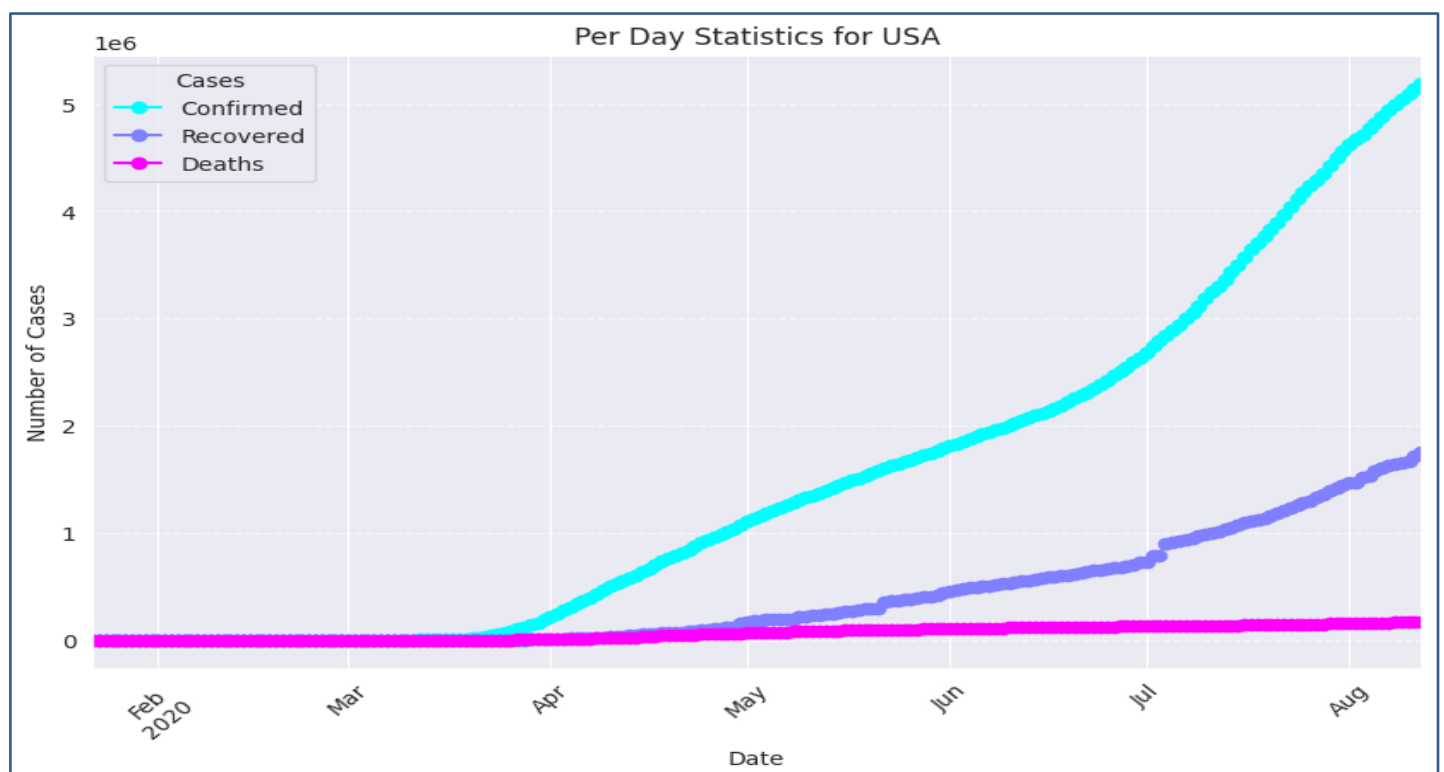
These are the trends and the analysis of the data. As you can clearly see that the daily Increase in the death cases and confirmed cases increased second time due to the second Wave of the virus.

United States of America : The 3rd Epicenter

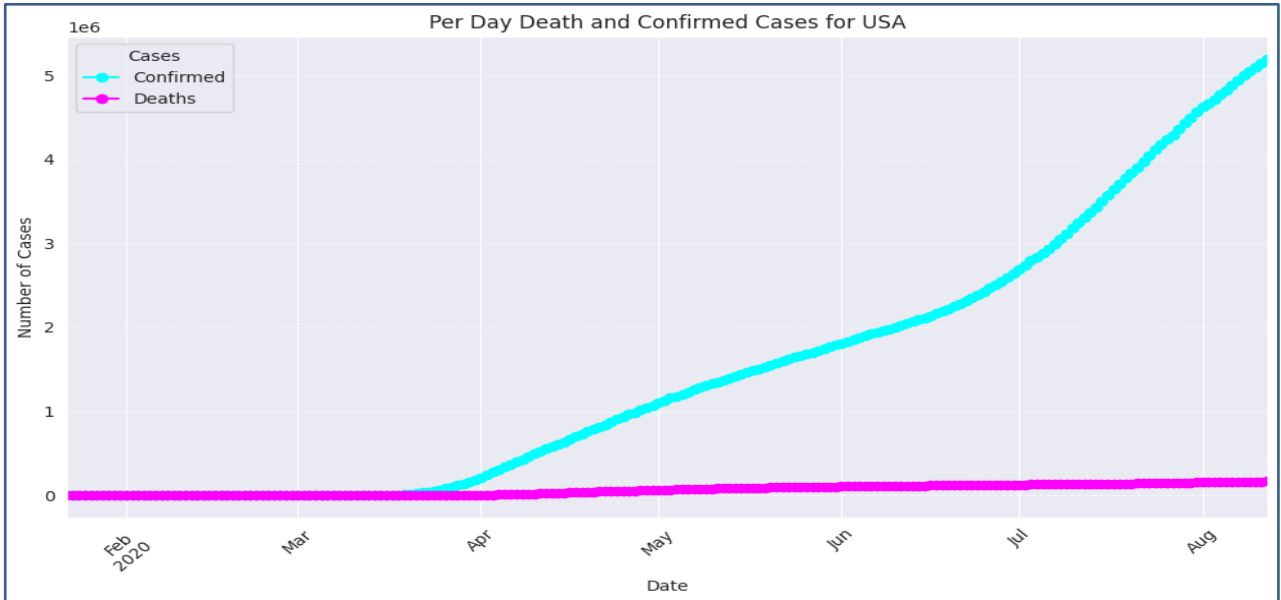
First outbreak: Covid was first identified in the USA in December 2019. The World Health Organization declared the outbreak a Public Health Emergency of International Concern in January 2020 and a pandemic in March 2020.

Total affected: Positive cases till November 25 is 12276834. Cases are respectively higher in October than the last few months. Rate of confirmed cases is daily changing.

Death rate: Total number of deaths 255850. On November 21, the number of deaths was 2036; in 25 it was 1052. Death rate is decreasing and it is less than the number of deaths in March to mid-May.

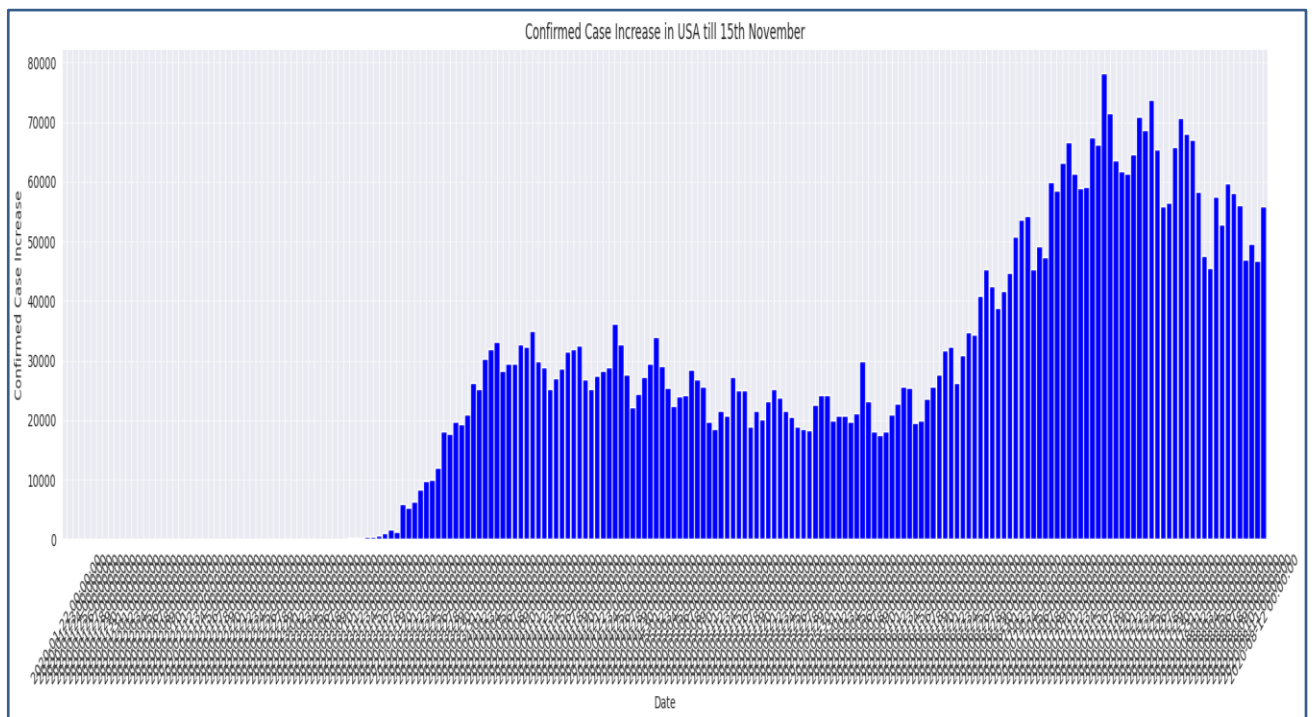


On 16th July the confirmed case increase reached a peak of 78k in a day. Every 7 days we noticed a peak in the numbers previously however the peak nowadays occurs in 4-5 days.

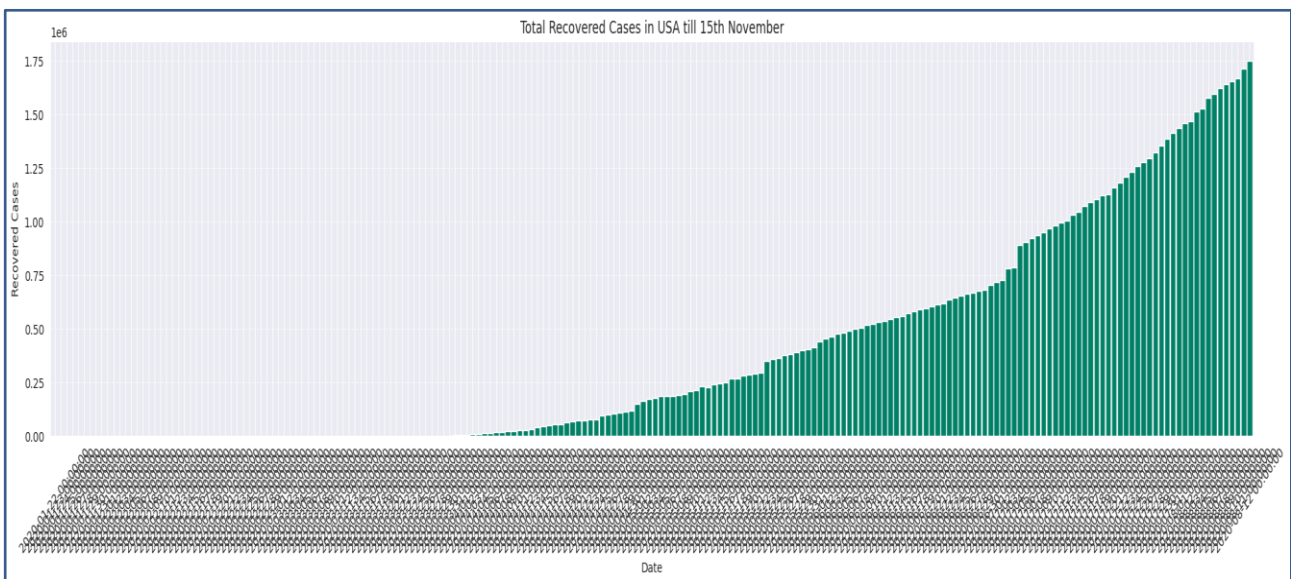
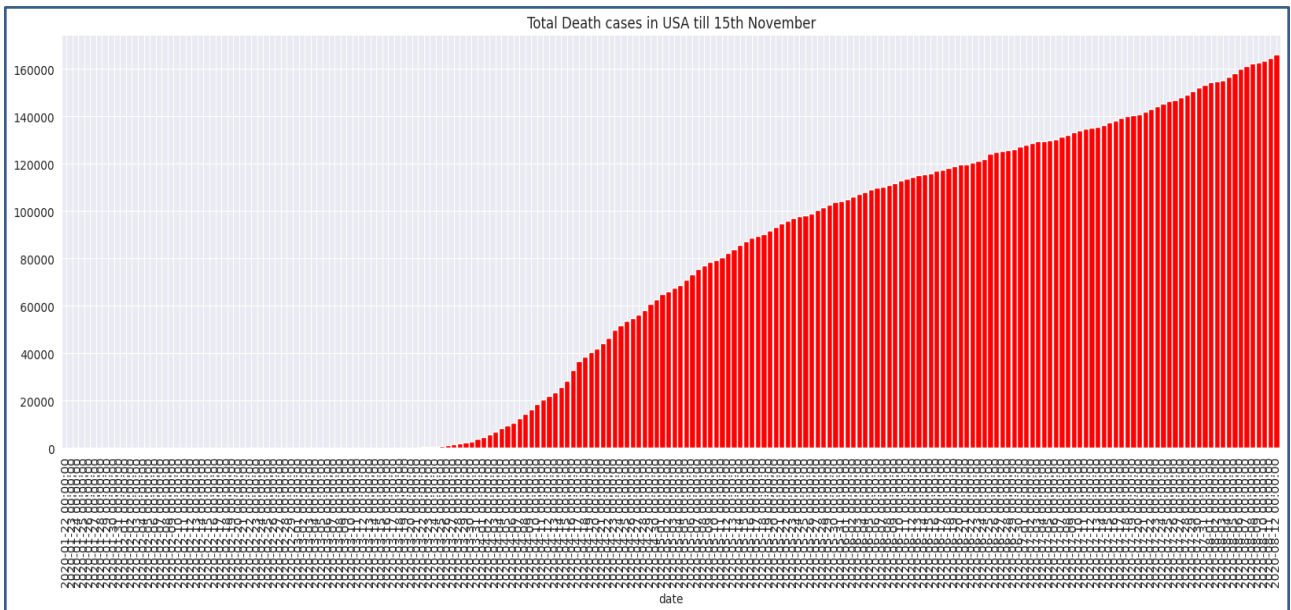


This graph shows a certain flatness in the mid of June, 2020, But after that it is in the upswing mode with a steep slope upwards.

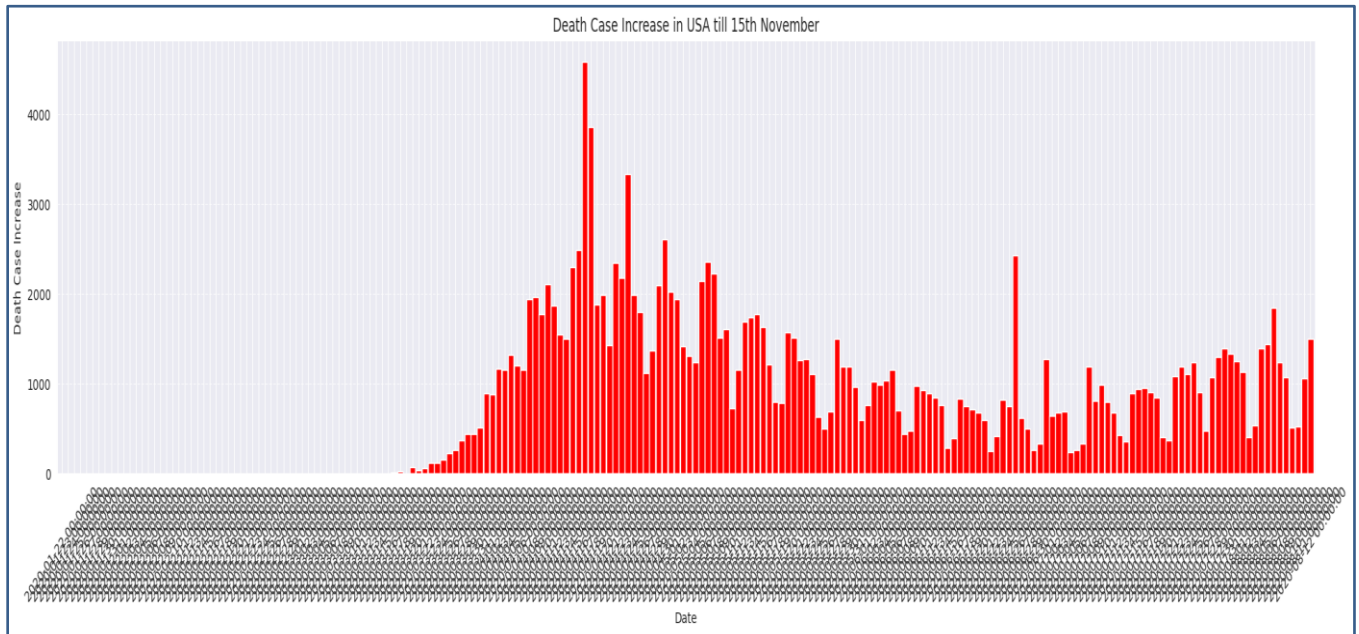
Analysis of USA based on the confirmed cases, death cases, recovered cases:



The Death toll in United States is not flatten any more rather it is increasing day by day in a exponential manner with a steep slope upwards. It does not have any kind of peak till now so that it can flatten after a particular peak.



In the mid of June the confirmed cases are decreased and the incrimination is flattened by a certain moment. But unfortunately in the last week July the incrimination in the confirmed cases are increased suddenly with a steep slope. After a week of steep slope in the increment of the confirmed cases there is a decrement in the confirmed cases in a daily basis manner. As we can see in the graph, that after a week of high rise the peaks are coming lower



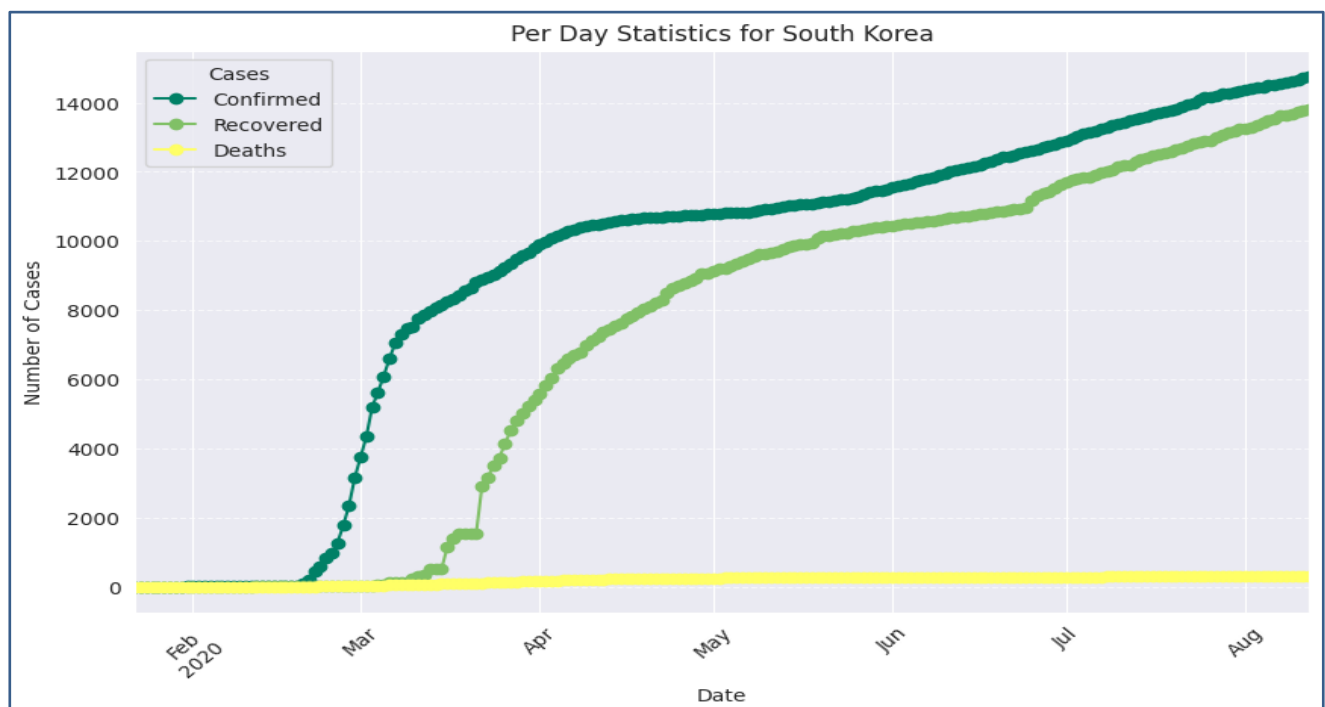
As the graph shows that on the particular day of 16th April 2020 USA reported a highest no. of death cases. After that the death cases are slowly coming down to a certain limit but suddenly in the recent times the Death cases are increasing day by day which are very much annoying for USA.

South Korea : The Role-model -

First outbreak: First confirmed case was founded in south korea on 20 January 2020. The number of confirmed cases increased after 10 February.

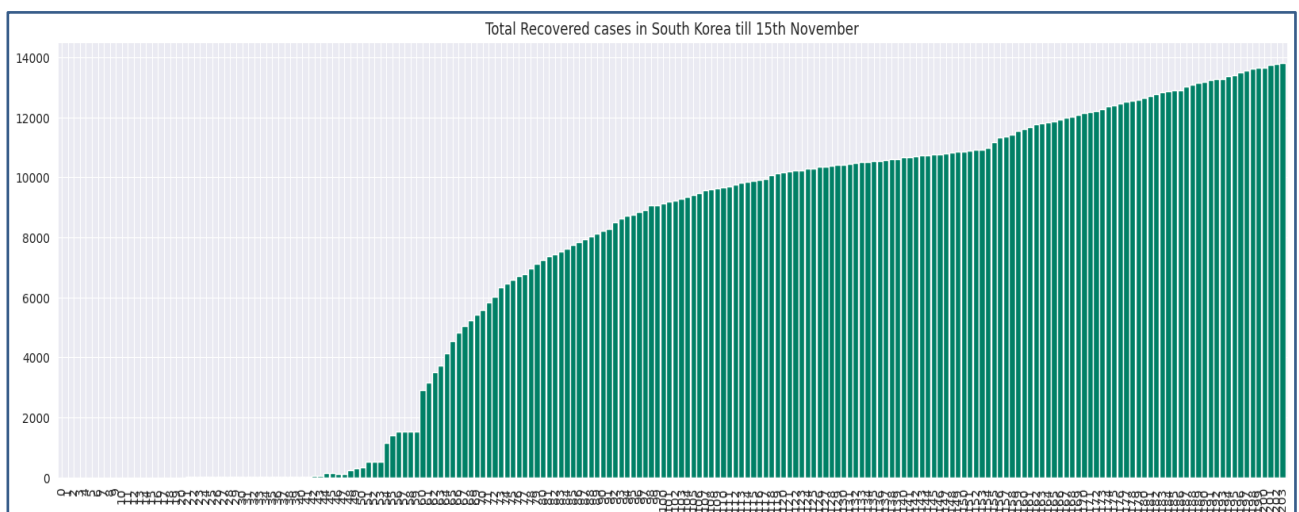
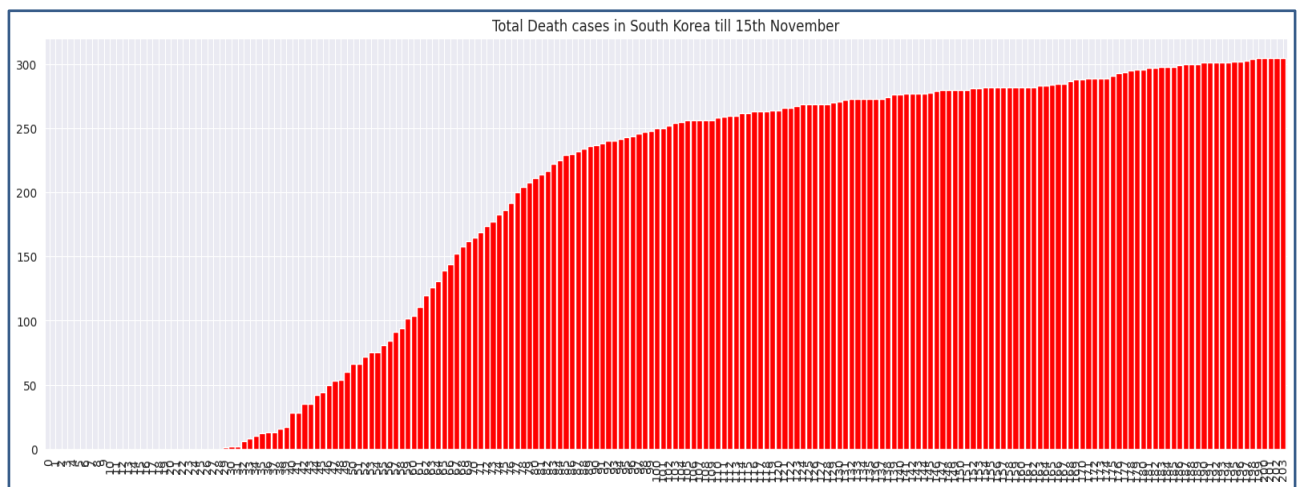
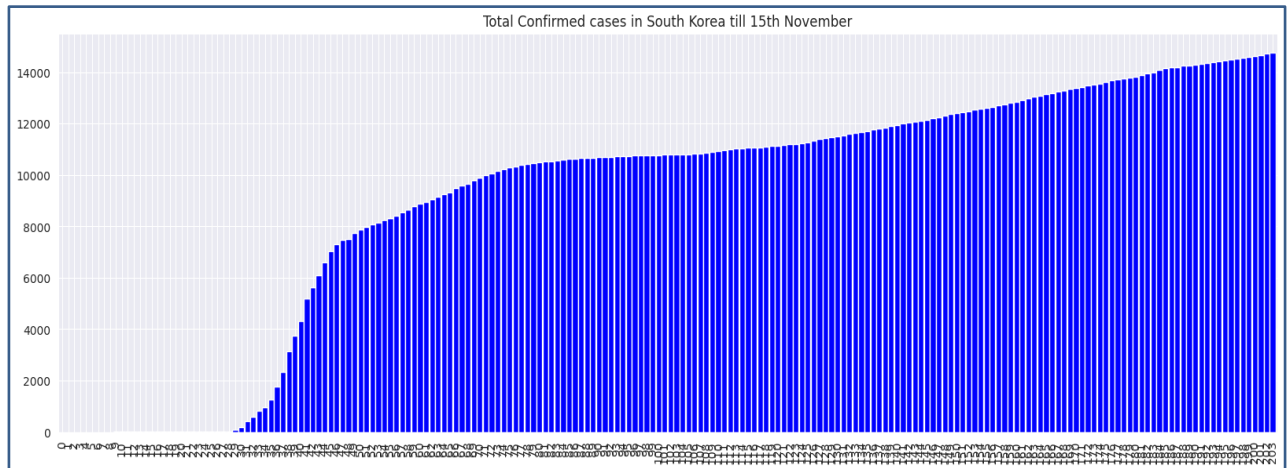
Total affected: In the present situation the total cases is 31735. Cases are increasing after February in a linear scale. Daily number of new cases was too high in February then it was reduced, after August 16 to September mid it was extremely high.

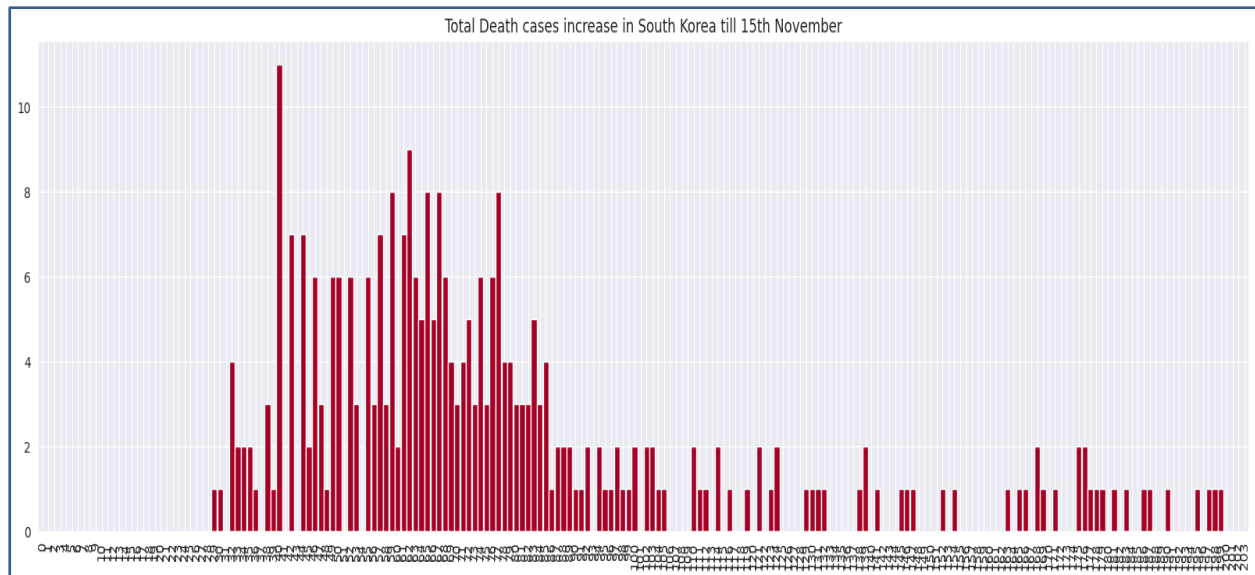
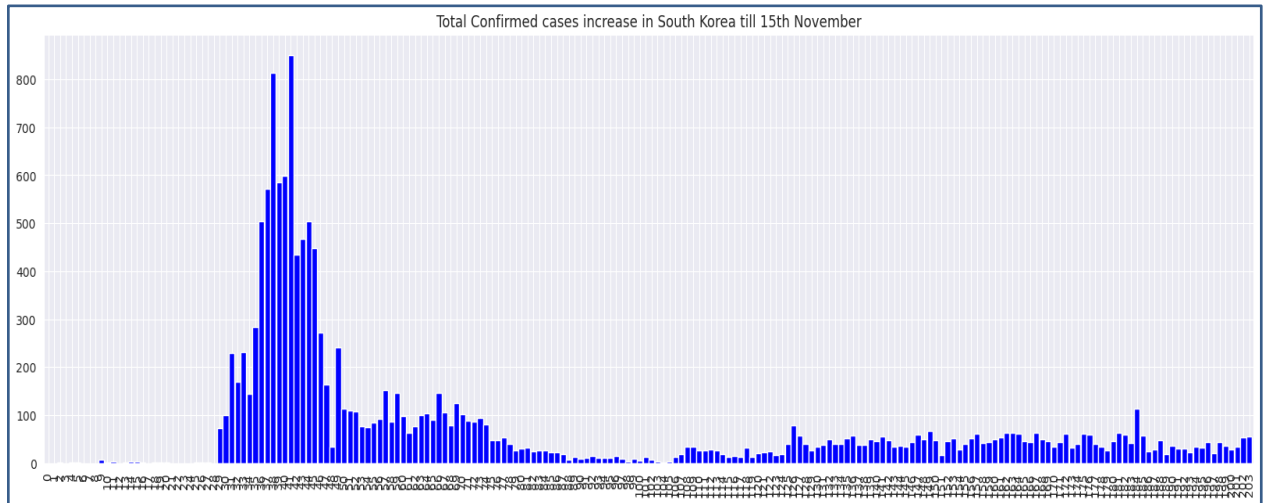
Death rate: Number of death cases are increasing from February but daily death was in the top level in mid of March. Total number of death is 513 till November 25, and recovery number is 26825



In the recent times if we check out the no. of Confirmed case Increase, we can find that the cases are barely minor as compared to the whole world. And the Death Case Increment is now down to 0 in the recent times. Which shows that SOUTH KOREA is able to fought against the virus with minimum casualties of 494 till 15th November, 2020.

Analysis of South Korea based on the confirmed cases, death cases, recovered cases:





As we can see that the confirm cases are lowered to merely 100 per day which signifies the great work of South Korea to fought against the virus
Also the Death Cases are lowered to 2 to 4 and from the recent few days it turns out to be 0.

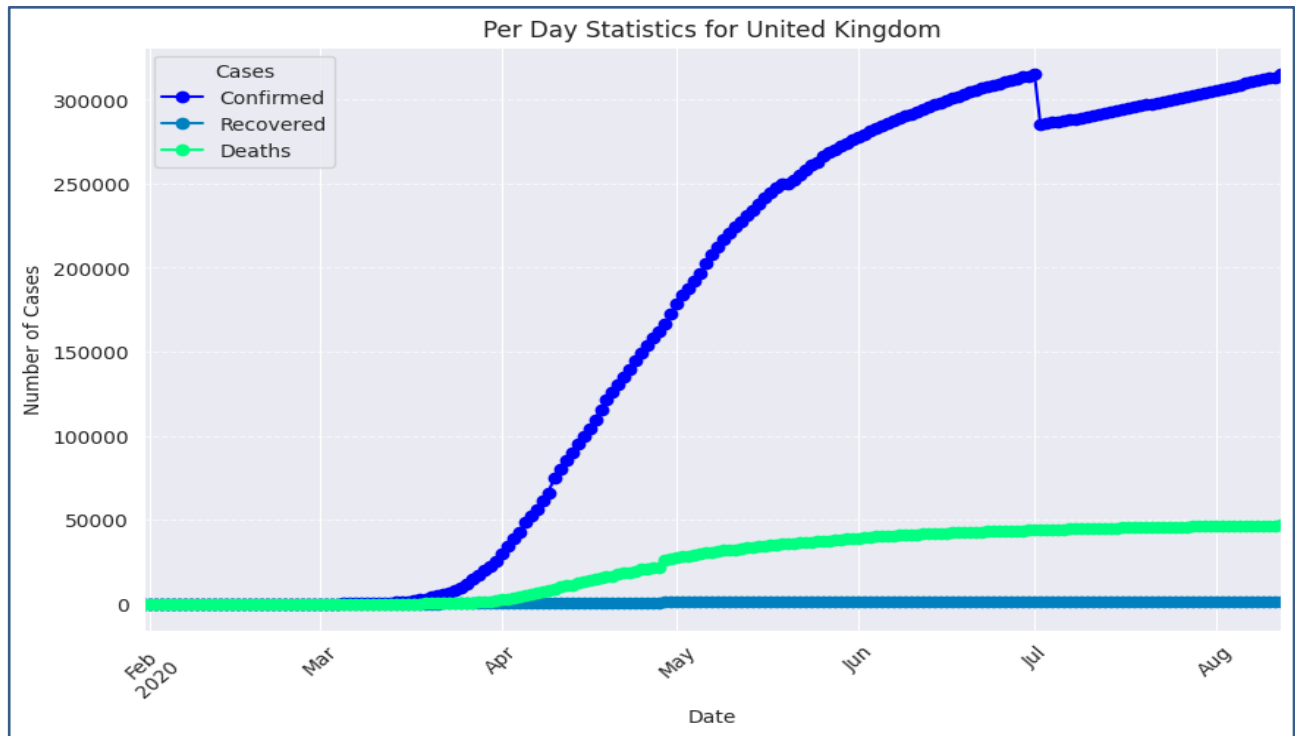
United Kingdom : Analysis-

First outbreak: The virus reached the country in late January 2020. The world's eighth-highest death rate per hundred thousand populations and the highest number overall

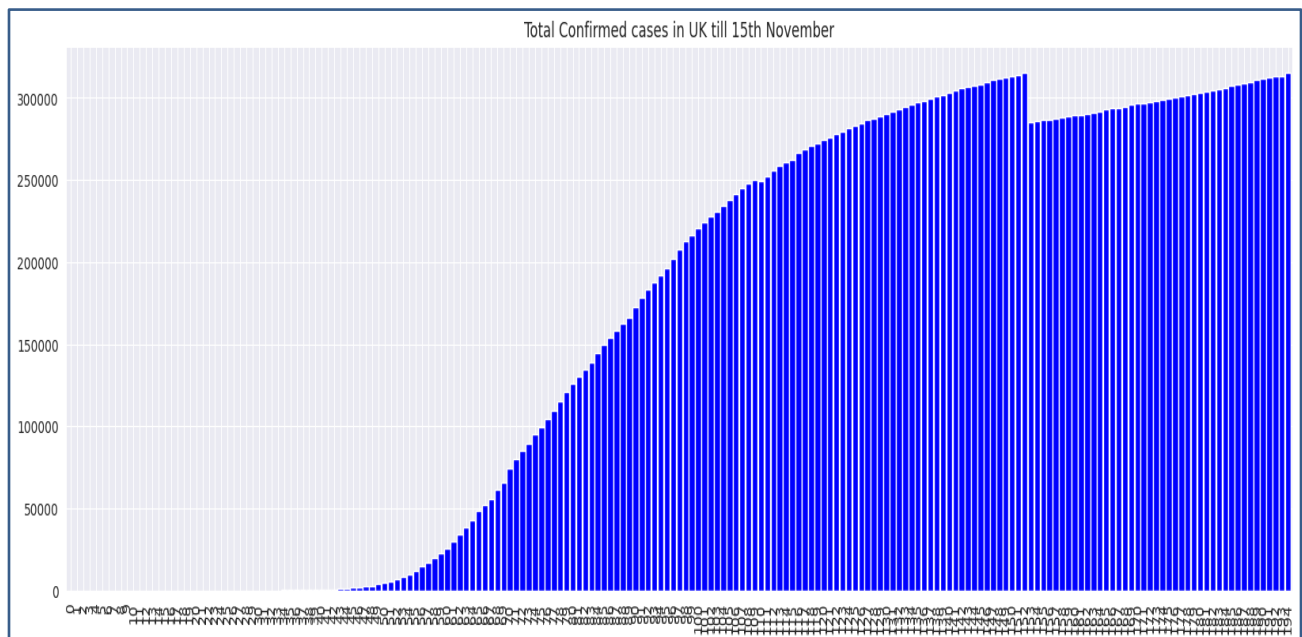
in Europe.

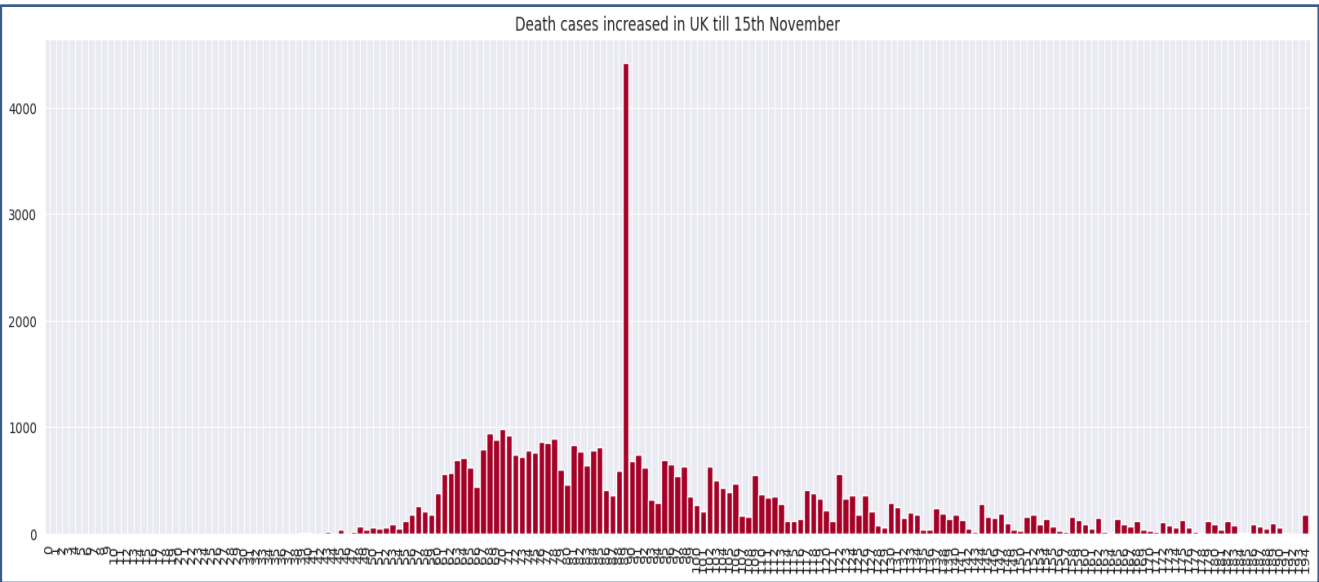
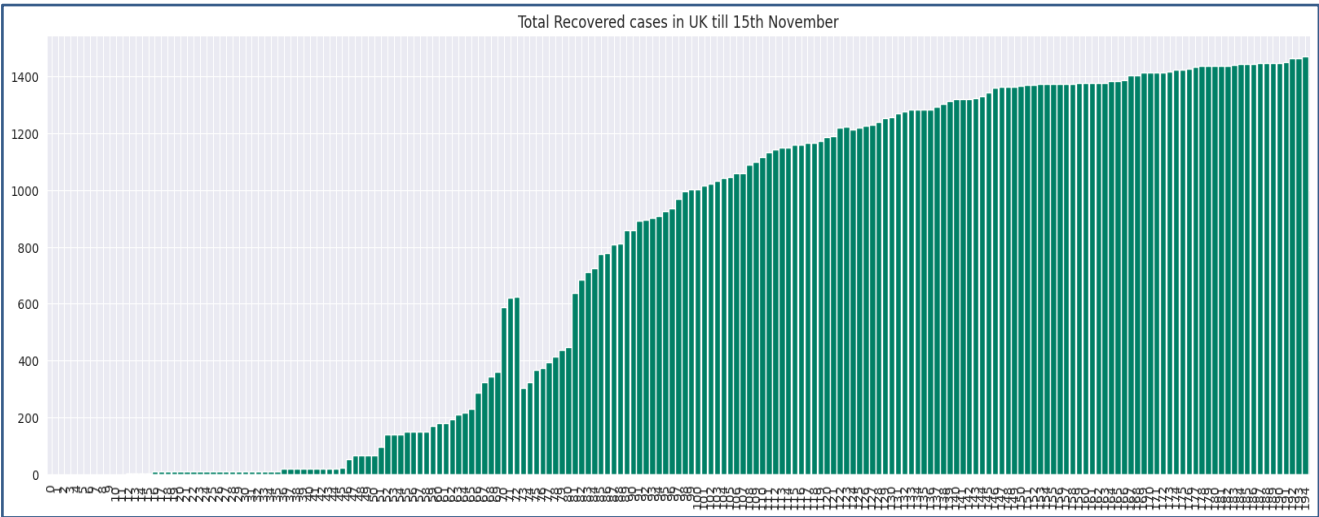
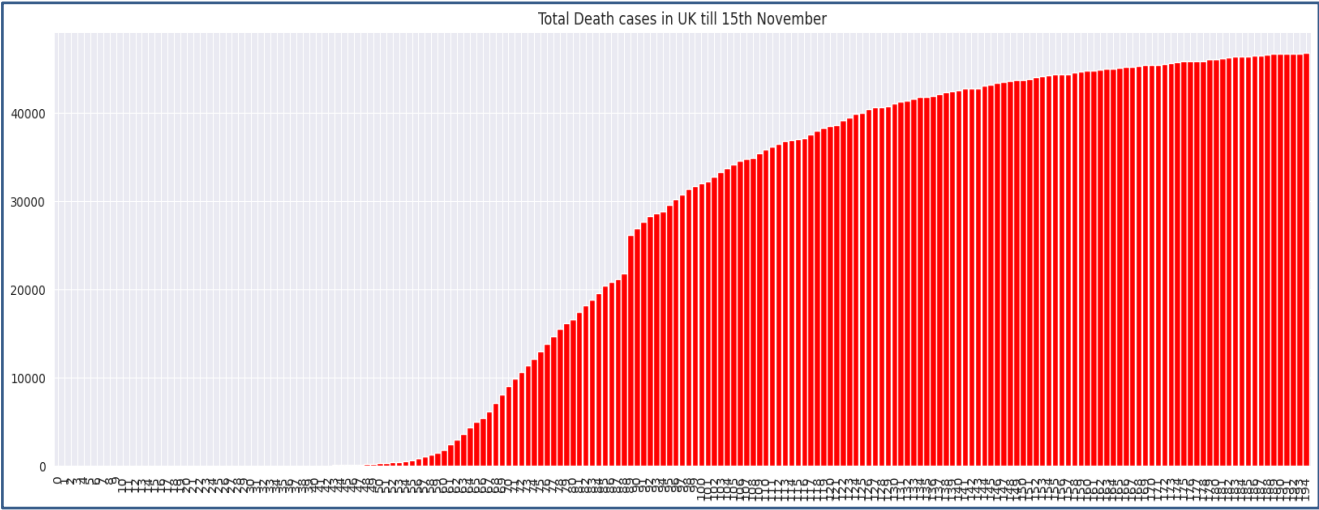
Total affected: As of 2 November 2020 there have been 1,256,725 confirmed cases. Number of cases is highly increasing after September 6(2020). Number of daily affected cases is much higher than mid-July-August.

Death rate: After March death rate was suddenly increased. Total death toll is 56533 till 25 November. On November 21 new death was 341 in 22 it was 398. Only on 25 November the Number of deaths was 696.



Analysis of United Kingdom based on the confirmed cases, death cases, recovered cases:





From the Plottings we can clearly observe that the confirmed, recovered and death tolls are flattened pretty much. Which signifies that UK is pretty much controlled the situation. From the analysis we can observe that after the certain stage of the pandemic situation the Number of confirmed cases are decreased and level up at a certain range. In the recent days The cases are reported near about 100.

From the above observation we can also find out that the death cases are not reported in a Huge manner in the recent times. The reported death cases are like under 10 per day, some days it is being reported as 0. From the above analysis it is clearly visible to us that the United Kingdom is fought against the pandemic situation quite brilliantly!

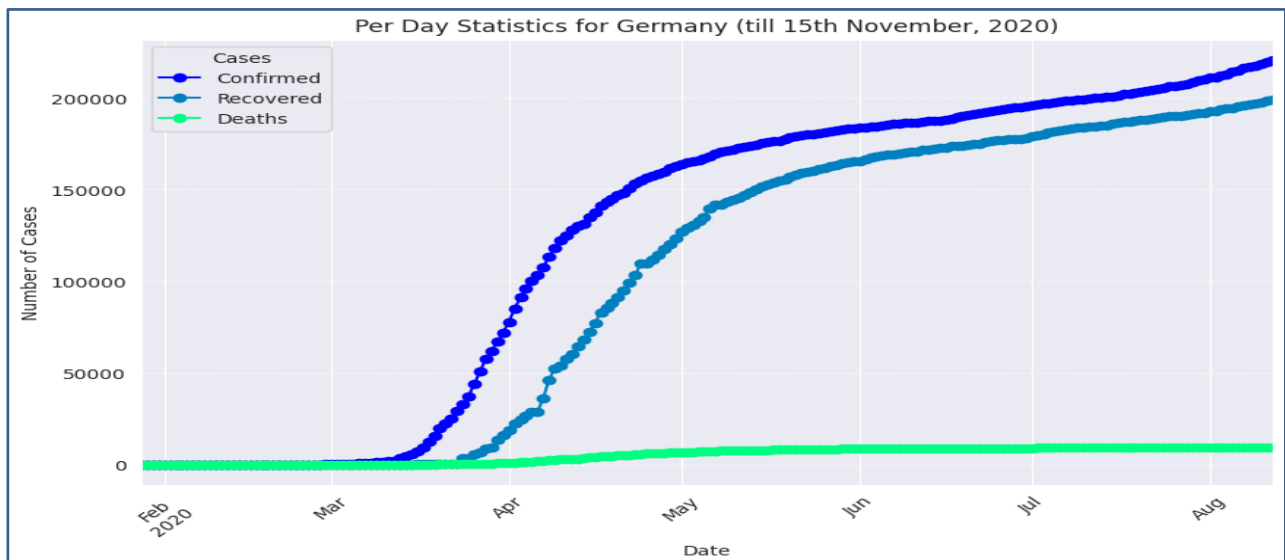
Conclusion: United Kingdom has fought well against the Covid-19!

Germany : Did they do well?

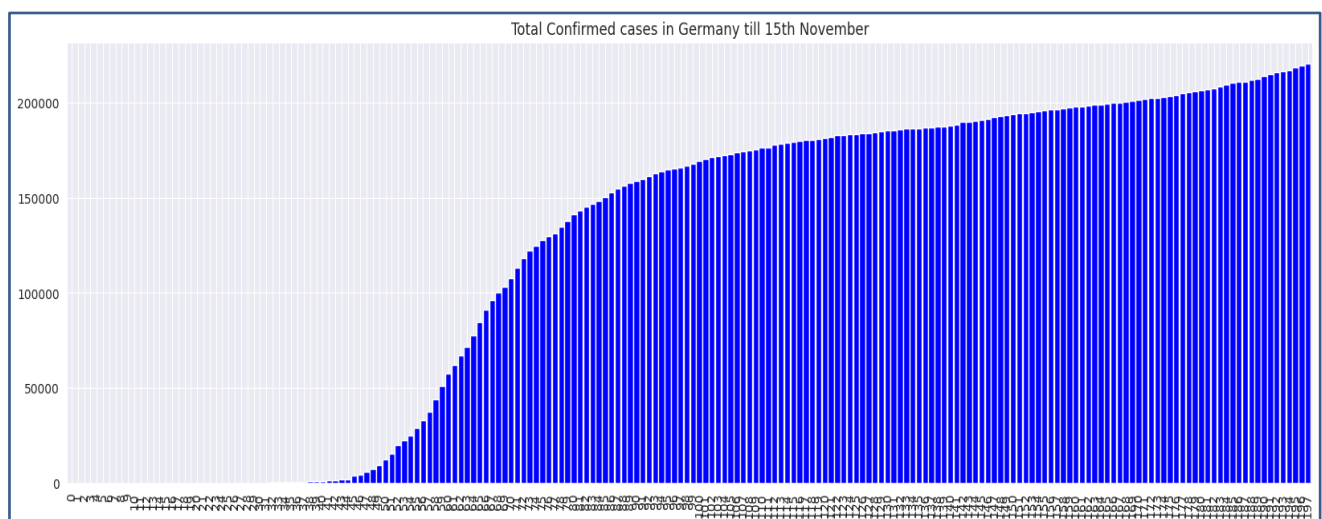
First outbreak: On 27 January 2020, the first case in Germany was confirmed near Munich, Bavaria. By mid February, the arising cluster of cases had been fully contained. On 25 and 26 February, multiple cases related to the Italian outbreak were detected in Baden-Württemberg.

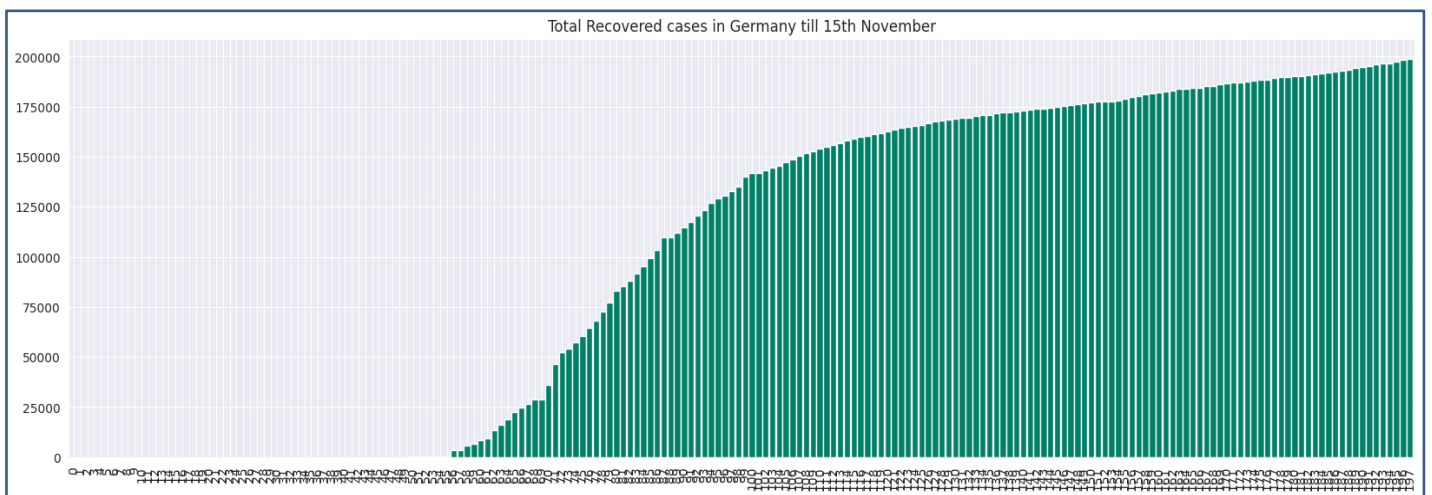
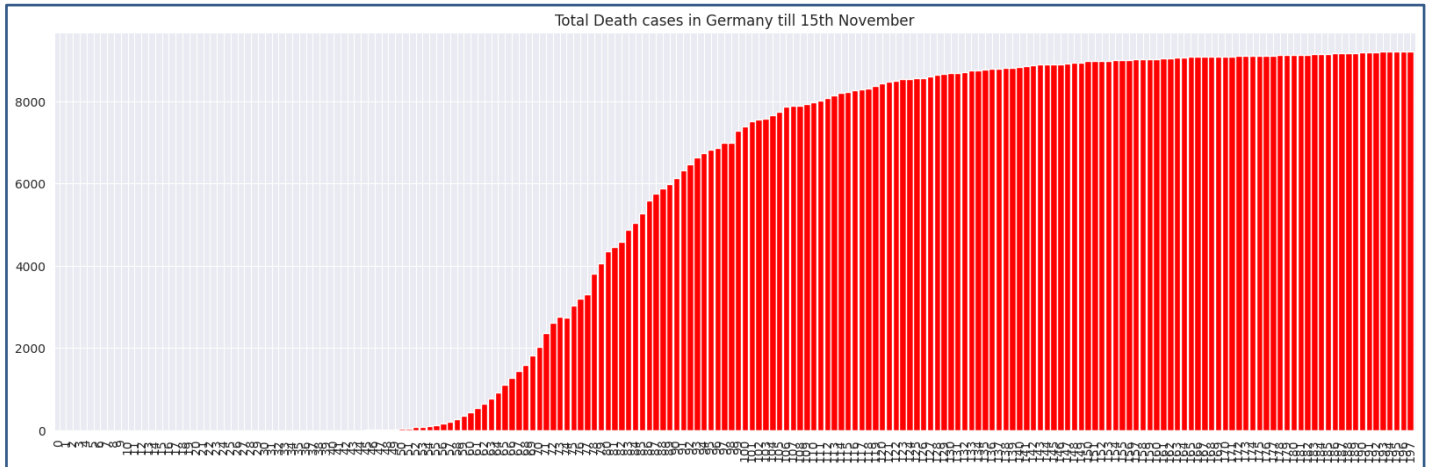
Total affected: Total positive cases are 983731 till now. Daily number of new cases is highly increasing after September 26.

Death rate: Death rate is very high after April 3. Total death till 25 November is 15381. But in Germany recovery rate is also high, till now total recovered number is 656400.

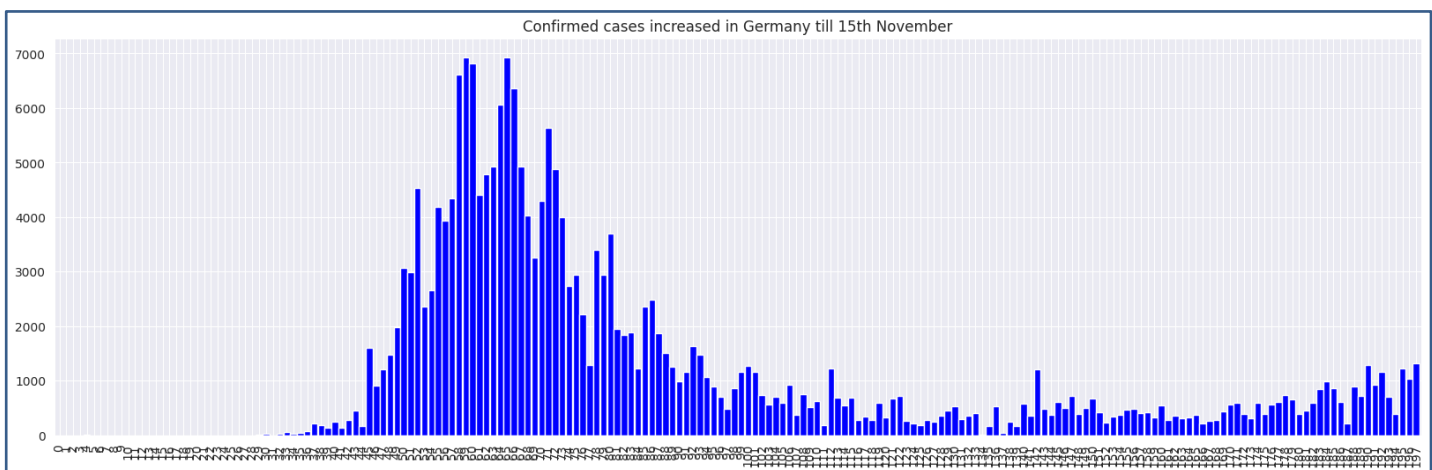


Analysis of Germany based on the confirmed cases, death cases, recovered cases:

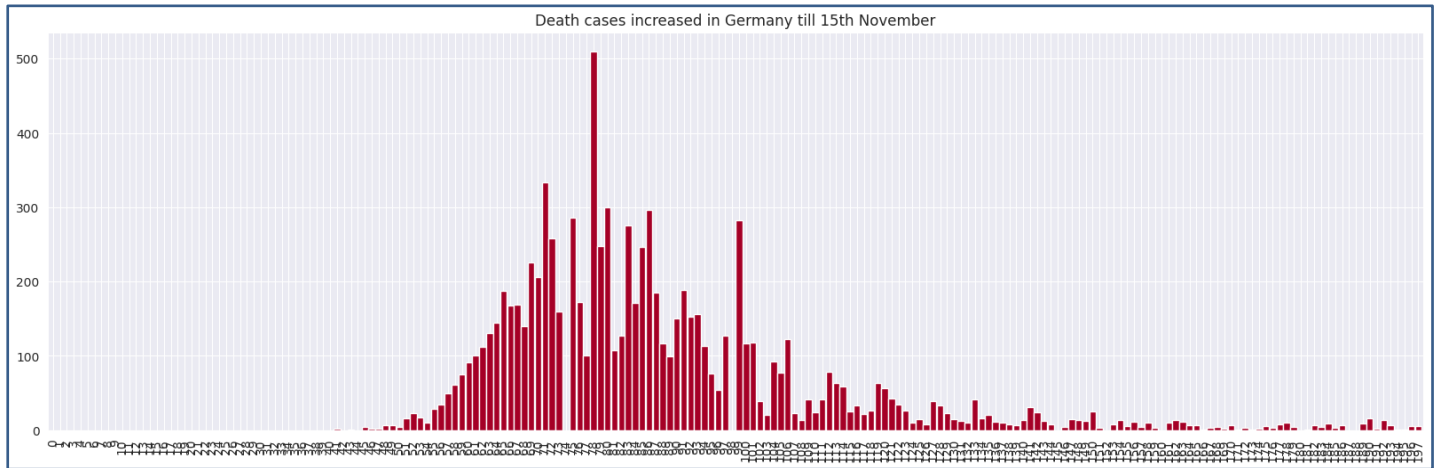




From the graphical analysis of the Total Confirmed cases, Recovered cases and death cases we have seen that the graph is flattened after the peak occurred. Which shows that the country is doing quite great job to fight against the virus



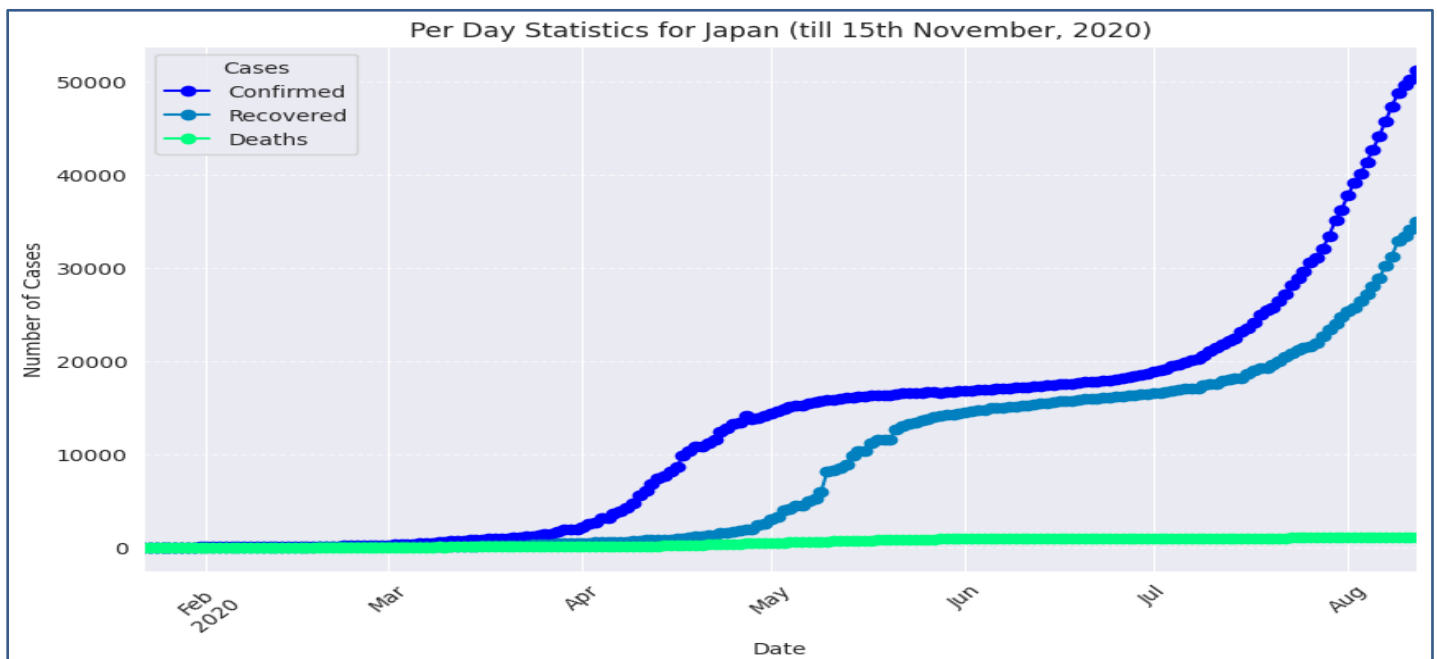
From the graph analysis we have got that the confirmed cases are now lowered to 1000 whereas the peak was at the 7000 mark a few months back. Which seems that the country is healing! But in the recent times unfortunately the condition is getting worsen for the country as it saw per day 30000 rise suddenly.



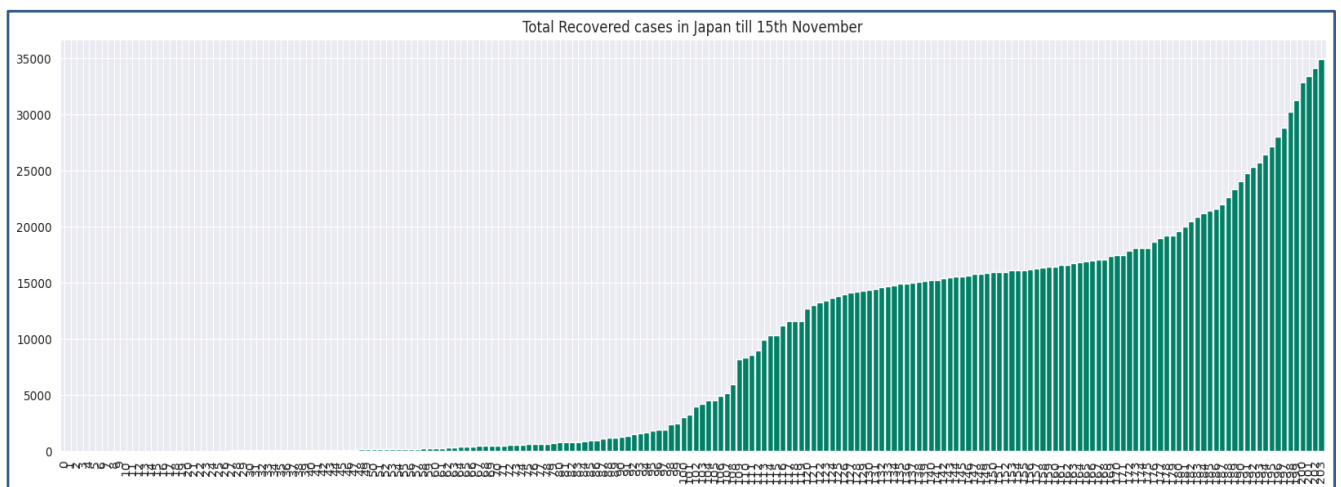
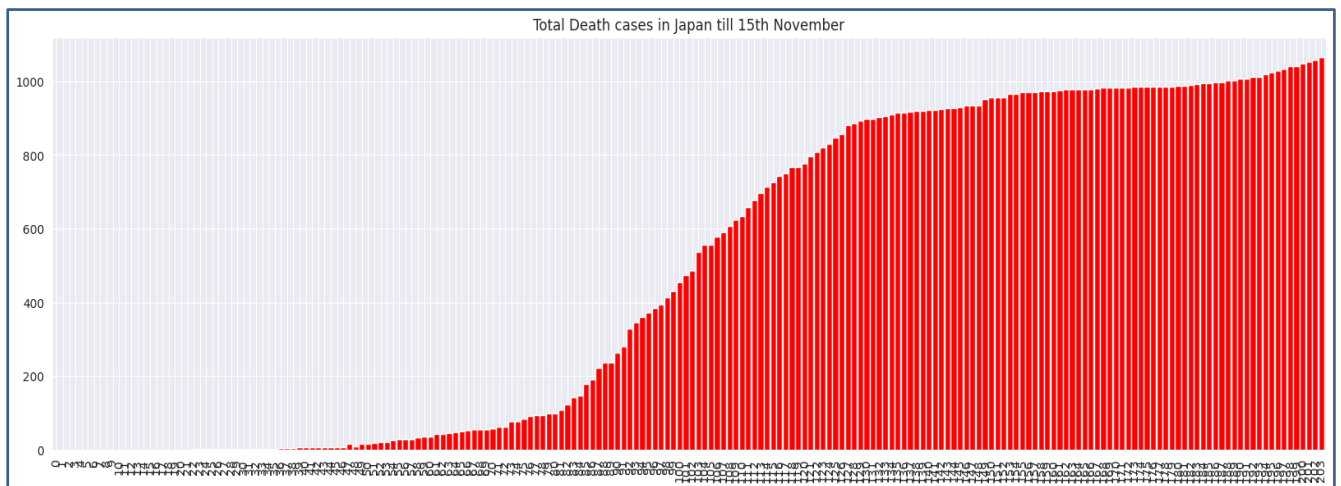
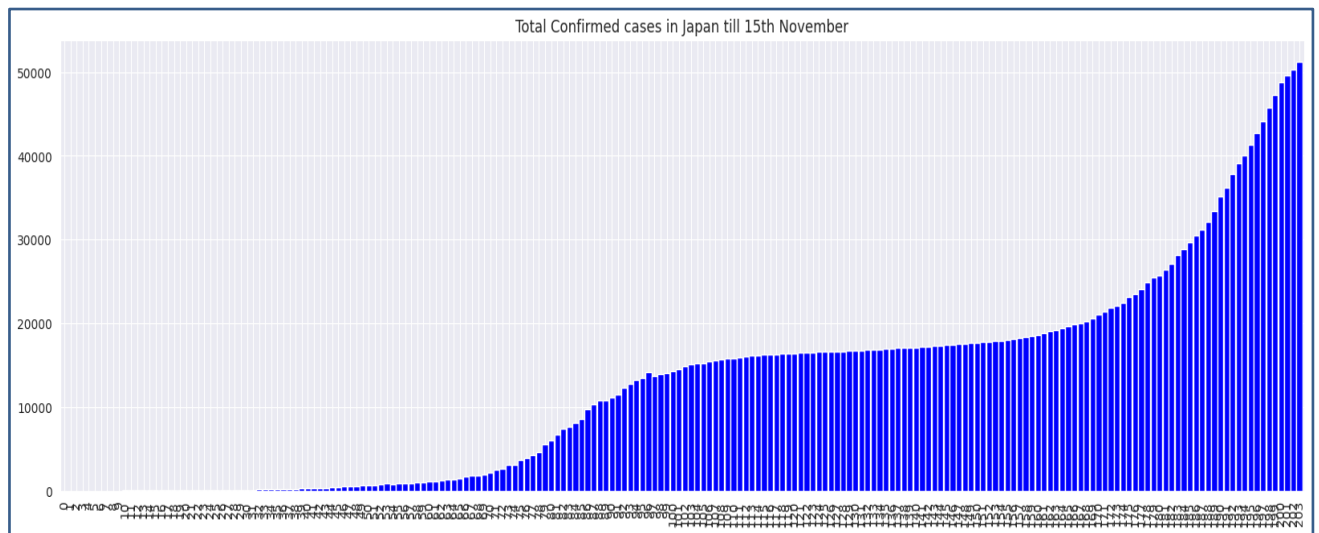
As per the analysis the Death tolls decreased to minimal numbers and it is clearly shows that Germany has fought against the virus pretty much
From the analysis we can say that Germany has overcome the pandemic situation positively!

Tech Country Japan against Coid-19 -

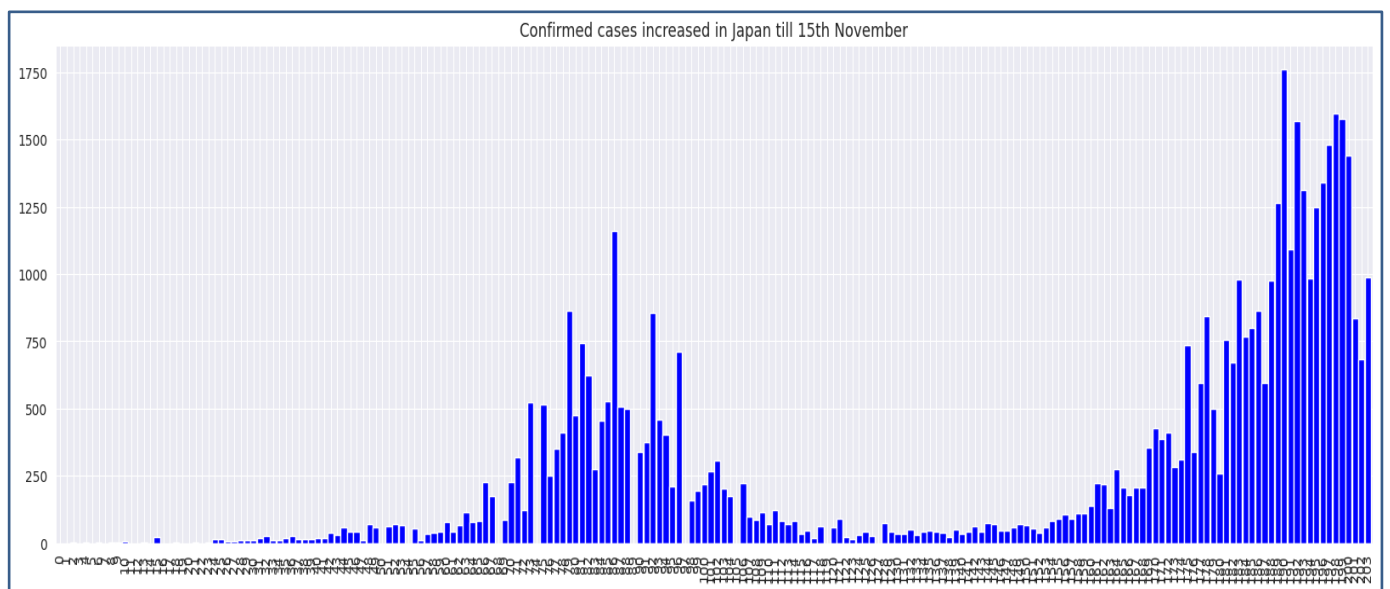
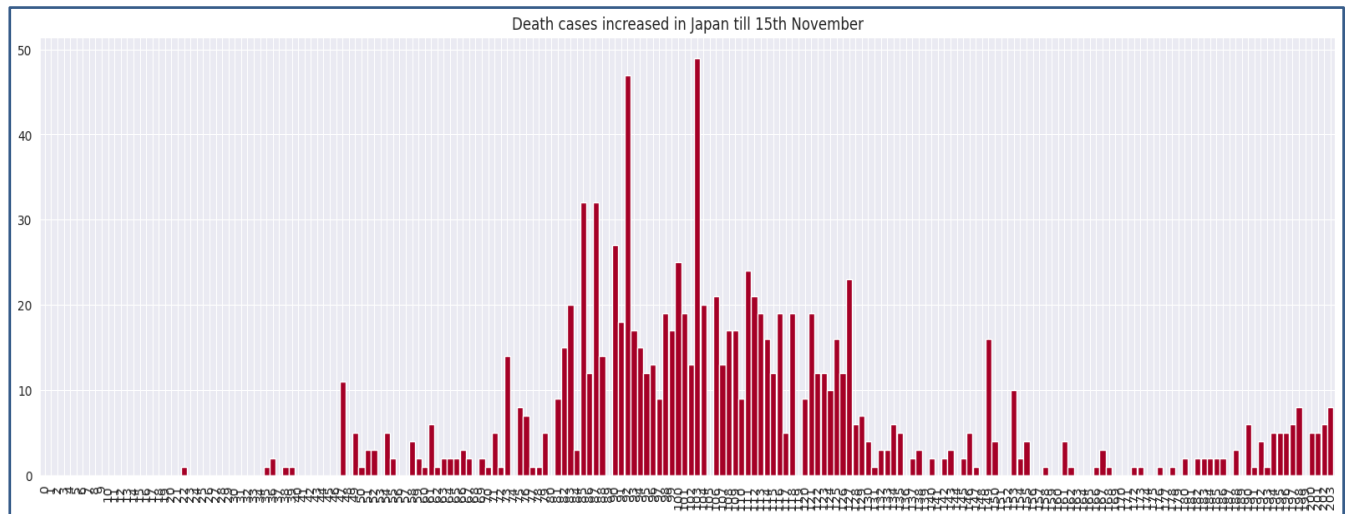
First outbreak: On 15 January 2020, the Ministry of Health, Labour and Welfare, Japan (MHLW) reported an imported case of laboratory-confirmed 2019-novel coronavirus (2019-nCoV) from Wuhan, Hubei Province, China. Total affected: Cases increasing rate is very high from February 15. Total affected is 31735. Number of daily new cases is lesser than number of cases in mid February but it was in level from September 12 to November 7. Death Rate: Number of death rate is creasing but daily death cases are reduced from February-March time. Number of death is 2001 till 25 November.



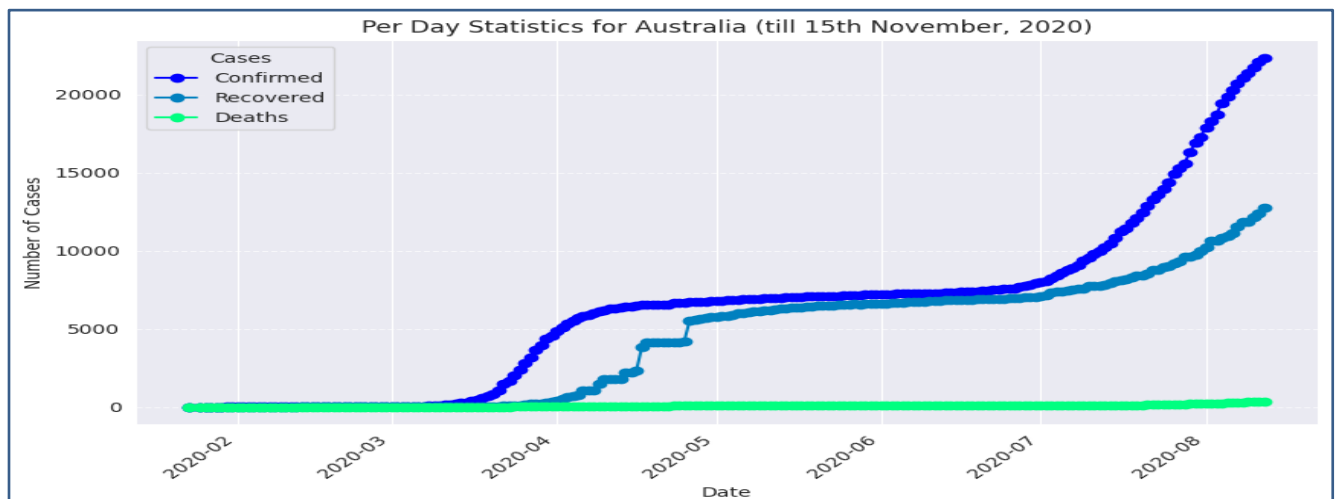
Analysis of Japan based on the confirmed cases, death cases, recovered cases:



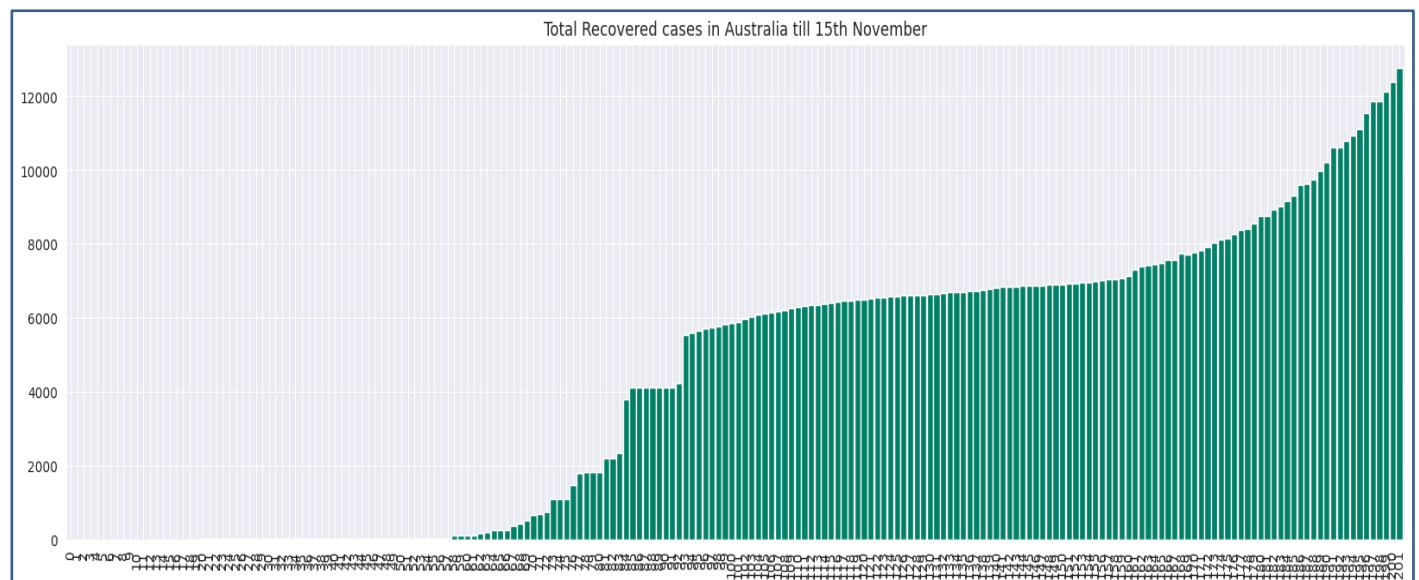
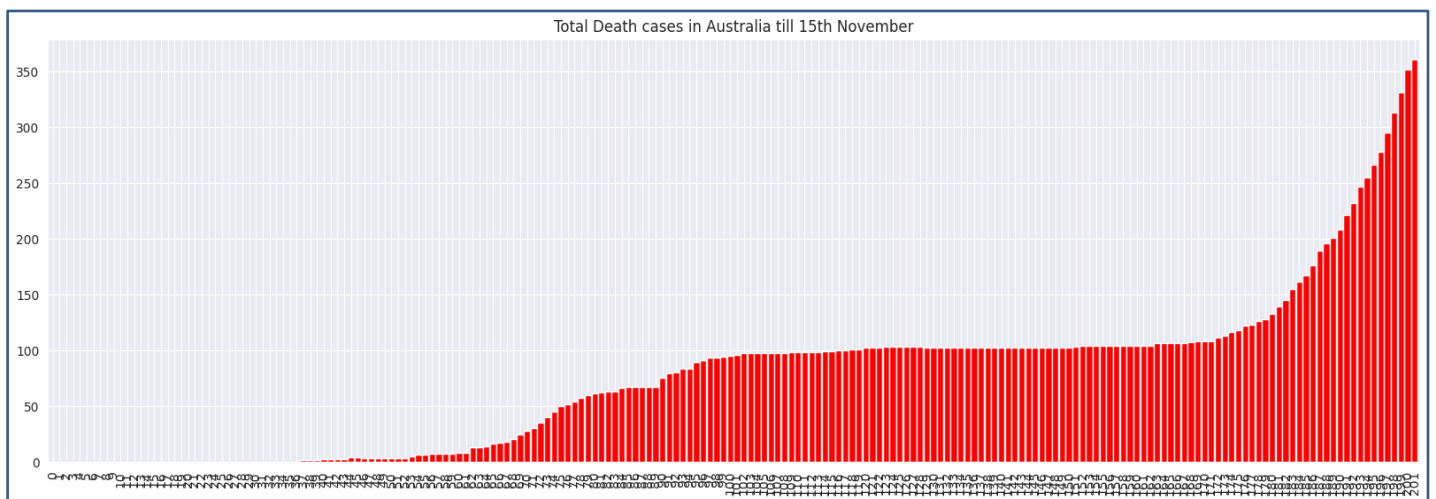
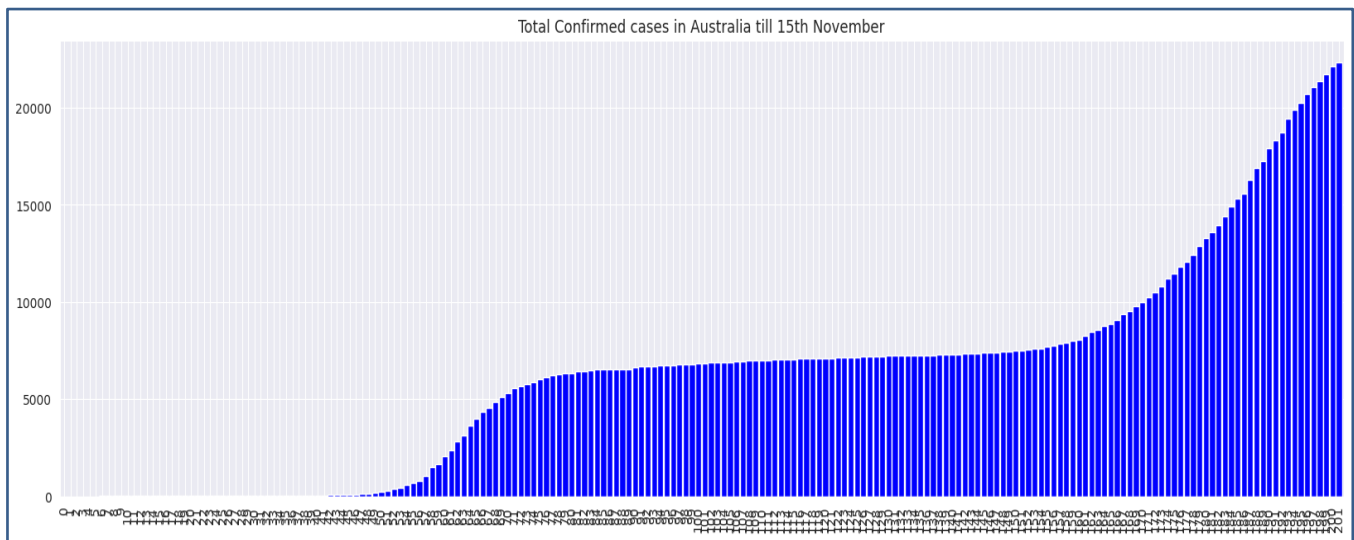
The Death toll is also increasing with the increment of the total confirmed cases. Which may cost the country after a certain relaxation period of flattened curve?



Australia: Analysis-



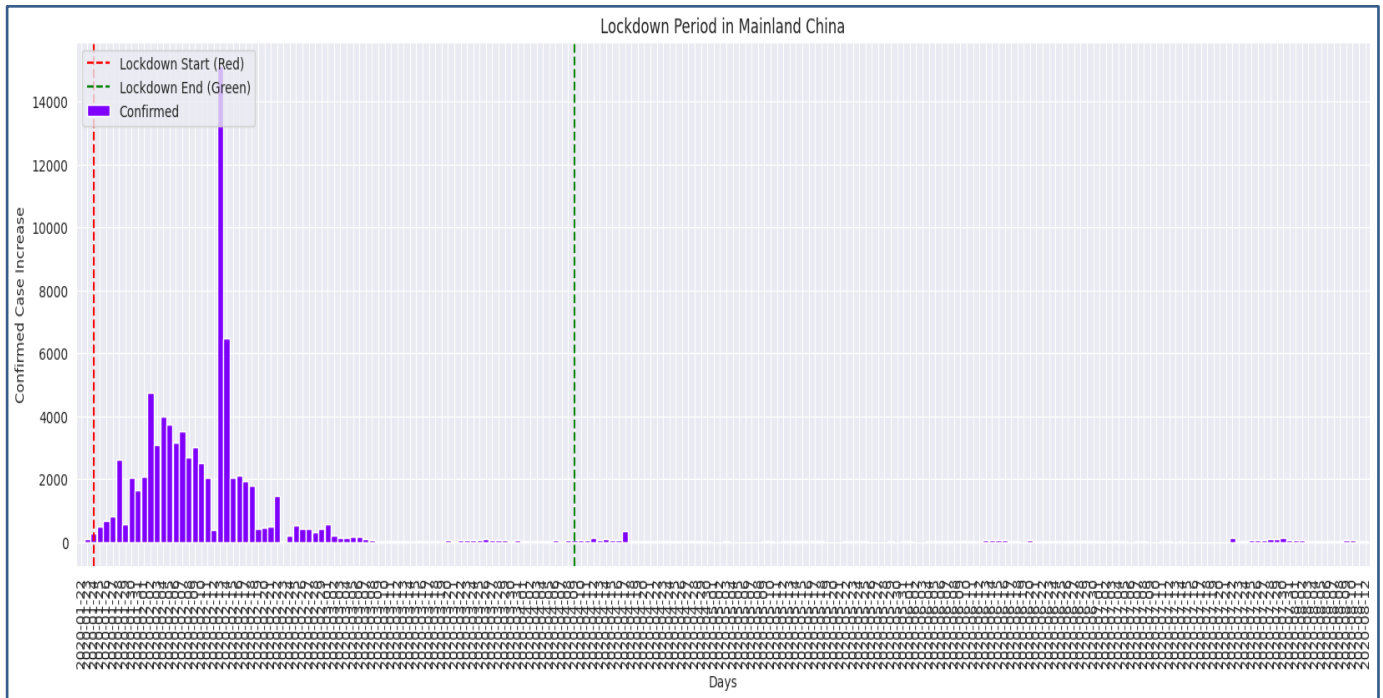
Analysis of Australia based on the confirmed cases, death cases, recovered cases:



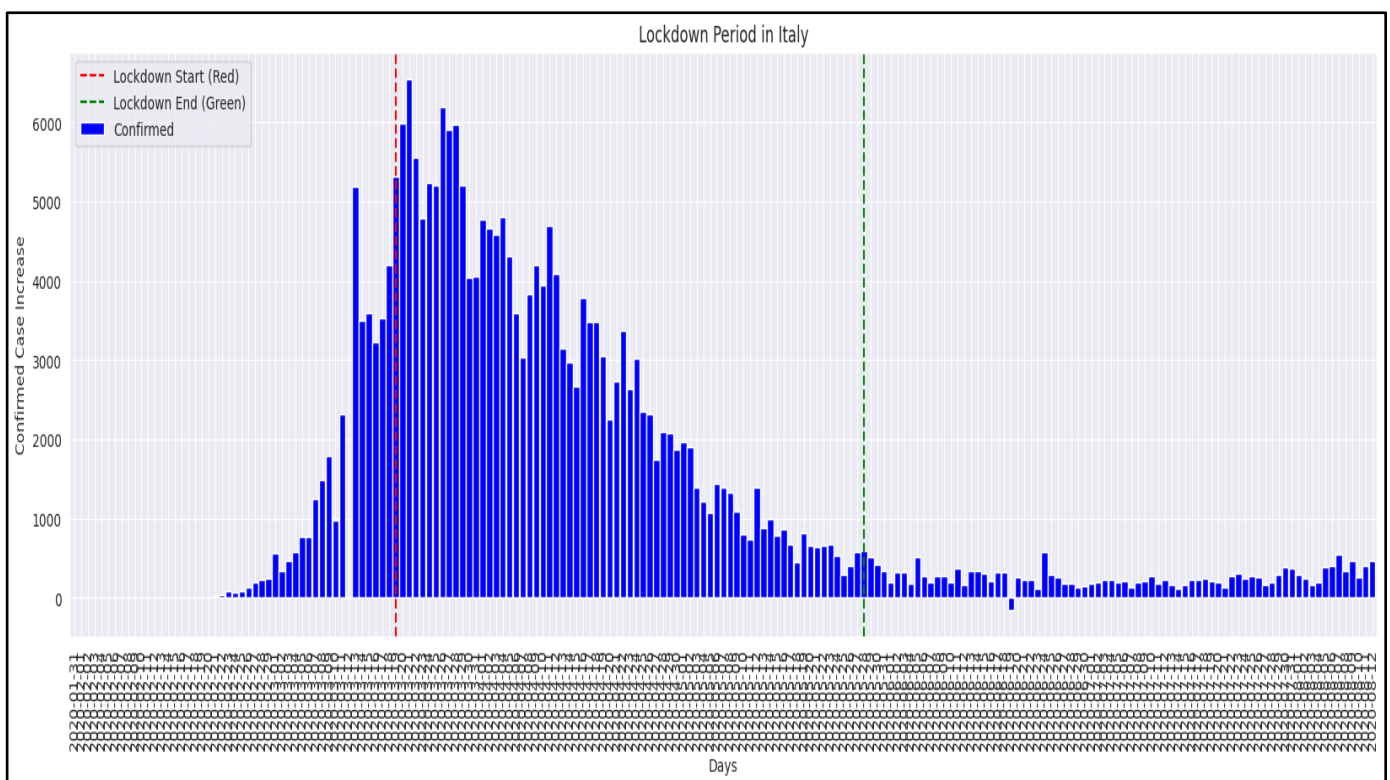
LOCKDOWN: The Savior or, not?

- RED LINE shows the date of IMPOSING LOCKDOWN
- GREEN LINE shows the date of REVOKING LOCKDOWN

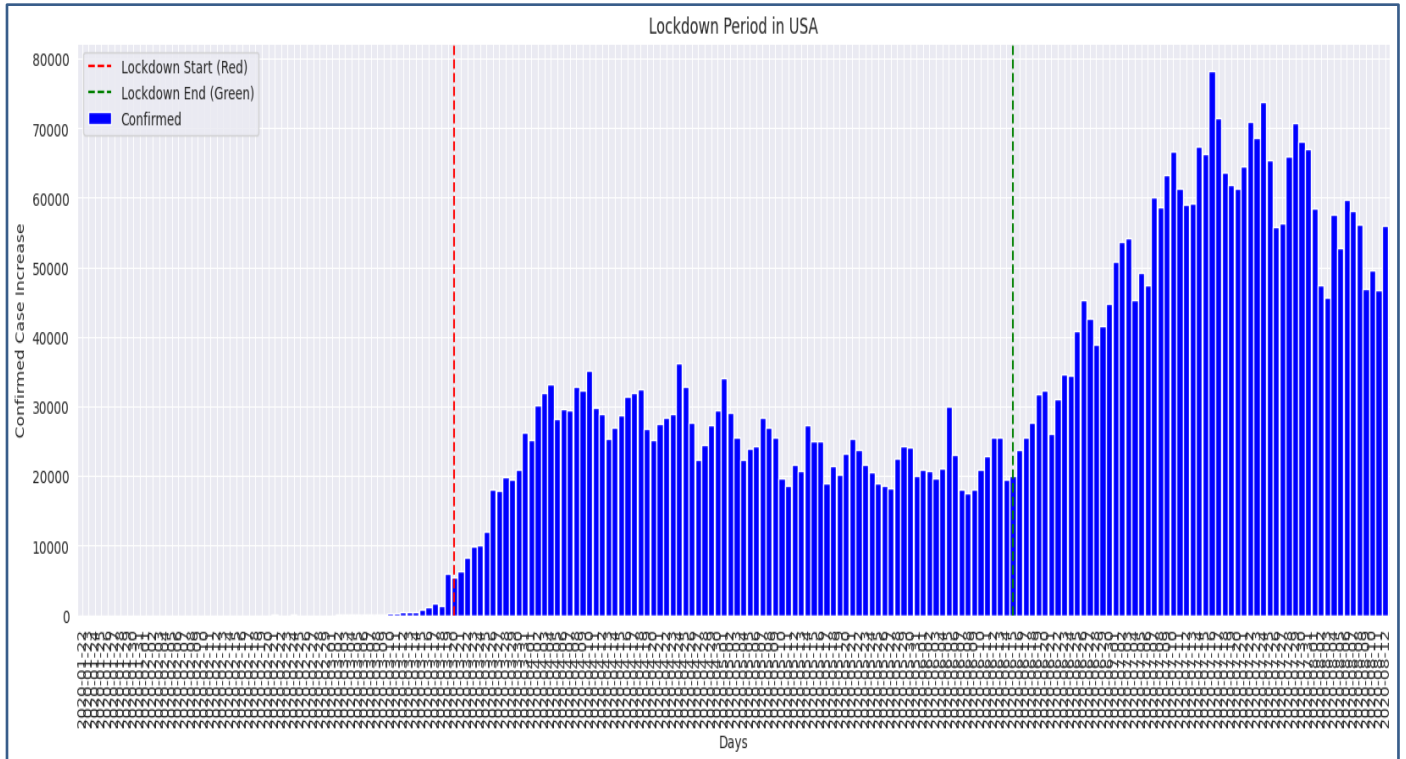
China's Lockdown Period – SUCCESSFUL



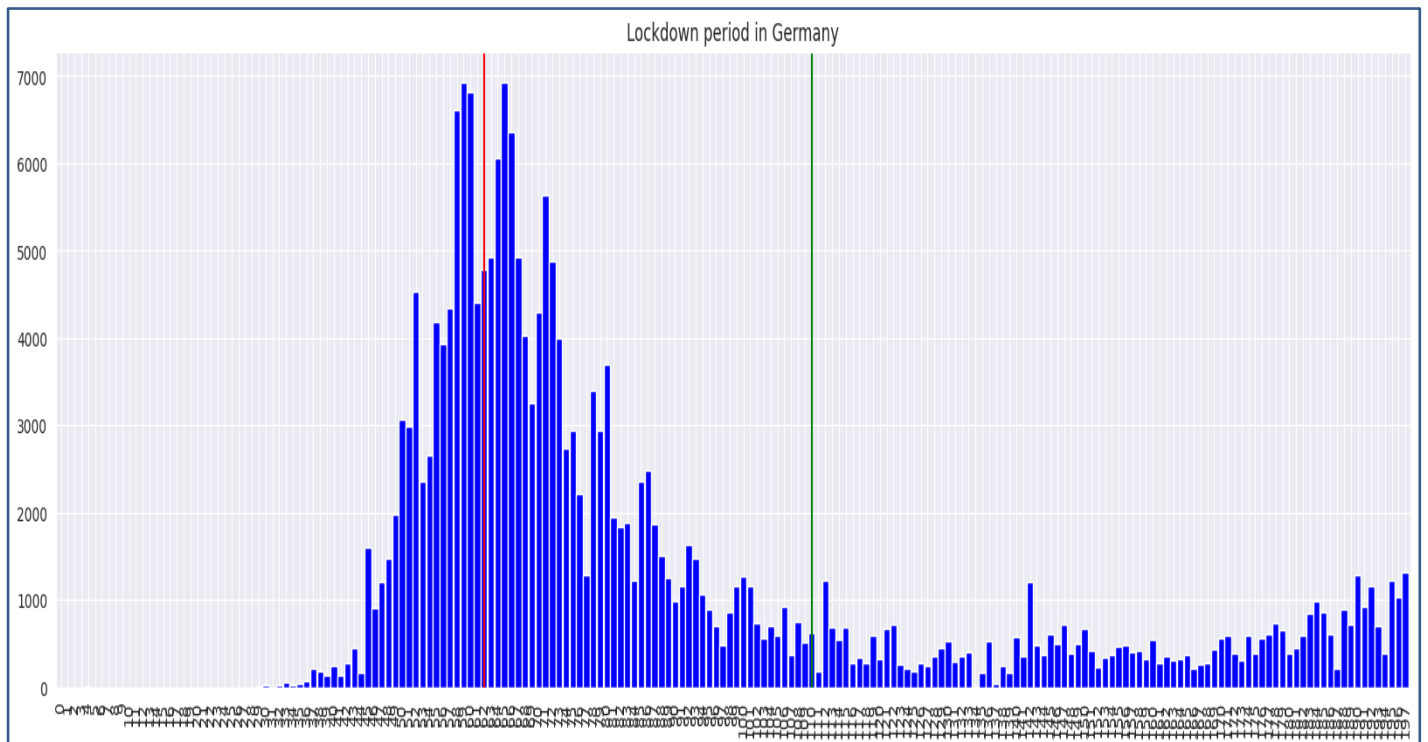
Italy's lockdown Period – SUCCESSFUL



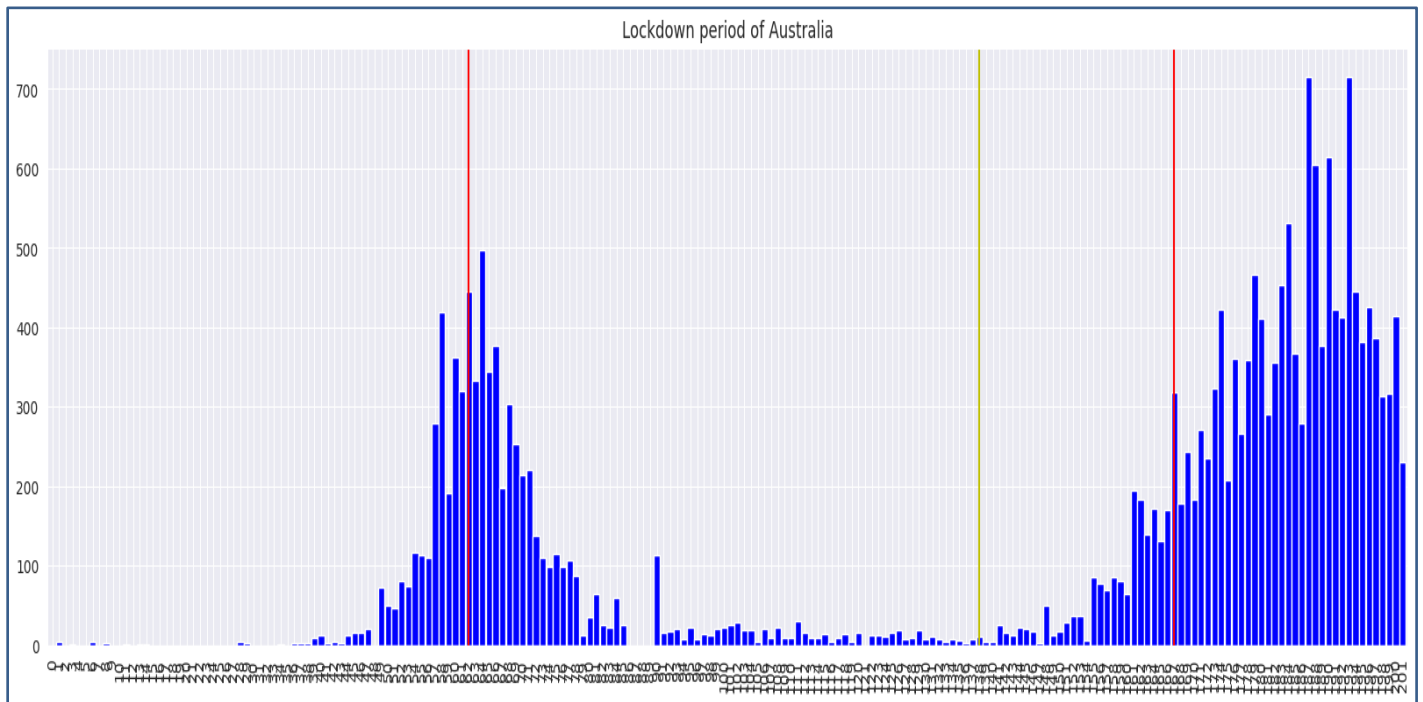
United States of America Lockdown Period – UNSUCCESSFUL



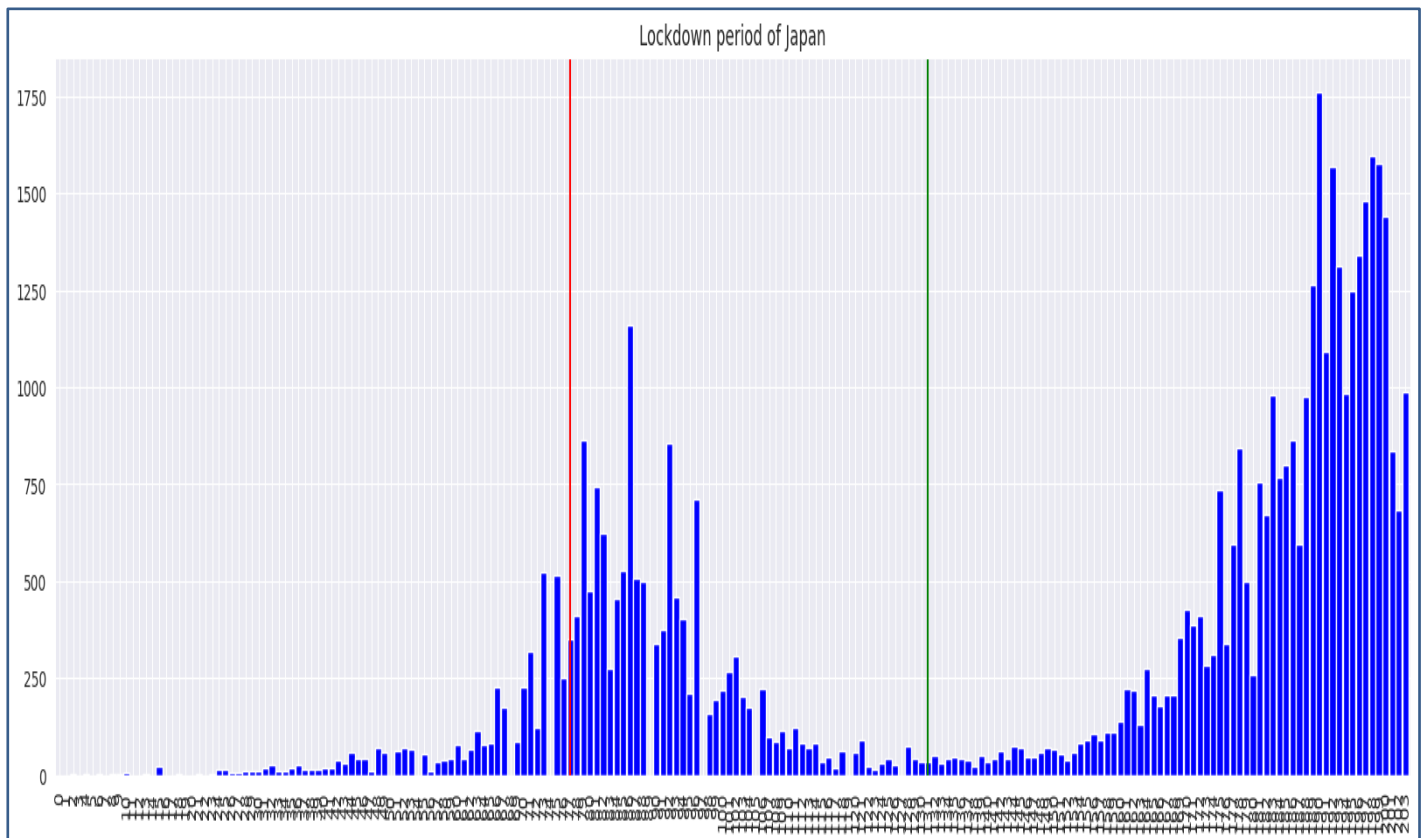
Germany Lockdown Period – SUCCESSFUL



AUSTRALIA Lockdown - SUCCESSFUL to some extent



JAPAN Lockdown Period - SUCCESSFUL to some extent

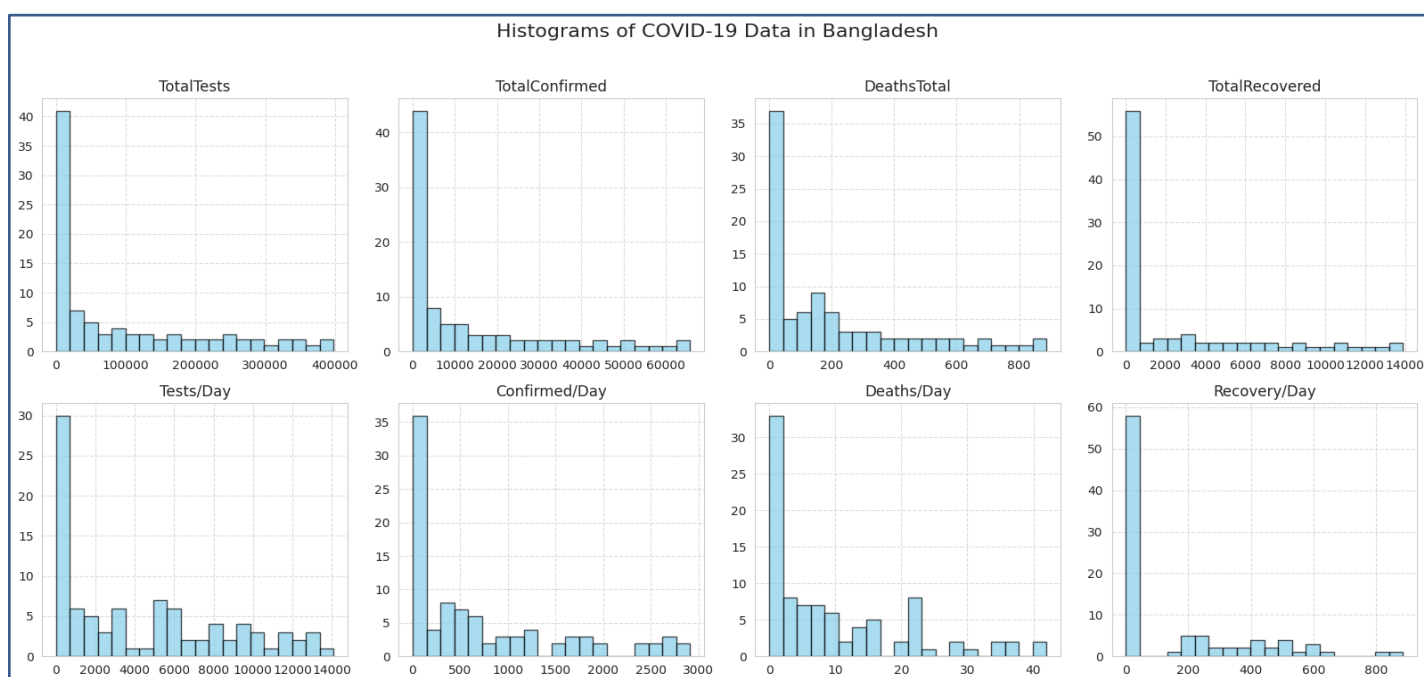
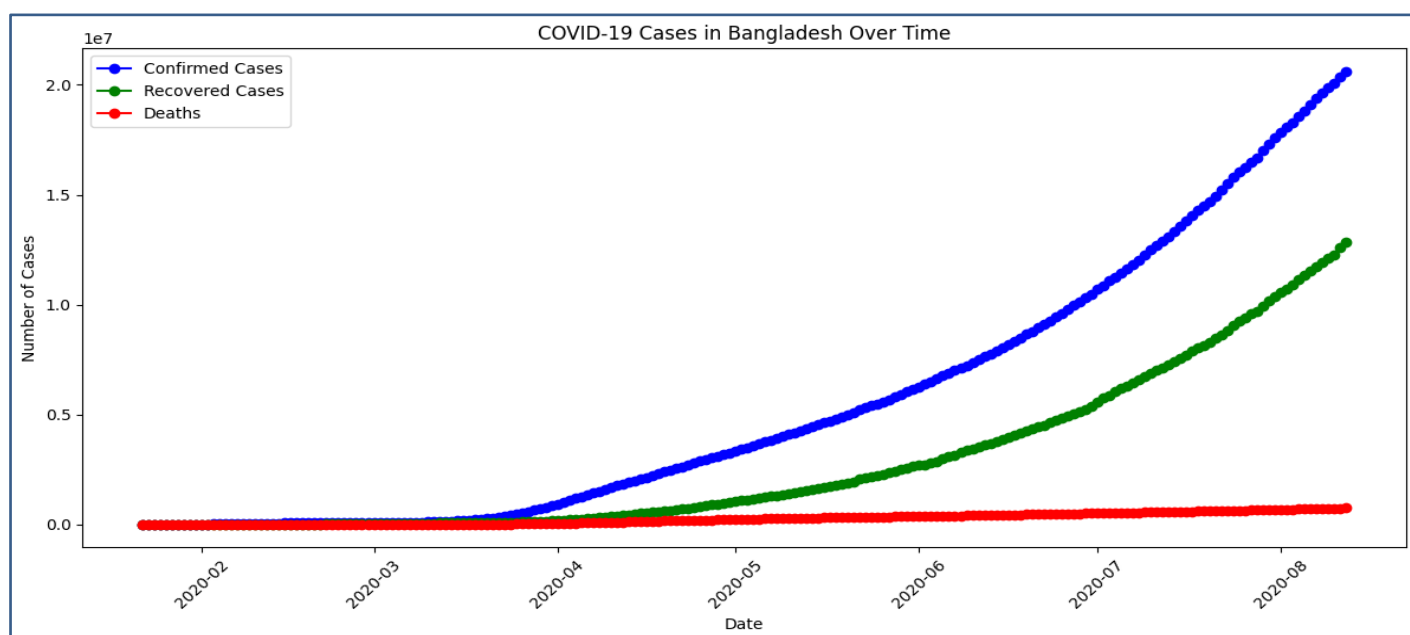


BANGLADESH COVID 19 ANALYSIS:

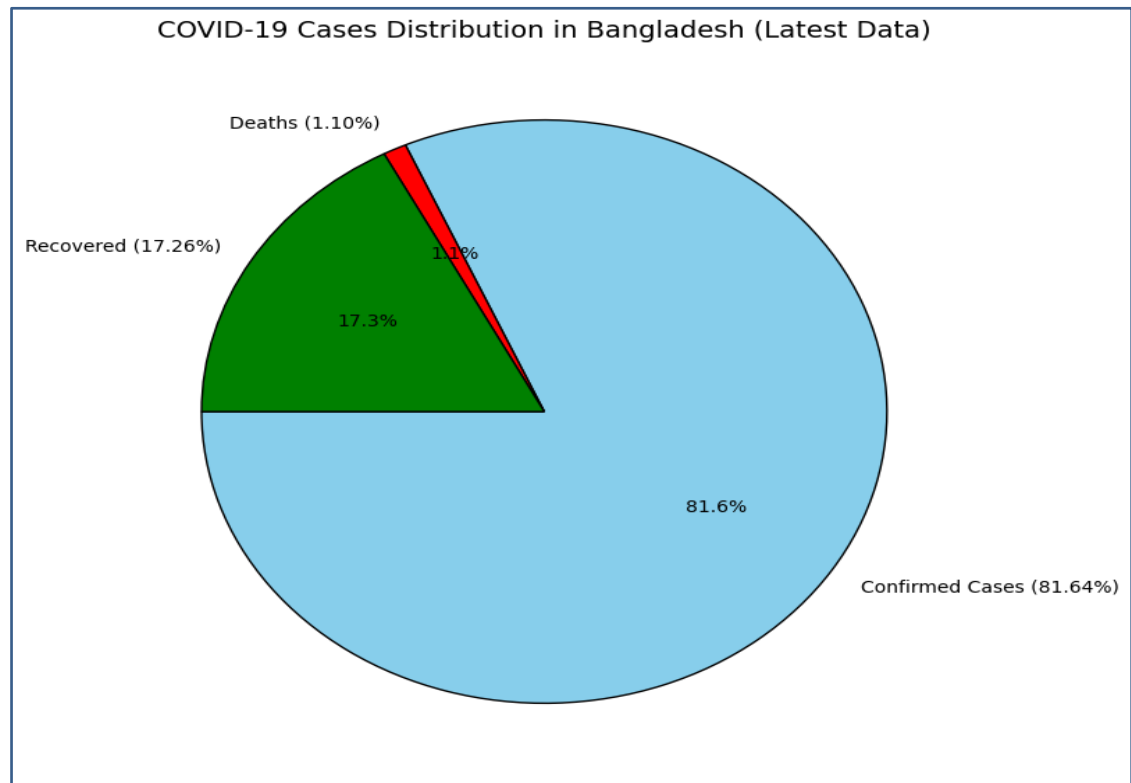
Bangladesh COVID-19 Description:

The COVID-19 pandemic reached Bangladesh in March 2020. The virus spread rapidly across the country, affecting public health and the economy. The government implemented nationwide lockdowns, health safety guidelines, and vaccination programs to control the outbreak. Major cities like Dhaka were heavily impacted. Bangladesh relied on international support, local healthcare efforts, and awareness campaigns to manage the crisis. By 2022, cases had significantly decreased due to widespread vaccination and public health measures, although monitoring and preventive actions continued.

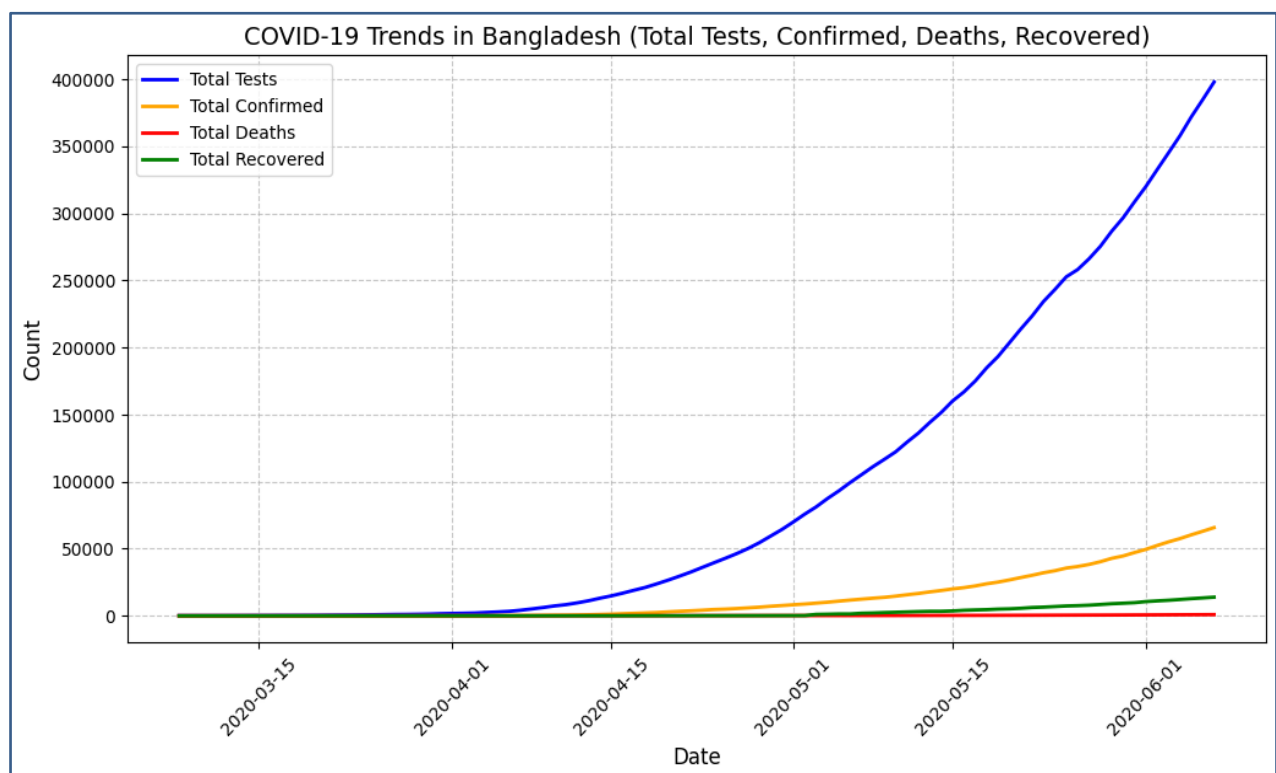
Bangladesh Current situation and analysis:



Bangladesh COVID 19 analysis in pie chart and percent:

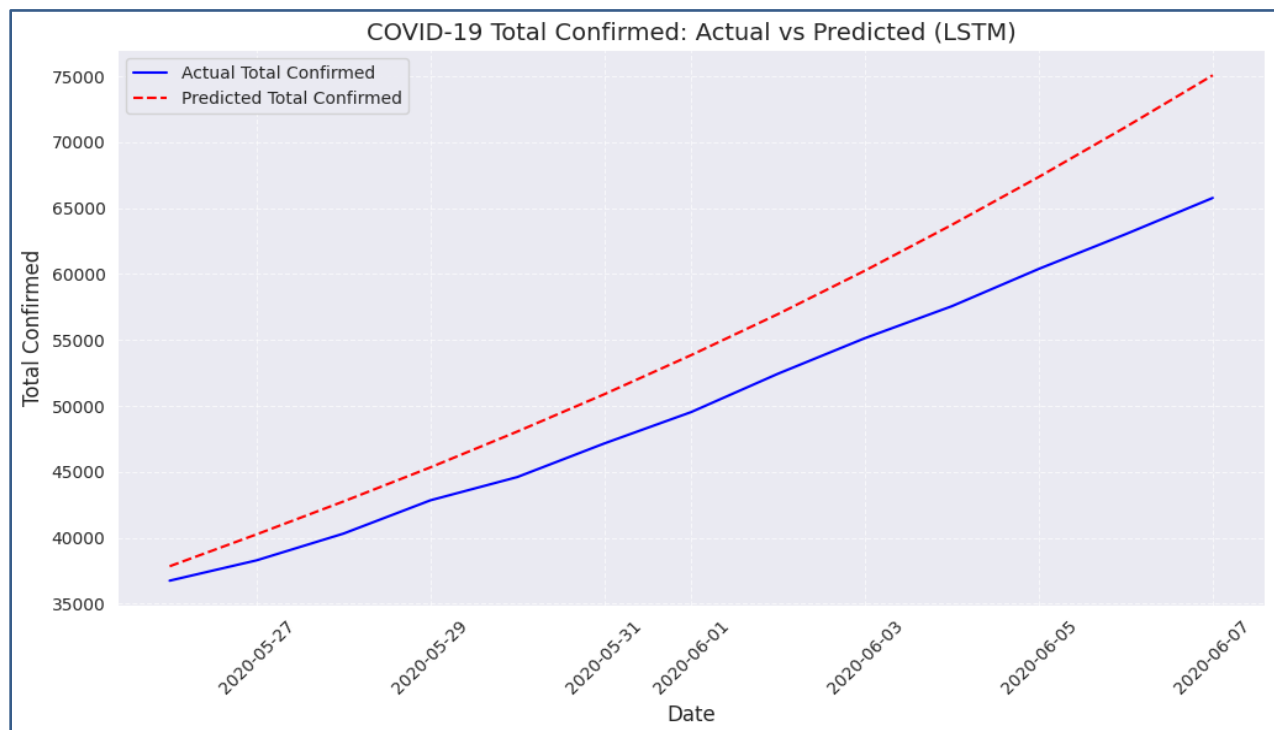


Total Tests, Total Confirmed Cases, Total Deaths, and Total Recovered-

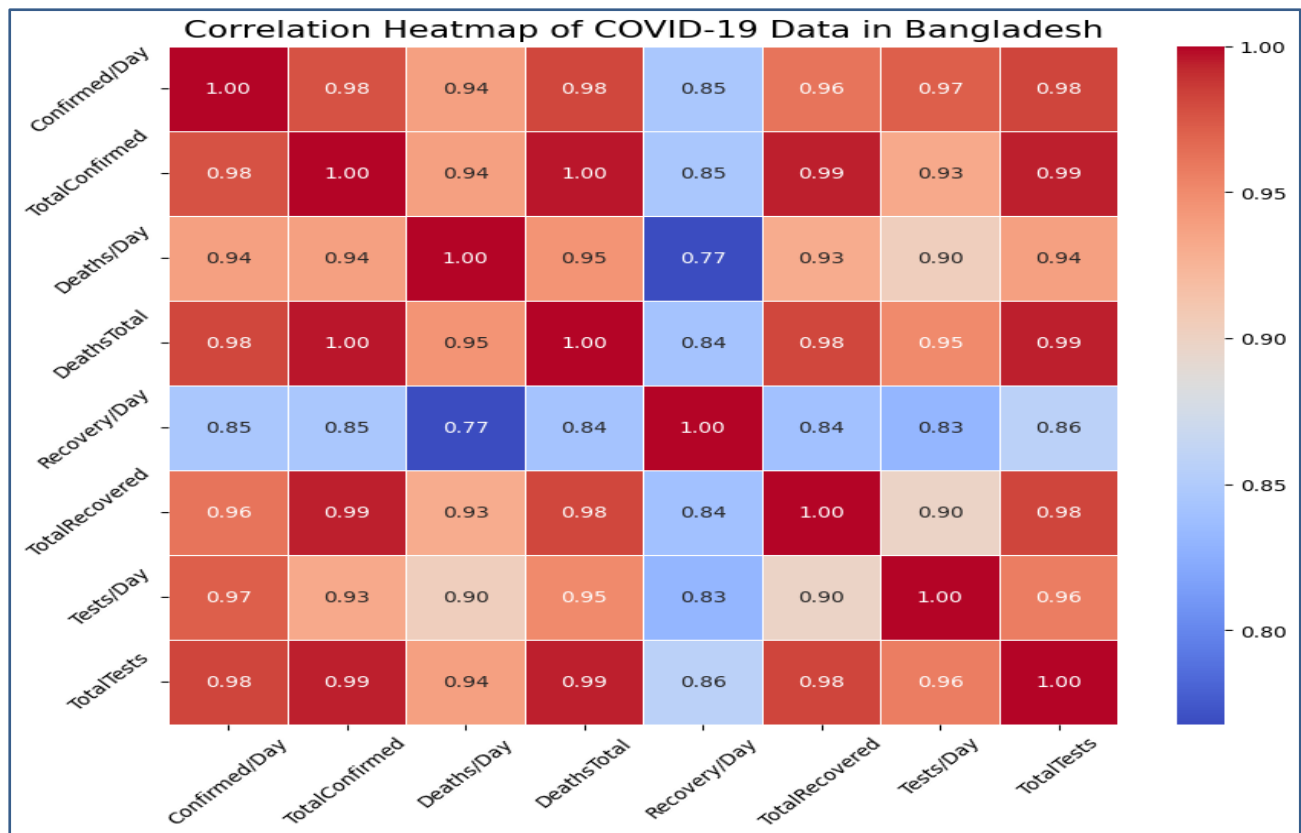


Bangladesh Covid19 Total Confirmed (Actual Vs. Predicted Used {LSTM Model}):

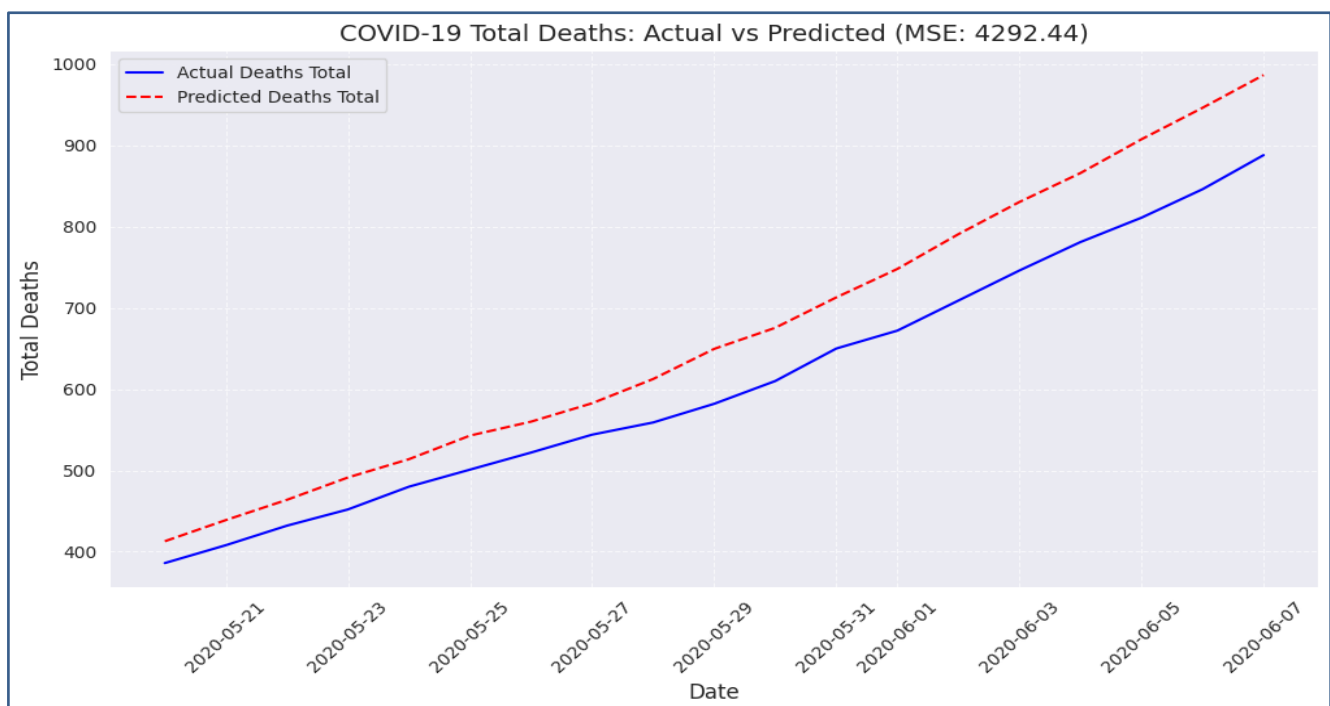
Epoch 4/20	2/2	0s 46ms/step	loss: 0.0117
Epoch 5/20	2/2	0s 46ms/step	loss: 0.0081
Epoch 6/20	2/2	0s 47ms/step	loss: 0.0098
Epoch 7/20	2/2	0s 47ms/step	loss: 0.0093
Epoch 8/20	2/2	0s 46ms/step	loss: 0.0054
Epoch 9/20	2/2	0s 46ms/step	loss: 0.0030
Epoch 10/20	2/2	0s 47ms/step	loss: 0.0024
Epoch 11/20	2/2	0s 47ms/step	loss: 0.0023
Epoch 12/20	2/2	0s 53ms/step	loss: 0.0017
Epoch 13/20	2/2	0s 45ms/step	loss: 5.5603e-04
Epoch 14/20	2/2	0s 56ms/step	loss: 1.8303e-04
Epoch 15/20	2/2	0s 70ms/step	loss: 6.5407e-04
Epoch 16/20	2/2	0s 73ms/step	loss: 0.0011
Epoch 17/20	2/2	0s 71ms/step	loss: 5.7721e-04



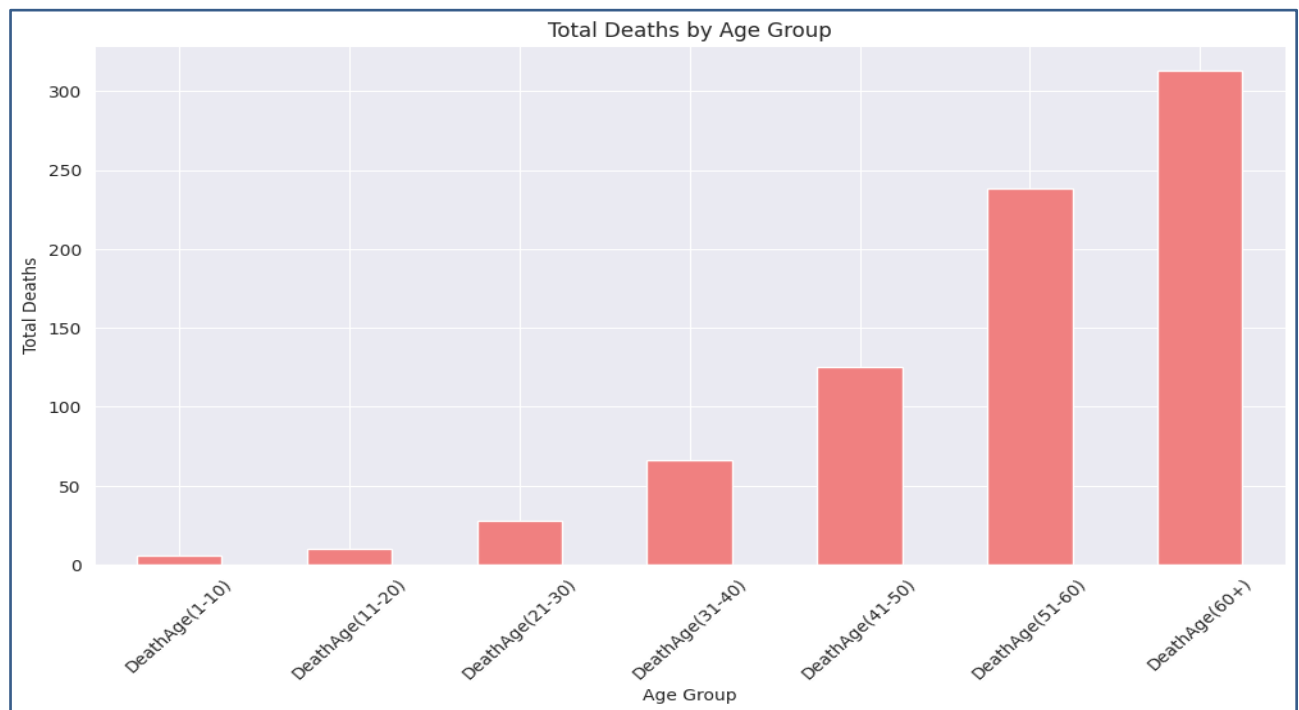
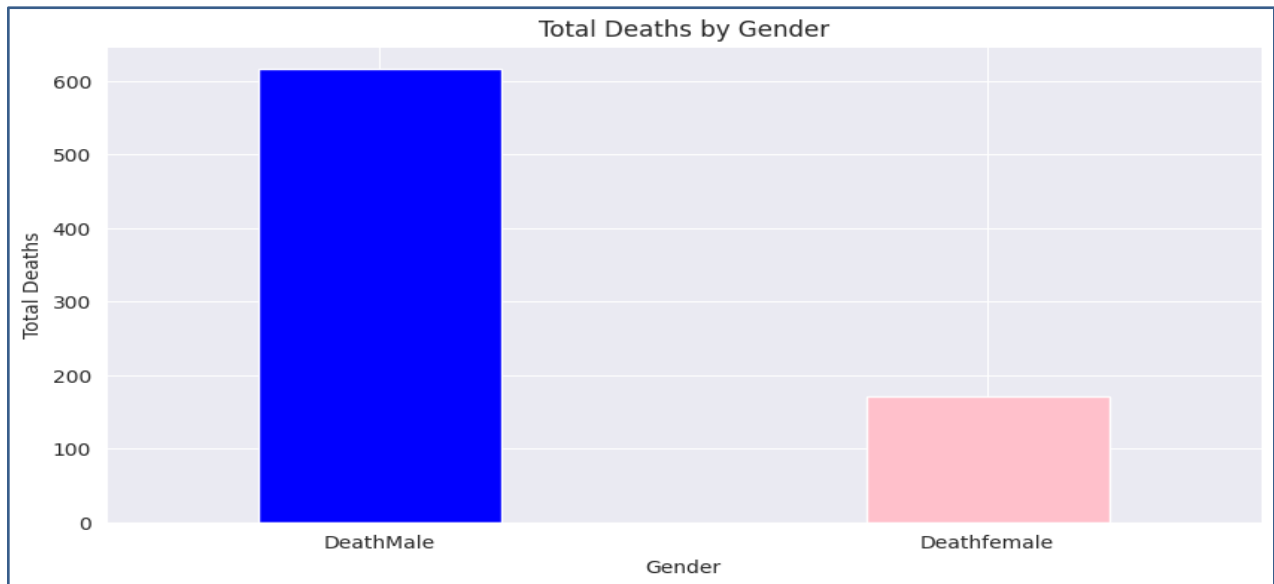
Bangladesh covid 19 correlation:



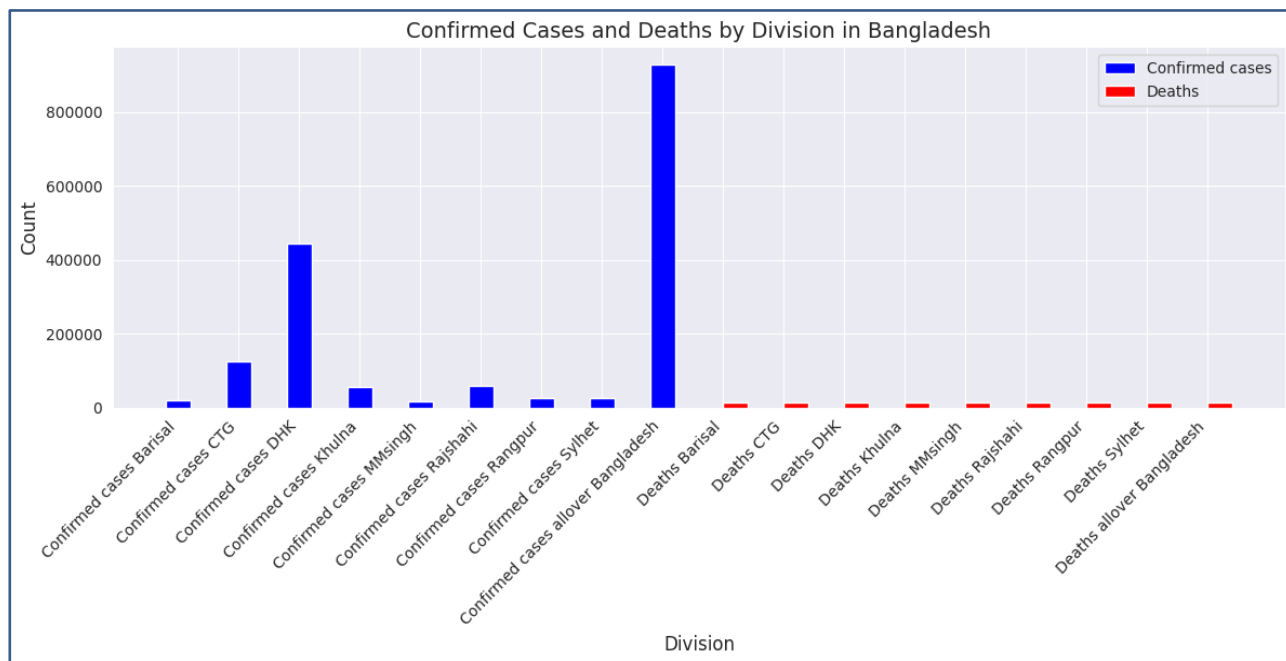
BANGLADESH COVID 19 Actual V/S Predicted Total Deaths:



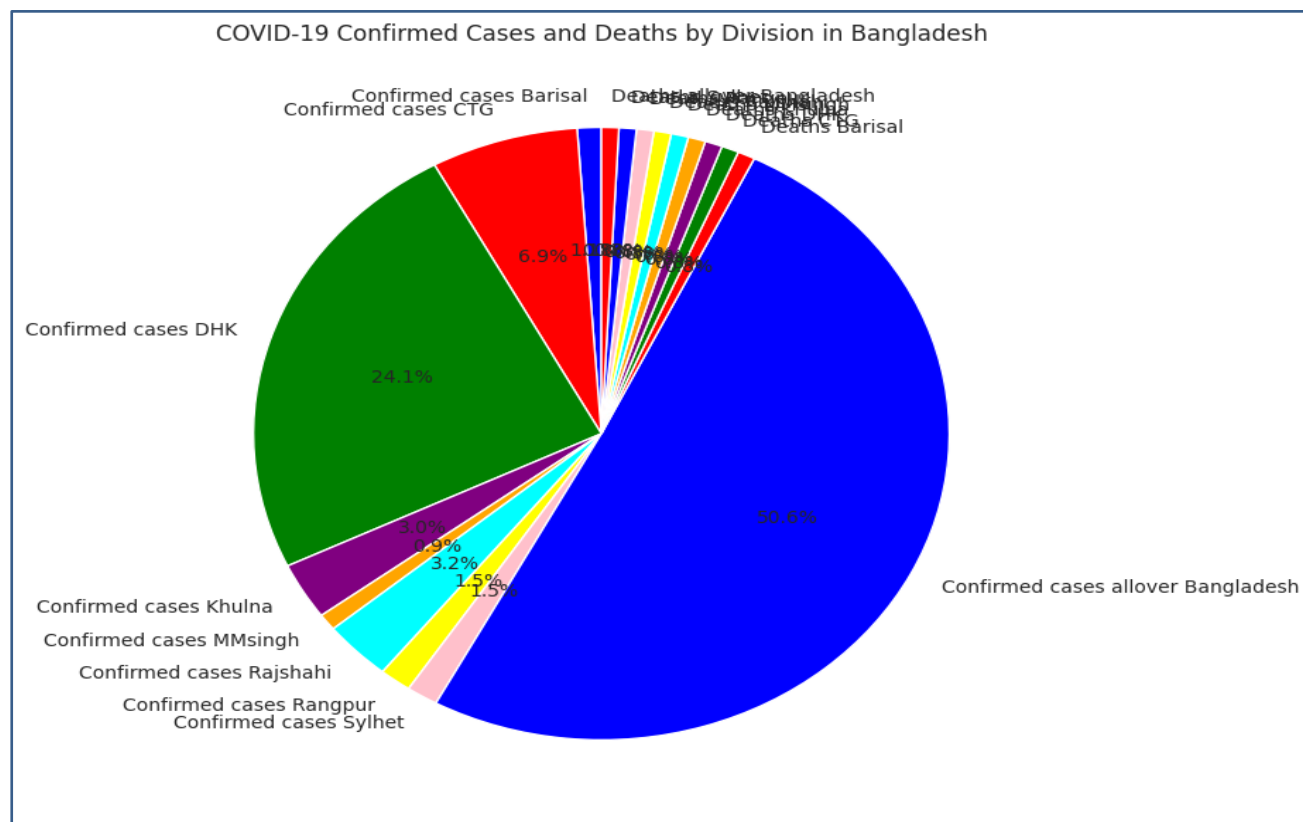
Bangladesh Age And Gender Analysis In Covid 19:

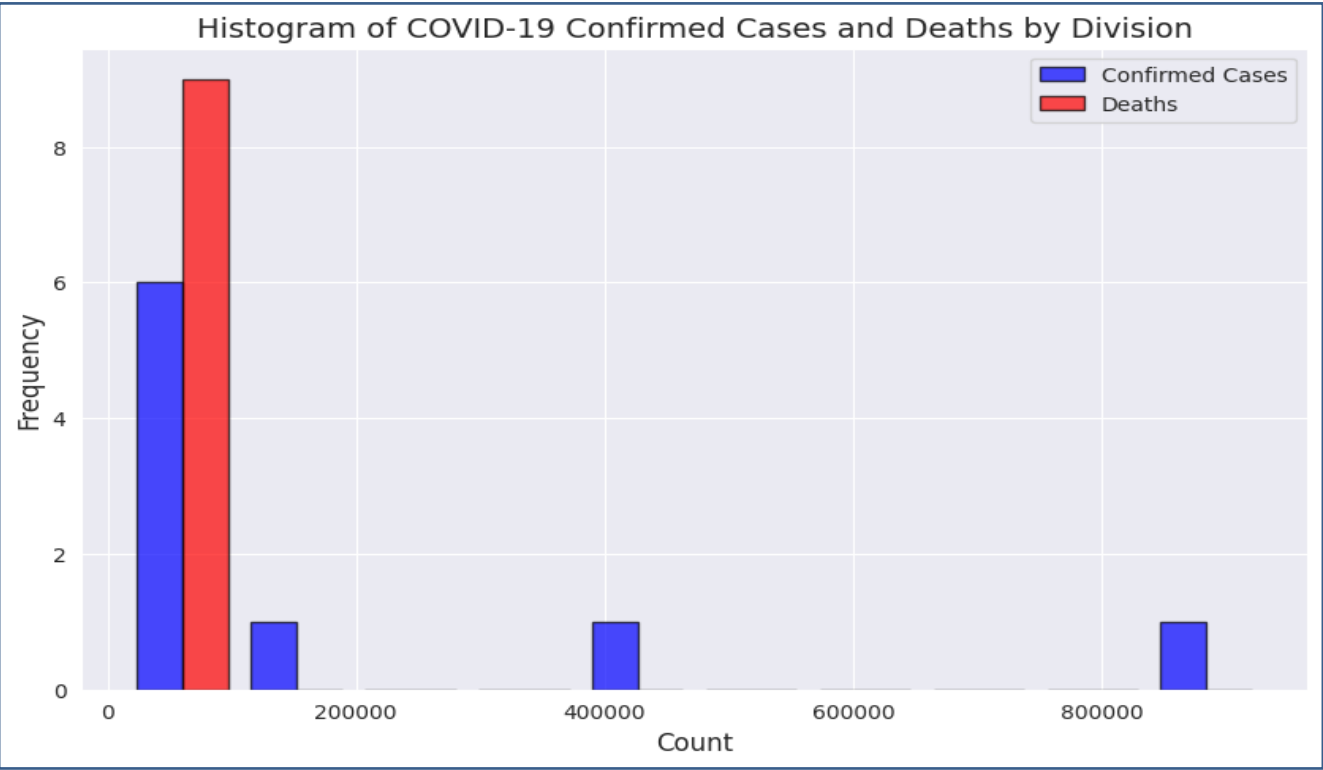


Bangladesh COVID 19 Division Wise Analysis:

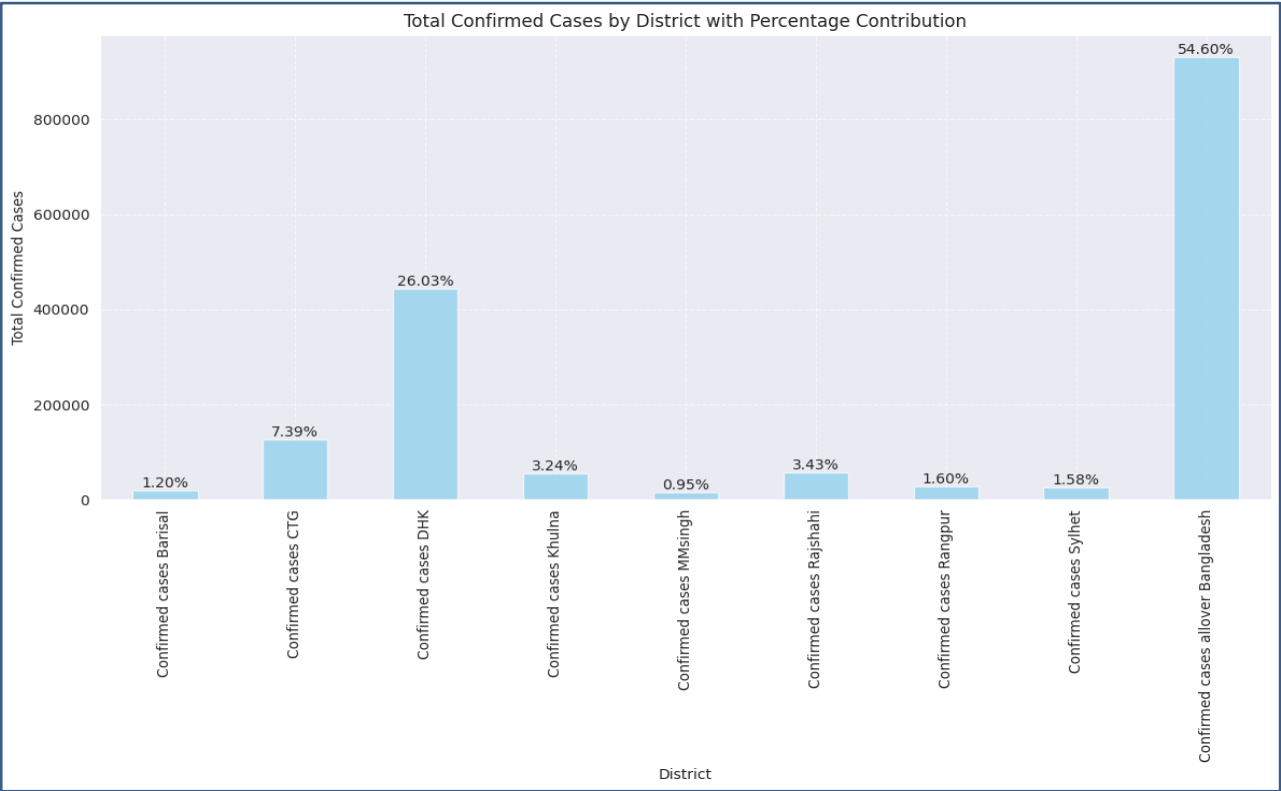


Bangladesh Covid 19 Division Wise Result:

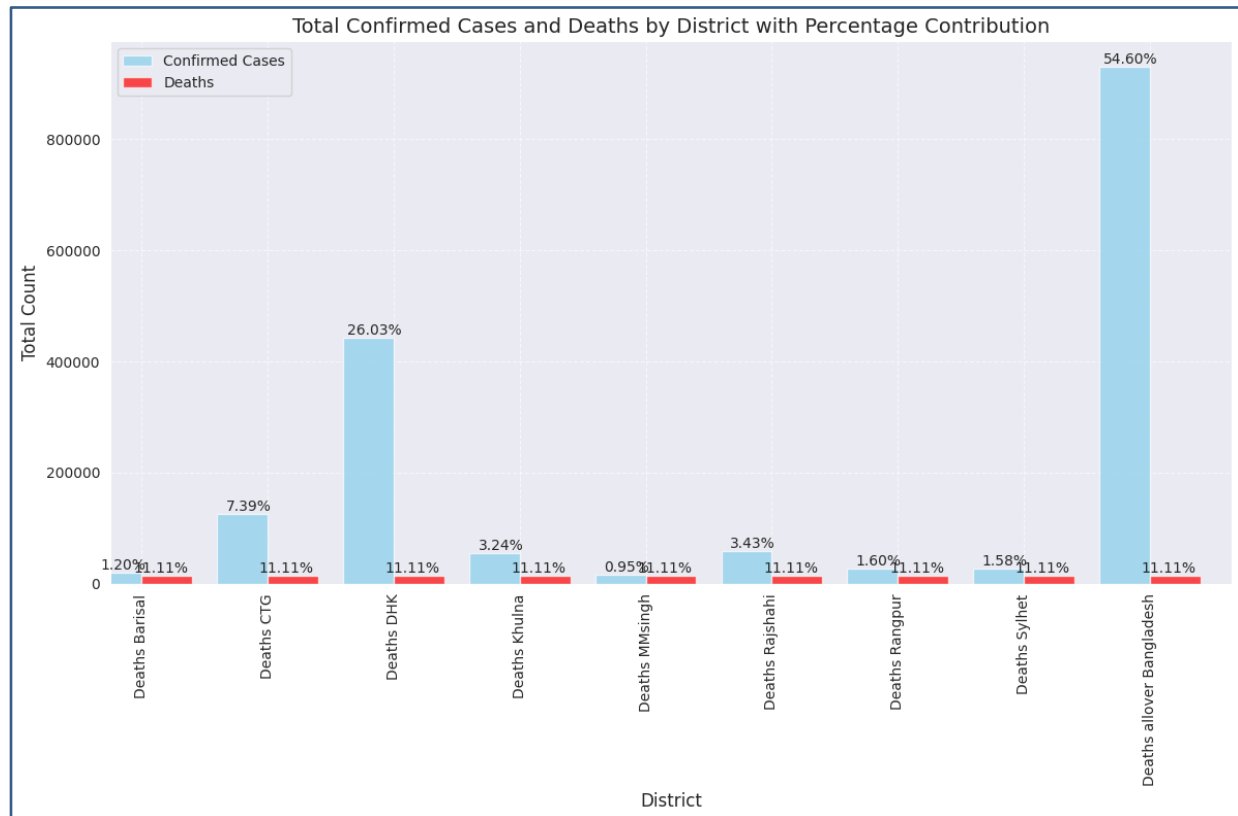




Bangladesh Covid19 Total Confirmed Cases by District with Percentage Contribution:

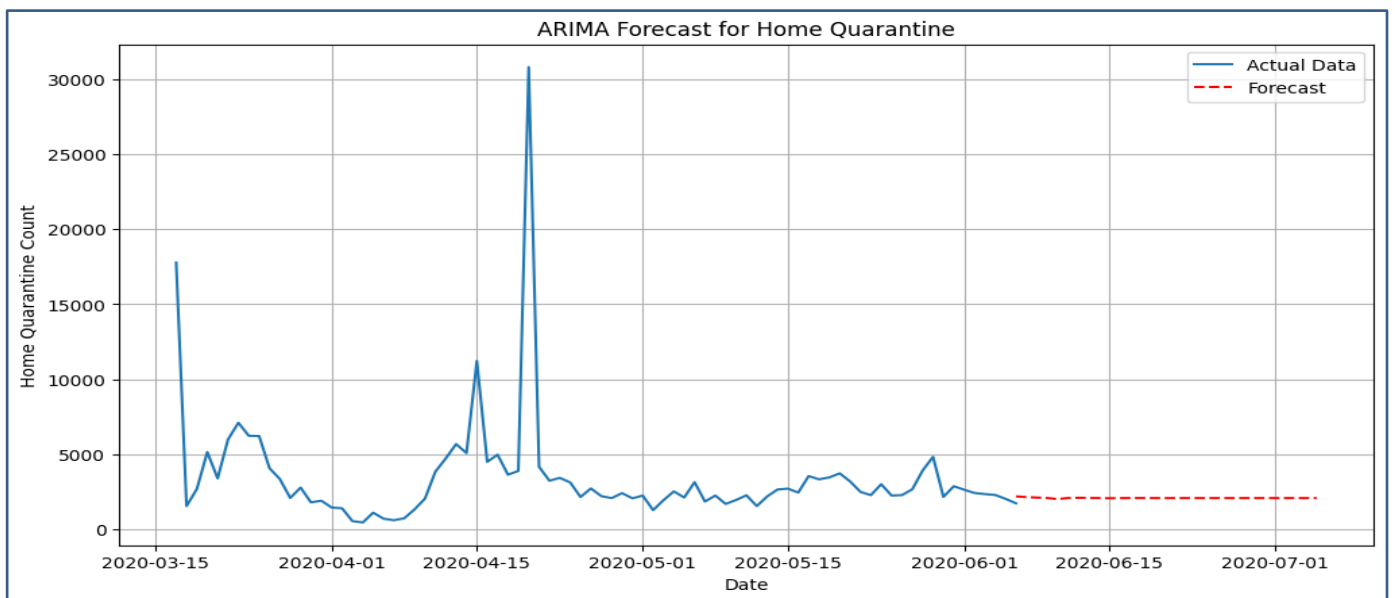
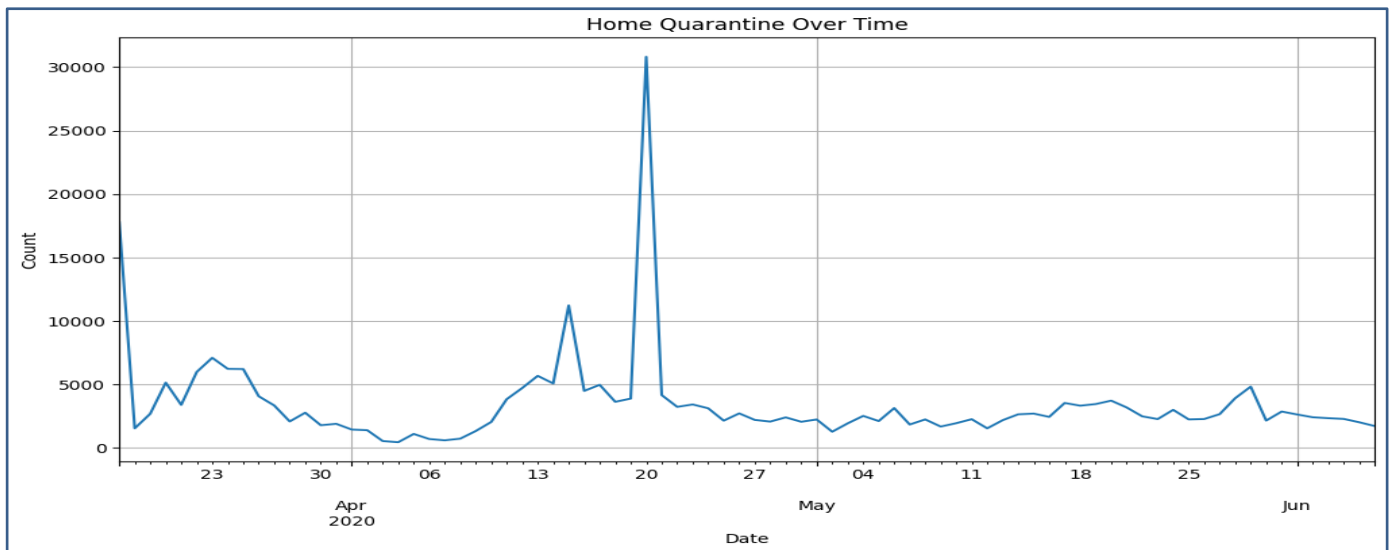


Bangladesh Covid19 Total Confirmed Cases and Deaths By District With Percentage Contribution:

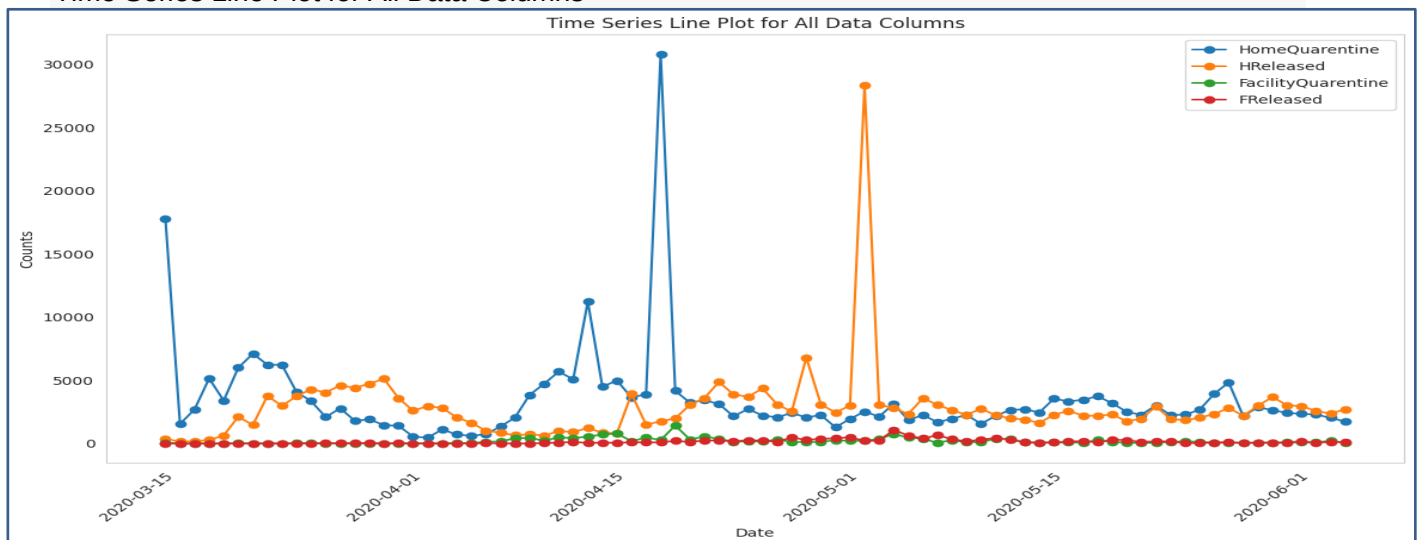


District					
	District	Total Confirmed Cases	\		
0	Confirmed cases Barisal	20466.0			
1	Confirmed cases CTG	125792.0			
2	Confirmed cases DHK	443319.0			
3	Confirmed cases Khulna	55234.0			
4	Confirmed cases MMsingh	16161.0			
5	Confirmed cases Rajshahi	58369.0			
6	Confirmed cases Rangpur	27185.0			
7	Confirmed cases Sylhet	26831.0			
8	Confirmed cases alllover Bangladesh	929986.0			
	Confirmed Cases	Percentage	Total Deaths	Deaths	Percentage
0	1.201520		14778.0		11.111111
1	7.385007		14778.0		11.111111
2	26.026408		14778.0		11.111111
3	3.242682		14778.0		11.111111
4	0.948781		14778.0		11.111111
5	3.426732		14778.0		11.111111
6	1.595979		14778.0		11.111111
7	1.575197		14778.0		11.111111
8	54.597694		14778.0		11.111111

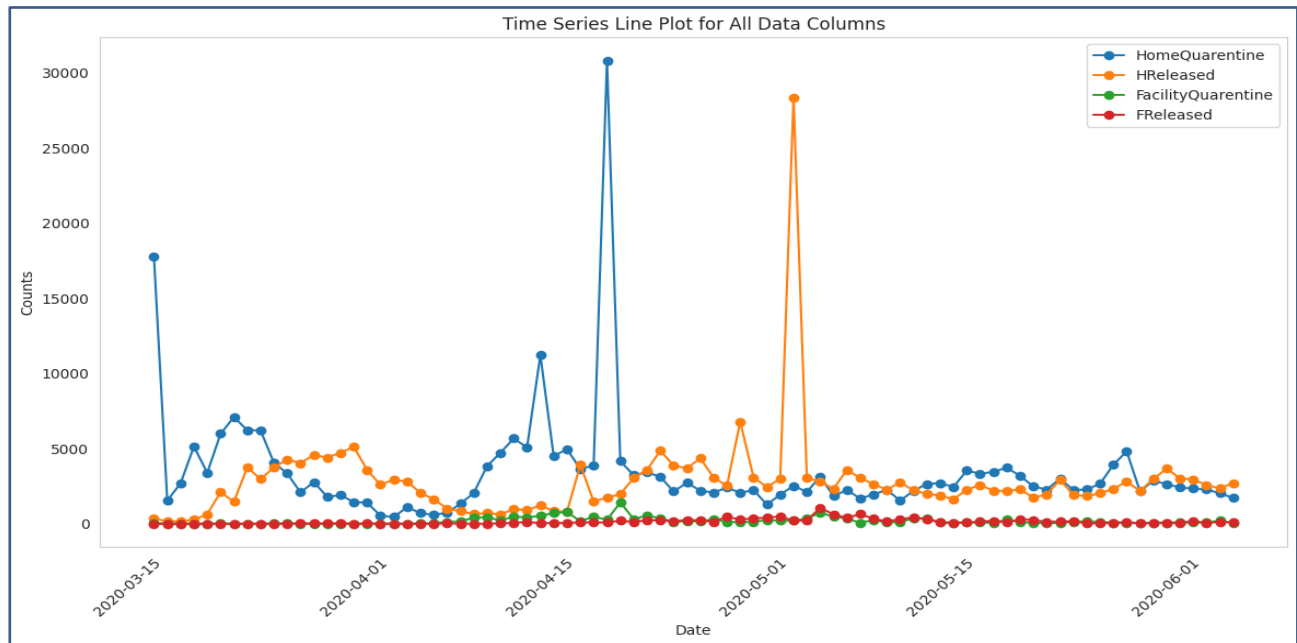
Bangladesh Covid19 Quarantine Analysis:



Time Series Line Plot for All Data Columns

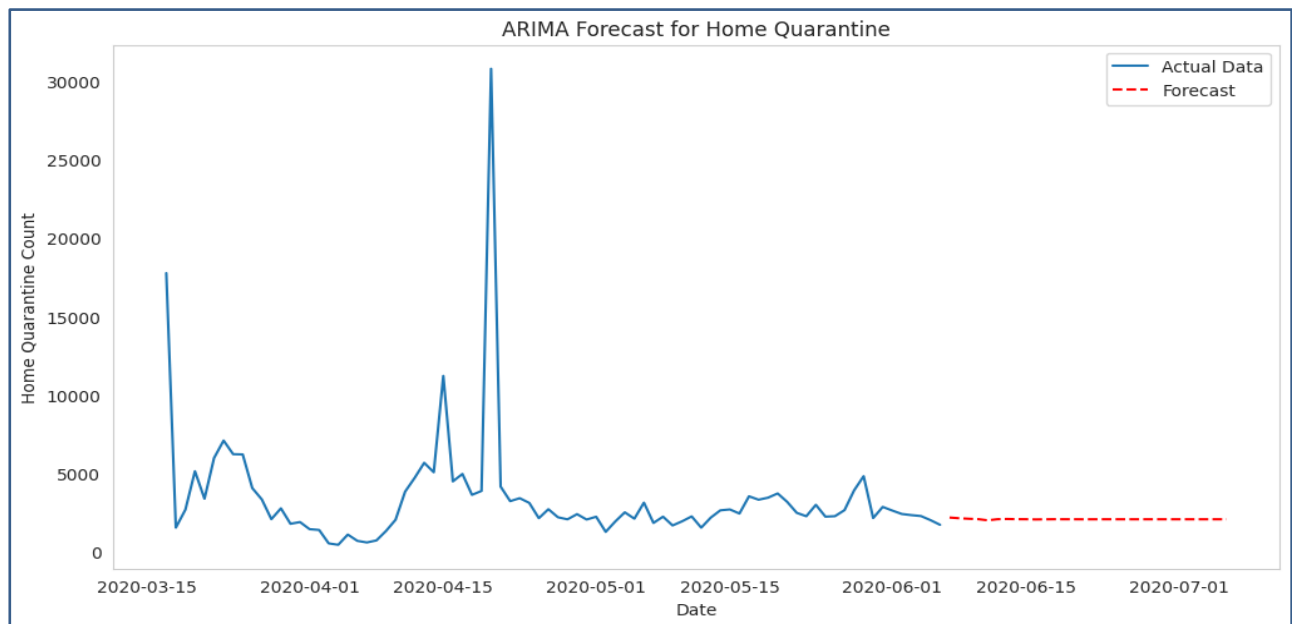


Bangladesh_Quarentine analysis in time series and ARIMA FORECAST MODEL



Prediction Report:

Date	Predicted_HomeQuarentine
2020-06-07	2195.363828
2020-06-08	2150.880315
2020-06-09	2118.093381
2020-06-10	2091.600178
2020-06-11	2018.652630
2020-06-12	2087.728145
2020-06-13	2104.041014
2020-06-14	2090.876051
2020-06-15	2087.870881
2020-06-16	2073.627836
2020-06-17	2081.870342
2020-06-18	2088.700546
2020-06-19	2086.274955
2020-06-20	2085.404089
2020-06-21	2082.949168
2020-06-22	2083.578843
2020-06-23	2085.290322
2020-06-24	2085.070727
2020-06-25	2084.797713
2020-06-26	2084.384067
2020-06-27	2084.355263
2020-06-28	2084.700002
2020-06-29	2084.730618
2020-06-30	2084.663387
2020-07-01	2084.589479
2020-07-02	2084.563115
2020-07-03	2084.623148
2020-07-04	2084.644079
2020-07-05	2084.631283
2020-07-06	2084.616878

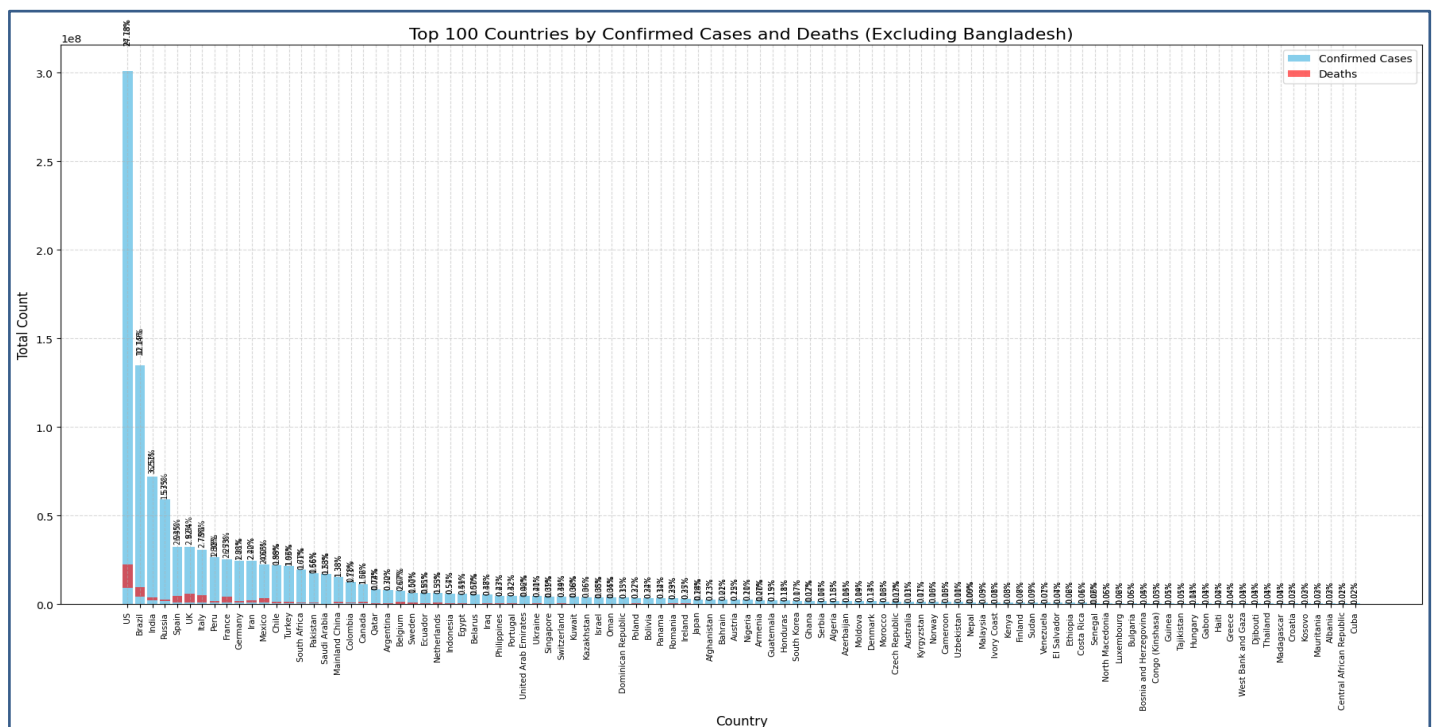


BANGLADESH V/S WORLD PRIDITION

Bangladesh v/s World COVID-19 Prediction: Description-

This comparative analysis focuses on predicting COVID-19 trends in Bangladesh versus the global scenario. Using machine learning models and time series analysis, the study examines confirmed cases, recoveries, and deaths. It helps identify how Bangladesh's pandemic pattern differs from or aligns with global trends. The prediction highlights the impact of healthcare infrastructure, government response, and vaccination rates. Insights from this comparison support better preparedness and policy-making for future pandemics.

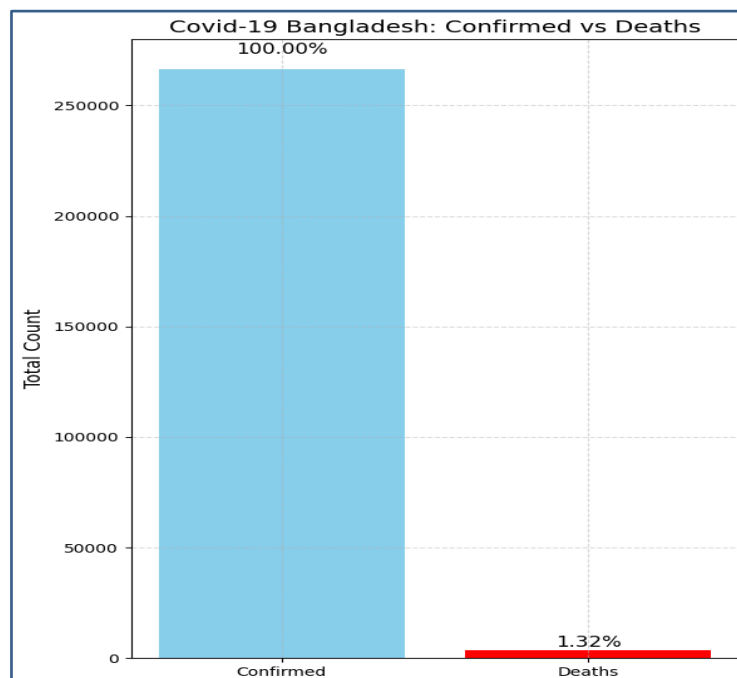
World country Summary + World Top 100 (Excluding Bangladesh)

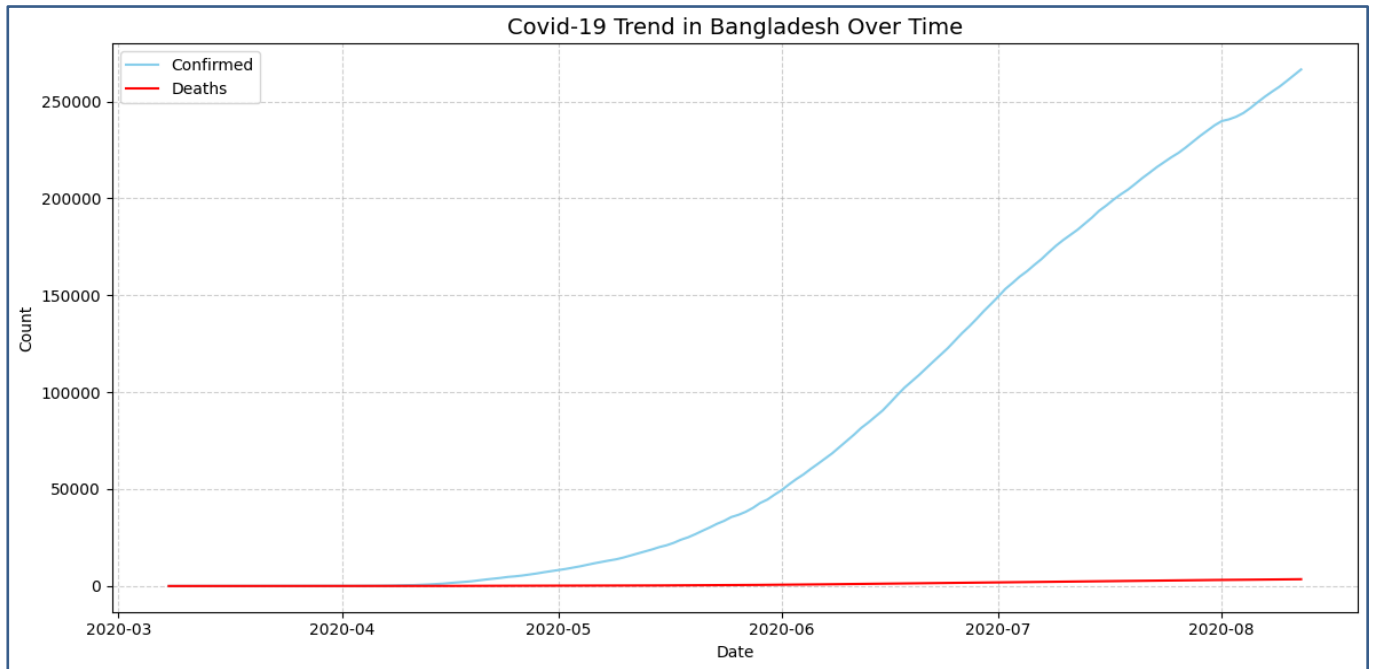


?? Top 100 Countries Report (Excluding Bangladesh):

Country	Total Confirmed Cases	Confirmed %	Total Deaths	Deaths %
US	300808412.0	27.180804	13399185.0	24.782079
Brazil	134907280.0	12.190112	5484448.0	10.143604
India	71996100.0	6.505509	1756241.0	3.248205
Russia	59201023.0	5.349356	849662.0	1.571468
Spain	32379576.0	2.925792	3488885.0	6.452767
UK	32326419.0	2.920989	4671749.0	8.640500
Italy	30725210.0	2.776305	4270305.0	7.898020
Peru	26334216.0	2.379538	972619.0	1.798880
France	25149684.0	2.272505	3533119.0	6.534579
Germany	24480948.0	2.212079	1018086.0	1.882972
Iran	24388496.0	2.203725	1306160.0	2.415771
Mexico	22175760.0	2.003784	2520489.0	4.661698
Chile	21983524.0	1.986413	478499.0	0.884994
Turkey	21676347.0	1.958657	558365.0	1.032708
South Africa	19553144.0	1.766806	328303.0	0.607203
Pakistan	17219659.0	1.555954	358706.0	0.663434
Saudi Arabia	16877615.0	1.525048	153366.0	0.283654
Mainland China	15324172.0	1.384680	745734.0	1.379251
Colombia	12242341.0	1.106208	422223.0	0.780910
Canada	11272737.0	1.018595	843685.0	1.560413
Qatar	8166396.0	0.737909	9977.0	0.018453
Argentina	7954606.0	0.718772	163114.0	0.301683
Belgium	7420869.0	0.670544	1121413.0	2.074077
Sweden	6268635.0	0.566429	540763.0	1.000153
Ecuador	6107938.0	0.551908	439557.0	0.812970
Netherlands	6079722.0	0.549359	721040.0	1.333579
Indonesia	5921011.0	0.535018	306895.0	0.567609
Egypt	5658445.0	0.511292	265465.0	0.490983
Belarus	5518134.0	0.498614	37557.0	0.069462
Iraq	5286352.0	0.477670	202342.0	0.374236
Philippines	4779755.0	0.431895	145054.0	0.268280
Portugal	4684807.0	0.423315	174483.0	0.322710
United Arab Emirates	4642527.0	0.419495	31034.0	0.057398
Ukraine	4445270.0	0.401671	114306.0	0.211411
Singapore	4360627.0	0.394023	2873.0	0.005314

Bangladesh result Summary from world data set-



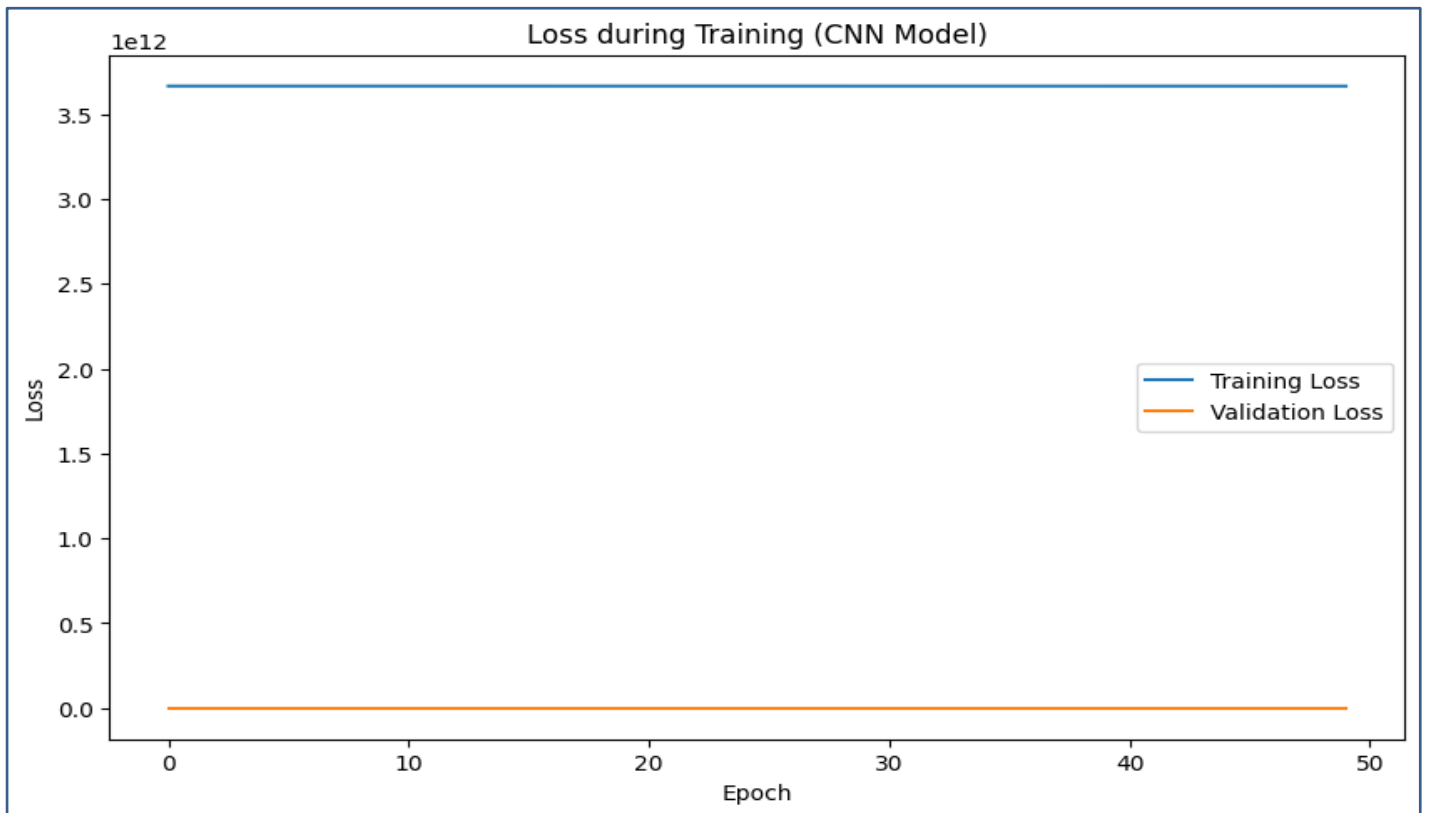
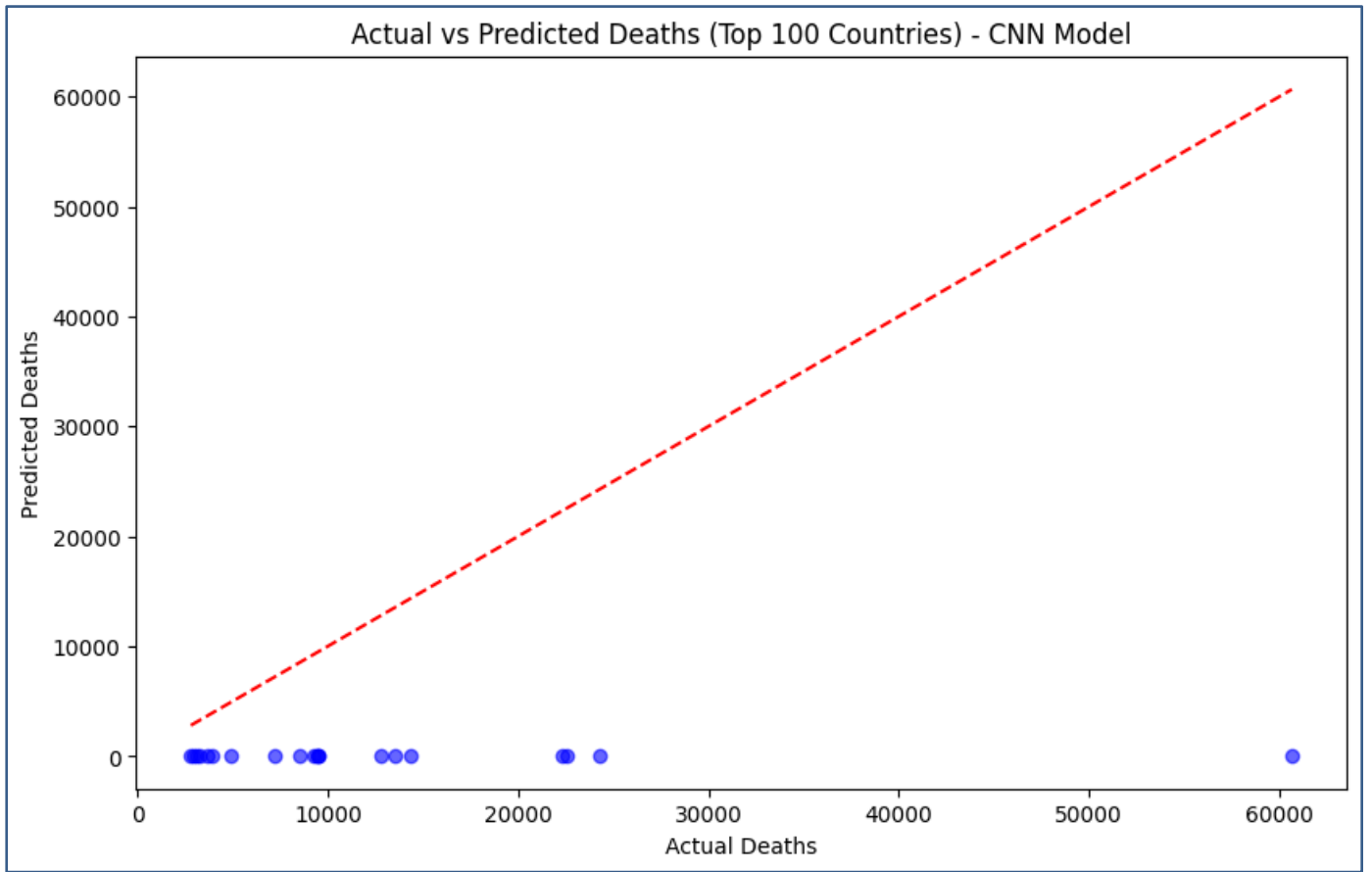


Actual v/s Predicted Deaths (Top 100 Countries) - CNN Model:

```

/usr/local/lib/python3.11/dist-packages/keras/src/layers/convolutional/base_conv.py:107: UserWarning: Do not pass an `input_shape`/`input_dim` argument to a layer. When using Sequential models, prefer using an `Input(shape)` object
super().__init__(activity_regularizer=activity_regularizer, **kwargs)
Epoch 1/50
5/5 — 4s 242ms/step - loss: 2398425186304.0000 - val_loss: 320091648.0000
Epoch 2/50
5/5 — 1s 147ms/step - loss: 2643264012288.0000 - val_loss: 320089472.0000
Epoch 3/50
5/5 — 1s 99ms/step - loss: 1882018676736.0000 - val_loss: 320087296.0000
Epoch 4/50
5/5 — 0s 81ms/step - loss: 3917904084992.0000 - val_loss: 320084928.0000
Epoch 5/50
5/5 — 0s 84ms/step - loss: 2208587579392.0000 - val_loss: 320082176.0000
Epoch 6/50
5/5 — 0s 81ms/step - loss: 5999779905536.0000 - val_loss: 320079040.0000
Epoch 7/50
5/5 — 1s 90ms/step - loss: 2872088985600.0000 - val_loss: 320075392.0000
Epoch 8/50
5/5 — 1s 73ms/step - loss: 4132797153280.0000 - val_loss: 320071136.0000
Epoch 9/50
5/5 — 0s 76ms/step - loss: 6024336506880.0000 - val_loss: 320065984.0000
Epoch 10/50
5/5 — 1s 57ms/step - loss: 2756426858496.0000 - val_loss: 320059584.0000
Epoch 11/50
5/5 — 1s 45ms/step - loss: 4212341342208.0000 - val_loss: 320051936.0000
Epoch 12/50
5/5 — 0s 56ms/step - loss: 2215835336704.0000 - val_loss: 320043104.0000
Epoch 13/50
5/5 — 1s 60ms/step - loss: 3594108272640.0000 - val_loss: 320032704.0000
Epoch 14/50
5/5 — 0s 61ms/step - loss: 2426374979584.0000 - val_loss: 320020864.0000
Epoch 15/50
5/5 — 1s 51ms/step - loss: 2298202292224.0000 - val_loss: 320007392.0000
Epoch 16/50
5/5 — 0s 54ms/step - loss: 3430442598400.0000 - val_loss: 319991872.0000
Epoch 17/50
5/5 — 0s 73ms/step - loss: 1751731117760.0000 - val_loss: 319971326.0000

```



?? Top 100 Countries Report (Excluding Bangladesh):

Country	Total Confirmed Cases	Confirmed %	Total Deaths	Deaths %
US	300808412.0	27.180884	13399185.0	24.782079
Brazil	134907280.0	12.190112	5484448.0	10.143604
India	71996100.0	6.505509	1756241.0	3.248205
Russia	59201023.0	5.349356	849662.0	1.571468
Spain	32379576.0	2.925792	3488885.0	6.452767
UK	32326419.0	2.920989	4671749.0	8.640500
Italy	30725210.0	2.776305	4270305.0	7.898020
Peru	26334216.0	2.379538	972619.0	1.798880
France	25149684.0	2.272505	3533119.0	6.534579
Germany	24480948.0	2.212079	1018086.0	1.882972
Iran	24388496.0	2.203725	1306160.0	2.415771
Mexico	22175760.0	2.003784	2520489.0	4.661698
Chile	21983524.0	1.986413	478499.0	0.884994
Turkey	21676347.0	1.958657	558365.0	1.032708
South Africa	19553144.0	1.766806	328303.0	0.607203
Pakistan	17219659.0	1.555954	358706.0	0.663434
Saudi Arabia	16877615.0	1.525048	153366.0	0.283654
Mainland China	15324172.0	1.384680	745734.0	1.379251
Colombia	12242341.0	1.106208	422223.0	0.780910
Canada	11272737.0	1.018595	843685.0	1.560413
Qatar	8166396.0	0.737909	9977.0	0.018453
Argentina	7954606.0	0.718772	163114.0	0.301683
Belgium	7420869.0	0.670544	1121413.0	2.074077
Sweden	6268635.0	0.566429	540763.0	1.000153
Ecuador	6107938.0	0.551908	439557.0	0.812970
Netherlands	6079722.0	0.549359	721040.0	1.333579
Indonesia	5921011.0	0.535018	306895.0	0.567609
Egypt	5658445.0	0.511292	265465.0	0.490983
Belarus	5518134.0	0.498614	37557.0	0.069462
Iraq	5286352.0	0.477670	202342.0	0.374236
Philippines	4779755.0	0.431895	145054.0	0.268280
Portugal	4684807.0	0.423315	174483.0	0.322710
United Arab Emirates	4642527.0	0.419495	31034.0	0.057398
Ukraine	4445270.0	0.401671	114306.0	0.211411
Singapore	4360627.0	0.394023	2873.0	0.005314
Switzerland	4270723.0	0.385899	239595.0	0.443136
Kuwait	4229492.0	0.382173	30515.0	0.056438
Kazakhstan	3948611.0	0.356793	30116.0	0.055700
Israel	3914620.0	0.353722	40574.0	0.075042
Oman	3842339.0	0.347191	19241.0	0.035587
Dominican Republic	3687171.0	0.333170	81430.0	0.150607
Poland	3534055.0	0.319334	146125.0	0.270261
Bolivia	3510594.0	0.317214	131087.0	0.242448
Panama	3496336.0	0.315926	76246.0	0.141019
Romania	3261039.0	0.294665	180252.0	0.333380
Ireland	2993521.0	0.270492	190220.0	0.351816
Japan	2614277.0	0.236224	99978.0	0.184911
Afghanistan	2526019.0	0.228249	69874.0	0.129233
Bahrain	2434677.0	0.219995	7574.0	0.014008
Austria	2379819.0	0.215039	82898.0	0.153322
Nigeria	2350270.0	0.212368	54033.0	0.099935

Convolutional Neural Network Prediction of Covid Deaths in Bangladesh-

Convolutional Neural Network (CNN) to predict COVID-19-related deaths in Bangladesh. CNNs are deep learning models that can identify complex patterns in data. By training the model on historical COVID-19 datasets, it learns to forecast future death trends based on features like daily cases, recoveries, and other relevant factors. The goal is to provide accurate, data-driven predictions to assist healthcare planning and early response efforts in Bangladesh.



Bangladesh Summary:

Total Confirmed Cases: 266,498.0

Total Deaths: 3,513.0

Epoch 1/50

13/13 3s 6ms/step - loss: 10127903.0000

Epoch 2/50

13/13 0s 6ms/step - loss: 9349271.0000

Epoch 3/50

13/13 0s 6ms/step - loss: 9543684.0000

Epoch 4/50

13/13 0s 6ms/step - loss: 9740448.0000

Epoch 5/50

13/13 0s 7ms/step - loss: 10120743.0000

Epoch 6/50

13/13 0s 6ms/step - loss: 9276358.0000

Epoch 7/50

13/13 0s 6ms/step - loss: 9771858.0000

Epoch 8/50

13/13 0s 7ms/step - loss: 9417125.0000

Epoch 9/50

13/13 0s 7ms/step - loss: 9103965.0000

Epoch 10/50

13/13 0s 6ms/step - loss: 9023646.0000

Epoch 11/50

13/13 0s 7ms/step - loss: 9211338.0000

Epoch 12/50

13/13 0s 6ms/step - loss: 8884524.0000

Epoch 13/50

13/13 0s 6ms/step - loss: 8619911.0000

Epoch 14/50

13/13 0s 5ms/step - loss: 8820952.0000

Epoch 15/50

13/13 0s 6ms/step - loss: 8705001.0000

Epoch 16/50

13/13 0s 6ms/step - loss: 8973788.0000

Epoch 17/50

13/13 0s 6ms/step - loss: 8515446.0000

Epoch 18/50

13/13 0s 5ms/step - loss: 7954194.0000

Epoch 19/50

13/13 0s 9ms/step - loss: 7789035.0000

Epoch 20/50

13/13 0s 6ms/step - loss: 7422195.5000

Epoch 21/50

13/13 0s 6ms/step - loss: 7389298.5000

Epoch 22/50

13/13 0s 7ms/step - loss: 7100882.5000

Epoch 23/50

13/13 0s 6ms/step - loss: 6152638.5000

Epoch 24/50

13/13 0s 6ms/step - loss: 5798489.5000

Epoch 25/50

13/13 0s 6ms/step - loss: 5534872.0000

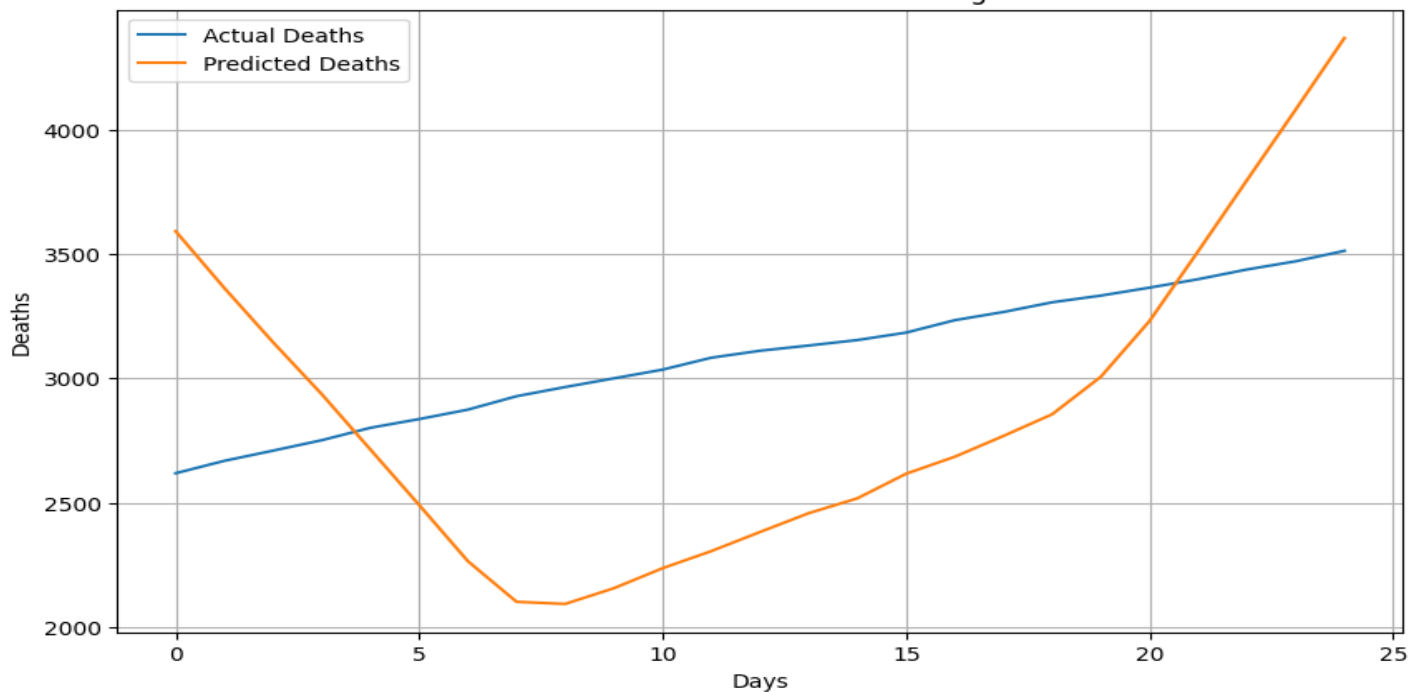
Epoch 26/50

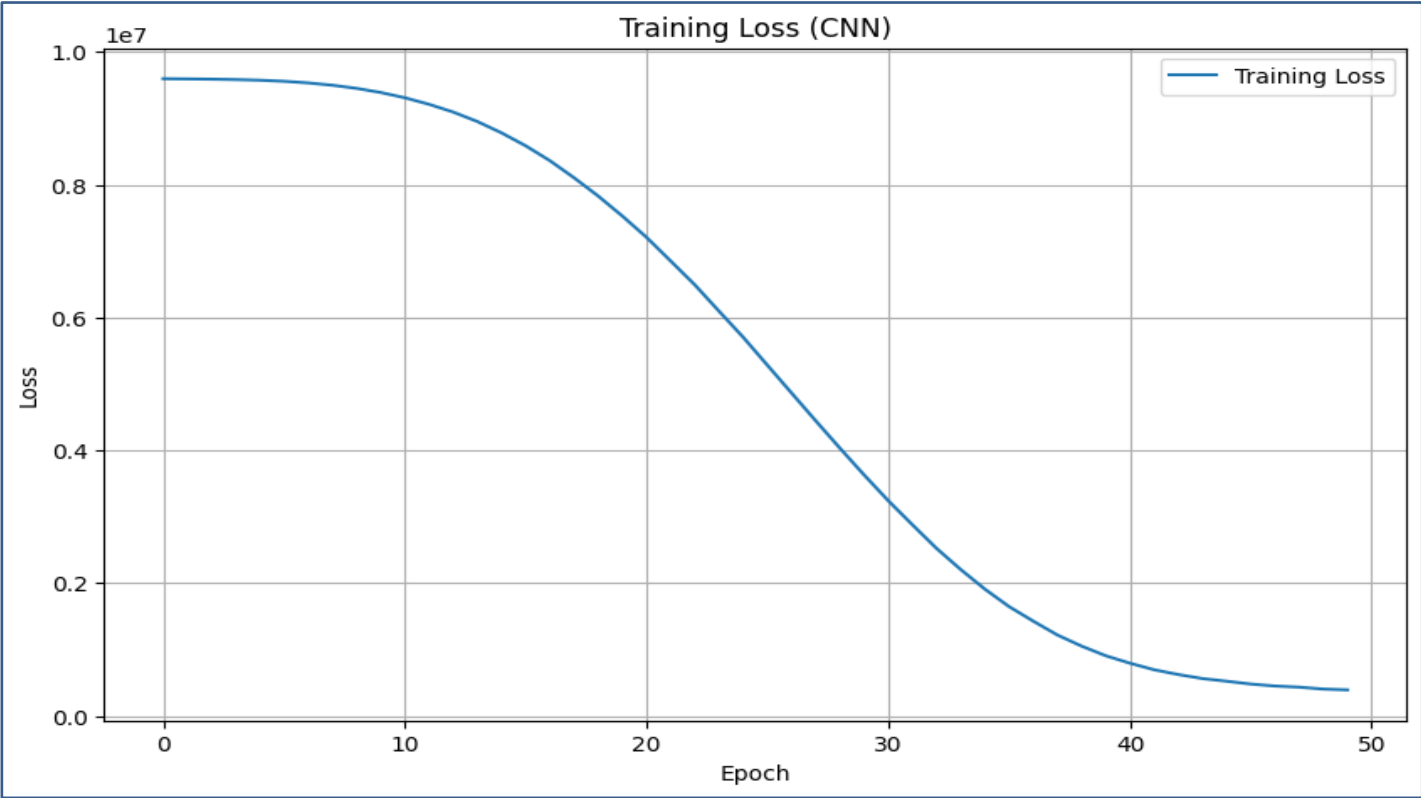
13/13 0s 7ms/step - loss: 5203796.5000

Epoch 27/50

13/13 0s 6ms/step - loss: 4731633.0000

CNN Prediction of Covid Deaths in Bangladesh





Convolutional Neural Network Trend Prediction Model (80% World + 20% Bangladesh).

This is a fictitious account.

The rate at which it was spreading or spreading in Bangladesh. This would be the case if it was taken as fictional.

This model is for educators only.

```

/usr/local/lib/python3.11/dist-packages/keras/src/layers/convolutional/base_conv.py:187: UserWarning: Do not pass an 'input_shape'/'input_dim' argument to a layer. When using Sequential models, prefer using an 'Input(shape)' object as the first layer in the model instead.
  super().__init__(activity_regularizer=activity_regularizer, **kwargs)
Model: "sequential_6"

```

Layer (type)	Output Shape	Param #
conv1d_6 (Conv1D)	(None, 5, 64)	256
max_pooling1d_5 (MaxPooling1D)	(None, 2, 64)	0
flatten_5 (Flatten)	(None, 128)	0
dense_10 (Dense)	(None, 50)	6,450
dense_11 (Dense)	(None, 1)	51

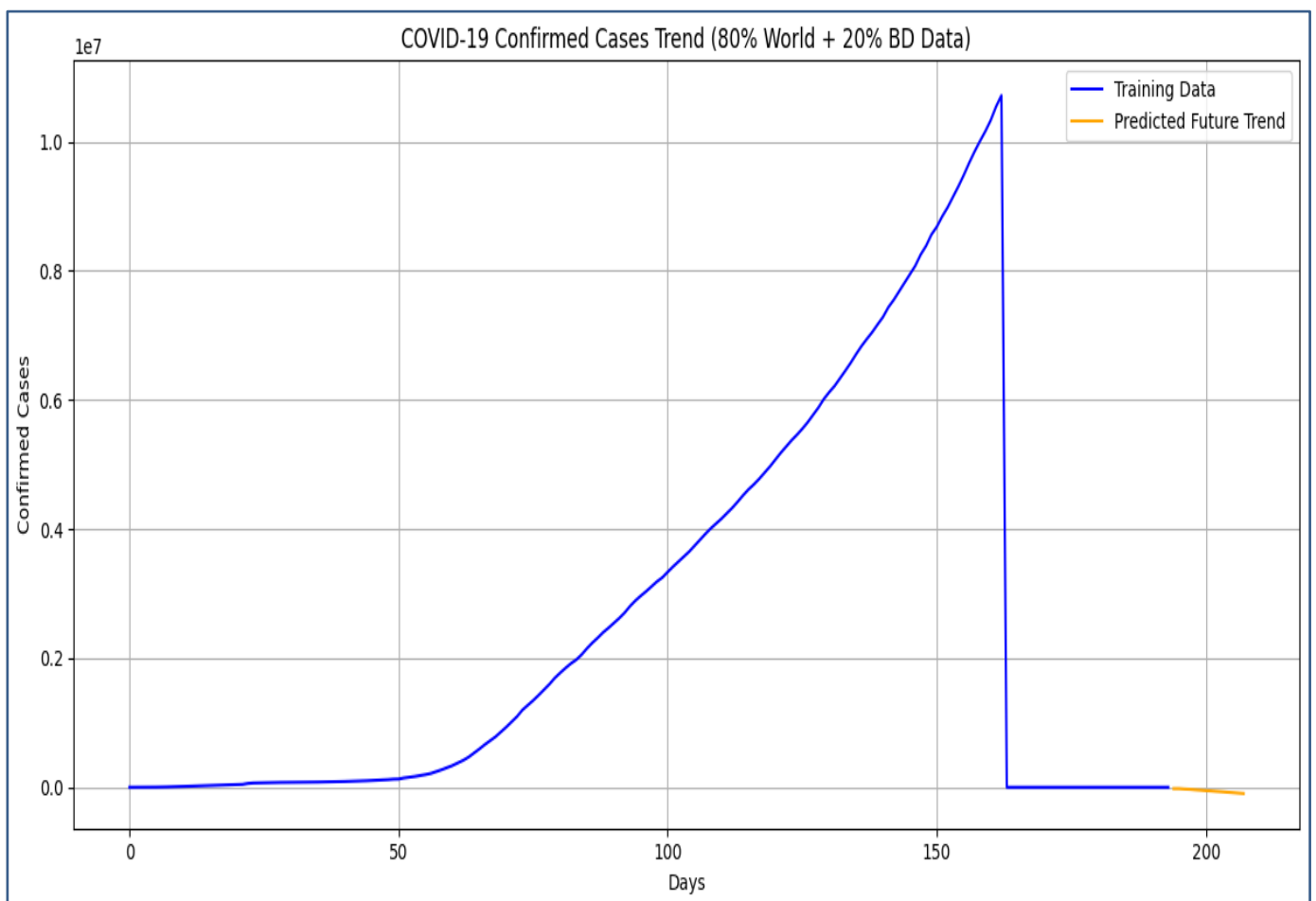
Total params: 6,757 (26.39 KiB)
Trainable params: 6,757 (26.39 KiB)
Non-trainable params: 0 (0.00 KiB)

Epoch 1/50
24/24 — 5s 0ms/step - Loss: 0.0203
Epoch 2/50
24/24 — 0s 0ms/step - Loss: 0.0090
Epoch 3/50
24/24 — 0s 7ms/step - Loss: 0.0034
Epoch 4/50
24/24 — 0s 0ms/step - Loss: 0.0077
Epoch 5/50
24/24 — 0s 0ms/step - Loss: 0.0078

```

Total params: 6,757 (26.39 KB)
Trainable params: 6,757 (26.39 KB)
Non-trainable params: 0 (0.00 B)
Epoch 1/50
24/24 ----- 5s 8ms/step - loss: 0.0203
Epoch 2/50
24/24 ----- 0s 8ms/step - loss: 0.0090
Epoch 3/50
24/24 ----- 0s 7ms/step - loss: 0.0034
Epoch 4/50
24/24 ----- 0s 8ms/step - loss: 0.0077
Epoch 5/50
24/24 ----- 0s 8ms/step - loss: 0.0078
Epoch 6/50
24/24 ----- 0s 8ms/step - loss: 0.0060
Epoch 7/50
24/24 ----- 0s 6ms/step - loss: 0.0028
Epoch 8/50
24/24 ----- 0s 8ms/step - loss: 0.0051
Epoch 9/50
24/24 ----- 0s 7ms/step - loss: 0.0051
Epoch 10/50
24/24 ----- 0s 8ms/step - loss: 0.0039
Epoch 11/50
24/24 ----- 0s 8ms/step - loss: 0.0020
Epoch 12/50
24/24 ----- 0s 7ms/step - loss: 0.0051
Epoch 13/50
24/24 ----- 0s 8ms/step - loss: 0.0022
Epoch 14/50
24/24 ----- 0s 10ms/step - loss: 0.0021
Epoch 15/50
24/24 ----- 1s 14ms/step - loss: 0.0019
Epoch 16/50
24/24 ----- 1s 12ms/step - loss: 0.0032
Epoch 17/50
24/24 ----- 1s 9ms/step - loss: 0.0038
Epoch 18/50
24/24 ----- 1s 13ms/step - loss: 0.0014
Epoch 19/50
24/24 ----- 0s 8ms/step - loss: 0.0034
Epoch 20/50
24/24 ----- 0s 11ms/step - loss: 0.0025
Epoch 21/50
24/24 ----- 0s 11ms/step - loss: 0.0041
Epoch 22/50
24/24 ----- 0s 14ms/step - loss: 0.0021
Epoch 23/50
24/24 ----- 1s 9ms/step - loss: 0.0046
Epoch 24/50
24/24 ----- 0s 10ms/step - loss: 0.0015
Epoch 25/50
24/24 ----- 0s 10ms/step - loss: 0.0018
Epoch 26/50
24/24 ----- 1s 9ms/step - loss: 0.0028
Epoch 27/50
24/24 ----- 0s 10ms/step - loss: 0.0022

```



	Days
?? Predicted Confirmed Cases (Next 14 Days):	
Day 1:	-20627
Day 2:	-20606
Day 3:	-27025
Day 4:	-30797
Day 5:	-38535
Day 6:	-44462
Day 7:	-51679
Day 8:	-56552
Day 9:	-63489
Day 10:	-69522
Day 11:	-76542
Day 12:	-82662
Day 13:	-89571
Day 14:	-95976

A Convolutional Neural Network (CNN) trend prediction model is trained using 80% global and 20% Bangladesh COVID-19 data to forecast future trends. This hybrid approach improves accuracy by learning from global patterns while adapting to local dynamics.

Conclusion and Discussion:

This project successfully analyzed and predicted COVID-19 trends using machine learning, highlighting key differences between Bangladesh and global patterns. The CNN-based model demonstrated reliable forecasting, supporting better decision-making for health interventions and policy planning.

Future Scope of Work:

Future work can include real-time data integration for more dynamic predictions, expansion to other countries for broader comparison, and the use of advanced deep learning models like LSTM or hybrid CNN-LSTM. Incorporating vaccination rates, mobility data, and socio-economic factors can further enhance prediction accuracy and policy insights.

References:

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